SIEMENS

SIMATIC Ident

RFID systems
SIMATIC RF690R, RF695R

Configuration Manual

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Legal information

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This manual contains notices you have to observe in order to ensure your personal safety, as well as to prevent damage to property. The notices referring to your personal safety are highlighted in the manual by a safety alert symbol, notices referring only to property damage have no safety alert symbol. These notices shown below are graded according to the degree of danger.

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indicates that minor personal injury can result if proper precautions are not taken.

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We have reviewed the contents of this publication to ensure consistency with the hardware and software described. Since variance cannot be precluded entirely, we cannot guarantee full consistency. However, the information in this publication is reviewed regularly and any necessary corrections are included in subsequent editions.

Siemens Aktiengesellschaft
Digital Industries
Postfach 48 48
90026 NÜRNBERG
GERMANY

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Introduction

1

Purpose of this document

This manual contains all the information required for the parameter assignment and commissioning of the SIMATIC RF690R and RF695R readers of the SIMATIC RF600 system.

This manual is intended for:

- Commissioning engineers
- Configuration engineers
- Service technicians

Scope of validity of this document

This documentation is valid for all supplied versions of the SIMATIC RF690R/RF695R readers and describes the delivery state as of 06/2025 and firmware version V1.2.

Registered trademarks

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SIMATIC ®, SIMATIC RF ® and MOBY ®

Documentation classification

You will find information on the properties, technical specifications and possible applications of the RF600 readers in the "SIMATIC RF600" system manual.

You can find all relevant information for XML configuration and programming in the "XML Programming for SIMATIC Ident" manual. You can find detailed information on configuring and programming over OPC UA in the "OPC UA for SIMATIC Ident" manual.

You can find the current versions of the various manuals on the pages of the Siemens Industry Online Support (https://support.industry.siemens.com/cs/ww/en/ps/14970/man).

Abbreviations and naming conventions

The following terms/abbreviations are used synonymously in this document:

Transponder	Tag, data carrier, mobile data storage (MDS)
Communications module (CM)	Interface module (ASM)

Identification Link

The QR code on the front of this product uniquely identifies the product. This code may be printed on the documentation accompanying the product and on its packaging.

The QR code is an ID link (Identification Link) according to IEC 61406-1.

By scanning the code or ID link (e.g. with a smartphone), you can open the Siemens website associated with the product with the digital name plate.

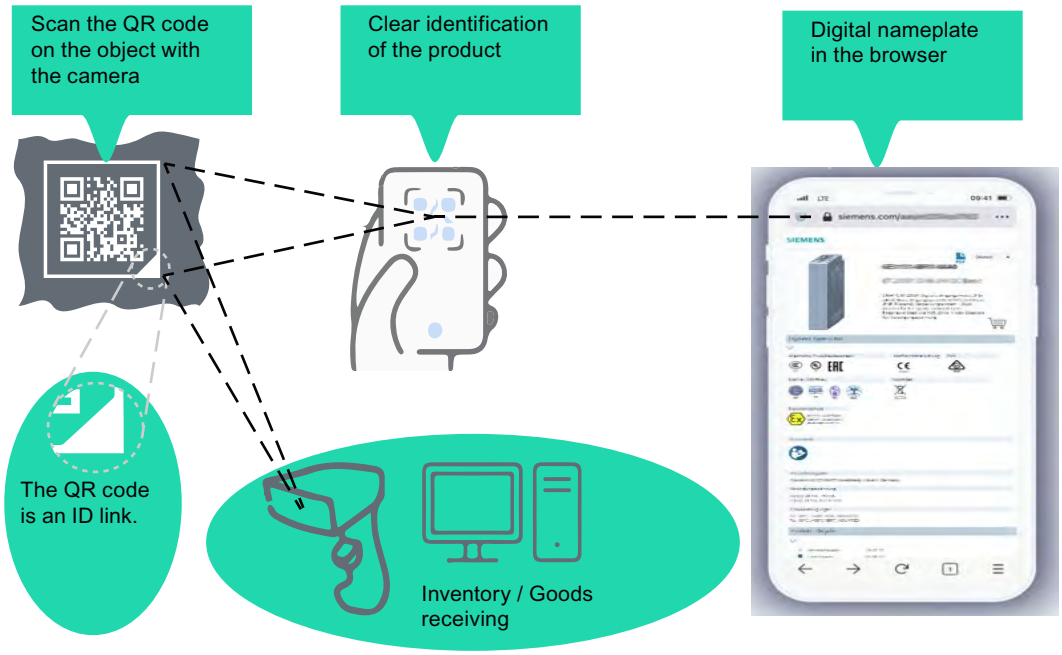


Figure 1-1 Application & functions of the Identification Link

Digital nameplate

The digital nameplate is:

- A contribution by Siemens to reduction of the CO_2 footprint and a means of saving resources.
- An efficient way to access the product information available on the Internet. This information is kept up to date over the entire product life.
  For your system documentation, you have the option to link or download the following information from the website:
  - Technical specifications
  - Certifications and declaration of conformity
  - Manufacturer information
  - Product identification
  - User documentation
  - Repurchase
  - Responsive web design
  - Other information

Security recommendations

2

To prevent unauthorized access, observe the following security recommendations when working with the reader and WBM (Web Based Management).

General

- Check regularly that the device complies with these recommendations and/or other internal security policies.
- Evaluate your plant as a whole in terms of security. Use a cell protection concept with suitable products.
- Keep the software up to date. Always use the latest firmware/software version of the device. Check regularly for security updates for the products and apply them. After the release of a new version, previous versions are no longer supported and are not maintained. Information regarding product news and new software versions is available at the following address: (https://support.industry.siemens.com/cs/ww/en/ps/15081)
- Do not connect the device directly to the Internet. Operate the device within a protected network area. Use a firewall to connect the internal protected network to external networks. Configure this with restrictive rules.
- For transmitting data over an non-secure network, use additional security components that provide an encrypted VPN tunnel (IPsec, OpenVPN).
- Disconnect connections properly (e.g. logout in WBM).

Physical access

- Restrict physical access to the device to qualified and authorized personnel.
- The memory card or the PLUG (CLP) contains sensitive data, such as certificates, keys, etc., that can be read out and modified. An attacker in possession of the replaceable media of the device can extract critical information such as certificates, keys, etc. or reprogram the media.

Security functions

- Only enable protocols that you actually need to use the device. Note that, in the factory setting, all protocols can be used. However, while the device has established a connection via a protocol, all other protocols are disabled.
- The XML protocols have no authentication and are sent unencrypted. Take suitable measures to ensure that only authorized persons can establish XML communication and that the XML communication is tap-proof.
- For optimal security, use the authentication and encryption mechanisms of SNMPv3. SNMPv1 is classified as non-secure and should only be used when absolutely necessary.
- Use the latest web browser version that is compatible with the product to ensure that the most secure available encryption method is being used.

- Limit access to the device using an external device through a firewall or rules in an access control list (ACL). The firewall and access list can only be configured via an external device.
- The device has protection against brute force attacks to secure the system against trying out numerous passwords. Limit the maximum number of failed login attempts permitted.
- The device has an automatic disconnection function ("Session timeout"). Define the amount of time after which the connection to the device is automatically disconnected.
- Lock unused physical ports on the device. Unused ports can be used to access the system without permission.
- Configuration files can be downloaded from the device. Make sure that the configuration files are appropriately protected. You can, for example, digitally sign and encrypt the files, store them at a safe location or transfer them only via secure communication channels.
- The device offers options for saving and restoring the configuration on a PC/PG. For security reasons, neither the IP address of the network interface nor local user administration data are backed up to a PC/PG. We recommend that you use the network management system "SINEC NMS" or a SCALANCE CLP to manage this data. All data is saved on the CLP.

Authentication

Note

Accessibility risk - Risk of data loss

Do not lose the passwords for the device. Access to the device can only be restored by resetting the device to factory settings, which removes all configuration data.

- Always use the user management and create new user profiles.
- Replace the default passwords for all user accounts, access modes and applications (if applicable) before using the device.
- Define rules for using devices and assigning passwords.
- Define password policies.
- Use strong passwords. Avoid weak passwords, such as "Password1", "123456789", "abcdefgh" and passwords with repeating characters ("abcabc"). This recommendation also applies to the symmetrical passwords/keys configured on the device.
- Make sure that all passwords are protected and inaccessible to unauthorized personnel.
- Do not use the same password for different users and systems.
- Store passwords at a secure location (not online) to have them at hand in case of loss.
- Change passwords and keys regularly to increase security.
- A password must be changed if it has been, or is suspected to have been, disclosed to unauthorized persons.

Certificates and keys

- The device contains an SSL/TLS certificate for accessing the WBM and an OPC UA certificate for access via OPC UA. Replace these certificates with higher-quality certificates with keys you have created yourself.
  Use a certificate signed by a reliable external or internal certificate authority.
- Use a certificate authority, including key revocation and management, to sign the certificates.
- Use certificates in format "PKCS #12".
- If possible, use RSA certificates with a key length of at least 4096 bits, or ECDSA certificates with a key length of at least 256 bits.
- Make sure that user-defined private keys are protected and inaccessible to unauthorized persons.
- Change all certificates and keys immediately if a security violation is suspected.
- Verify certificates using the fingerprint on the server and client sides to prevent "Man-in-the-middle" attacks. Use a second secure transmission path for this.
- Before sending the device to Siemens for repair, replace the current certificates and keys with temporary throwaway certificates and keys that can be destroyed when the device is returned.
- If protocols support both certificates and keys, you should favor certificates.
- With operation via OPC UA, always use the "Sign and encrypt" security method.
- The following keys and key sizes are supported:

Service	Supported key and size
Web server	RSA ≥ 3072 bits ECDH ≥ 256 bits
OPC UA server	RSA 4096 bit [1]

[1] RSA 2048 bit with use of an outdated security profile

- You can find information on the file formats supported by the various certificate types in the section "The "Certificates" menu item (Page 86)".
- You can find information on the encryption methods supported by the device in section "Encryption methods (ciphers) (Page 123)".

Firmware/software

The firmware itself is signed and encrypted. This ensures that only firmware created by Siemens can be loaded into the device.

Check regularly for new firmware/software versions or security updates and install them. After the release of a new version, previous versions are no longer supported and are not maintained.

Decommissioning

Decommission the device properly to prevent unauthorized persons from accessing confidential data in the device memory. To do this, reset the device to the factory settings.

Reset any storage media (CLPs) used to factory settings as well.

2.1 Protocols

Secure/non-secure protocols

- Use only secure protocols if possible. Avoid or disable non-secure protocols. If you need non-secure protocols and services, ensure that the device is operated in a protected network area. The following protocols provide secure alternatives:
  - HTTP → HTTPS
    HTTPS is enabled at the factory. HTTP is classed as non-secure, however, it can be activated again at a later time if necessary.
  - SNMPv1 → SNMPv3
    SNMPv1 is activated at the factory. Check whether it is necessary to use SNMPv1. SNMPv1 is classified as non-secure. Make use of the possibility to prevent write access. The product offers corresponding setting options.
    If SNMPv1 is enabled, change the community names. If no unrestricted access is necessary, restrict access over SNMP.
- Enable only services/protocols that you need for use of the system, including the built-in interfaces/ports. Unused ports could be used to access the network downstream from the device.
- Limit the services and protocols available to the outside to the minimum required.
- When using OPC UA, only use profiles with secure encryption and authentication.
- With NTP, the time is transferred without authentication and encryption. Take suitable measures to ensure that the communication to the NTP server is protected against unauthorized access.

List of available services/protocols

The following is a list of all available services/protocols and their ports through which the device can be accessed.

Table 2-1 List of available services/protocols

Service/ Protocol	Protocol/ Port number	Preset port status	Port status configurable	Port number configurable	Authentication	Encryption [1)]
DHCP	UDP/69	Closed	✓	--	--	--
HTTP	TCP/80	Closed	✓	--	--	--
HTTPS	TCP/443	Open	✓	--	✓	✓
NTP	UDP/123	Closed	✓	--	--	--
SNMP	UDP/161	Closed	✓	--	✓ (when configured)	✓ (when configured)
OPC UA	TCP/4840	Open	✓	✓	✓ (when configured)	✓ (when configured)

Service/ Protocol	Protocol/ Port number	Preset port status	Port status configurable	Port number configurable	Authentication	Encryption [1]
XML	TCP/10001-10004	Open [2]	✓	✓	--	--
SYSLOG	UDP/514	Closed	✓ [3]	--	--	--

[1] You can find more information on the encryption methods used in the appendix.

[2] Applies to protocol number 10001, all others are closed.

[3] Only outbound, when configured.

Explanation of the table:

- Authentication
  Specifies whether authentication of the communication partner takes place or whether authentication can be configured.
- Encryption
  Specifies whether the transfer is encrypted or whether the encryption can be configured.

The following is a list of all available Layer 2 services through which the device can be accessed.

Table 2-2 List of Layer 2 services/protocols

Service/ Protocol	Default status	Configurable
ARP	Active	--
DCP	Active	✓
LLDP	Active	✓

The following is a list of all available Layer 3 services through which the device can be accessed.

Table 2-3 List of Layer 3 services/protocols

Service/ Protocol	Default status	Configurable
ICMP	Active	--

2.2 Cybersecurity information

Siemens provides products and solutions with industrial cybersecurity functions that support the secure operation of plants, systems, machines and networks.

In order to protect plants, systems, machines and networks against cyber threats, it is necessary to implement – and continuously maintain – a holistic, state-of-the-art industrial cybersecurity concept. Siemens' products and solutions constitute one element of such a concept.

Customers are responsible for preventing unauthorized access to their plants, systems, machines and networks. Such systems, machines and components should only be connected

to an enterprise network or the internet if and to the extent such a connection is necessary and only when appropriate security measures (e.g. firewalls and/or network segmentation) are in place.

For additional information on industrial cybersecurity measures that may be implemented, please visit
https://www.siemens.com/cybersecurity-industry.

Siemens' products and solutions undergo continuous development to make them more secure. Siemens strongly recommends that product updates are applied as soon as they are available and that the latest product versions are used. Use of product versions that are no longer supported, and failure to apply the latest updates may increase customer's exposure to cyber threats.

To stay informed about product updates, subscribe to the Siemens Industrial Cybersecurity RSS Feed under
https://new.siemens.com/cert.

2.3 Cell protection concept

The following graphic shows an example of a cell protection concept for RFID modules.

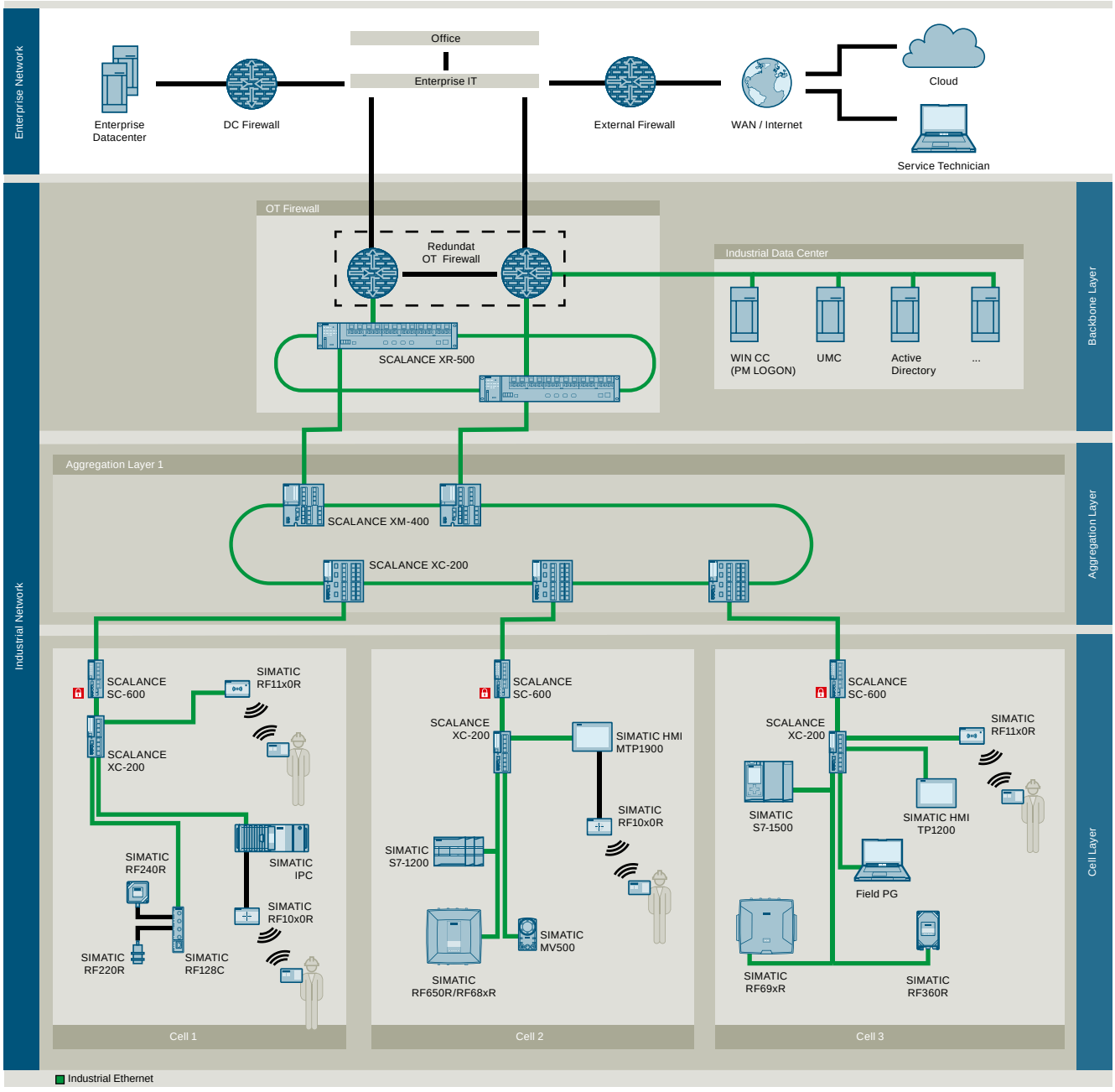


Figure 2-1 Schematic display of a cell protection concept for RFID devices

You can find information on a network/cell protection concept at the website of Siemens Industry Online Support (https://support.industry.siemens.com/cs/ww/en/view/109802750).

Description

3

3.1 Properties of the UHF readers

Area of application

The SIMATIC RF690R/RF695R readers are designed for use in logistics within an access-protected area. To meet these requirements, the readers were equipped with a high transmit power. All readers are equipped with extensive diagnostics options and can process ISO/IEC 18000-63 transponders.

The readers have one Ethernet interface (RJ45). This is used both to connect to PC/PG systems and for configuration and diagnostics, and it can also be used during operation. Higher-level software communicates with the reader using the XML protocol or OPC UA.

The WBM (Web Based Management) allows commissioning, configuration and diagnostics of the devices using a browser. This makes additional updates and installation of configuration and diagnostics software unnecessary.

For security reasons, you should only operate the device within a protected network area and not connect it directly to the Internet.

I/O interfaces

The RF690R/RF695R readers have one IO interface. You can use this interface either as a digital input or output or for the connection of an IO-Link module.

The IO interface can be addressed directly via the communication interfaces. Furthermore, the input/output can also be used to control the reader. The output can also be set, for example, due to internal events (antenna error, transponder detected, etc.). Note that the reaction time of the digital output (and input) of the reader depends on the reader load and is 100 - 200 ms on average.

If the I/O interface is used for connecting an IO-Link module, the reader can use up to 8 digital inputs/outputs of each of the modules. Only DI/DQ modules with a maximum of 32 bits of cyclic process data are supported (input and output data).

Note that connected IO-Link modules retain the set parameter assignment even in a de-energized state. As a consequence, an IO-Link module that was previously configured on a different IO-Link master and then used with an RF69xR can demonstrate unexpected behavior. In this case, you may need to re-assign the module parameters via the Port Configuration Tool (PCT) using a standard IO-Link master.

Characteristics

The following features characterize the reader:

Table 3-1 Characteristics of the readers

Characteristics	RF690R	RF695R
Antennas	4 x external antenna connectors	8x external antenna connectors
Transmit power (max.) [1]	2000 mW	
Digital inputs/ outputs	1x DI or DQ (or via IO-Link module)	
Ethernet interfaces	1x Industrial Ethernet (RJ45)	
Transmission speed (max.)	1000 Mbps	
Degree of protec-tion	IP30	
Configuration/ diagnostics op-tions	WBM (browser)	
Interfaces to PC/PG	OPC UA, XML	
Readable trans-ponder standards	ISO 18000-63 EPCglobal UHF Class 1 Gen 2	

[1] The maximum transmit power is restricted depending on the country of use.

NOTICE
Operation in VLANs Note that the readers cannot be operated in VLANs whose ID is ≠ 0.

Certificates

The readers support the following certificates and approvals:

- RF690R certificates (https://support.industry.siemens.com/cs/ww/en/ps/29638/cert)
- RF695R certificates (https://support.industry.siemens.com/cs/ww/en/ps/29639/cert)

Integration

The figure below shows the integration of the readers as an example.

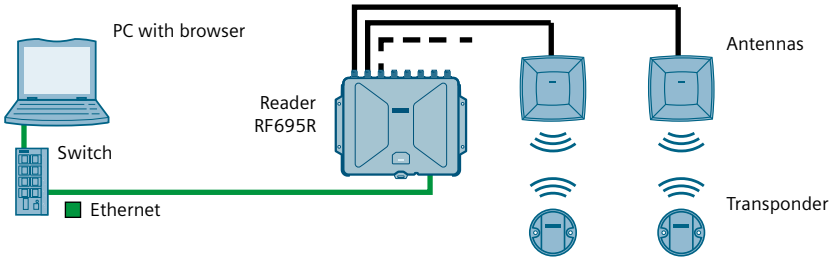


Figure 3-1 RF695R reader in an IT environment

3.2 User-specific procedures

The readers are pre-configured when shipped and can be immediately put into operation.

Pre-configuration

- Antenna port ANT1: RF650A
- Transmit power: 20 dBm
- IP address: 192.168.0.254

If you operate the readers in an XML environment, they are configured and programmed from the perspective of an XML user. If you operate the readers in an OPC UA environment, they are configured and programmed from the perspective of an OPC UA user.

If you want to adapt the readers to your requirements, we recommend the following user-specific procedure:

Procedure as XML user

1. Connect the hardware
   You can find information on this in the section "Connecting readers (Page 24)".
2. Assign the IP address / device name
   You can find information on this in the section "Assigning the IP address / device name (Page 27)".
3. Configure the reader
   You can find information on this in the section "Configuring with the WBM (Page 29)".
4. Program reader commands
   You can find information on this in the manual "XML programming for SIMATIC Ident (https://support.industry.siemens.com/cs/ww/en/ps/15081/man)".

Procedure as an OPC UA user

1. Connect the hardware
   You can find information on this in the section "Connecting readers (Page 24)".
2. Assign the IP address / device name
   You can find information on this in the section "Assigning the IP address / device name (Page 27)".
3. Configure the reader
   You can find information on this in the section "Configuring with the WBM (Page 29)".
4. Program reader commands
   You can find information on this in the manual "OPC UA for SIMATIC Ident (https://support.industry.siemens.com/cs/ww/en/ps/15081/man)".

Orientation in the document

In the rest of the document, the symbols shown in this section will help you to quickly find your way around and determine whether the respective section is of interest to you or not. Only the sections with user-specific content, in other words content that is tool/interface-specific contain these symbols. Sections without these symbols are general and relevant for both areas of application.

Commissioning

4

4.1 Important notes on using the device

Safety notices on the use of the device

The following safety notices must be adhered to when setting up and operating the device and during all work relating to it such as installation, connecting up, replacing devices or opening the device.

General notes

WARNING

Safety extra low voltage

The equipment is designed for operation with Safety Extra-Low Voltage (SELV) by a Limited Power Source (LPS). (This does not apply to 100 V ... 240 V devices).

This means that only SELV / LPS complying with IEC 60950-1 / EN 60950-1 / VDE 0805-1 must be connected to the power supply terminals. The power supply unit for the equipment power supply must comply with NEC Class 2, as described by the National Electrical Code (r) (ANSI / NFPA 70).

There is an additional requirement if devices are operated with a redundant power supply:

If the equipment is connected to a redundant power supply (two separate power supplies), both must meet these requirements.

NOTICE

SIMATIC RF680A antenna is not supported

The SIMATIC RF680A antenna is not supported by the SIMATIC RF690R/RF695R readers.

NOTICE

Max. permissible current with operation via PoE

With operation via PoE, only digital inputs/outputs that require a maximum of 200 mA in total can be connected.

NOTICE

Alterations not permitted

Alterations to the devices are not permitted. If this is not adhered to, the radio approvals, the relevant country approvals (e.g. CE or FCC) and the manufacturer's guarantee are invalidated.

Overvoltage protection

NOTICE
Protection of the external 24 VDC voltage supply If the module is supplied via extensive 24 V supply lines or networks, interference by strong electromagnetic pulses on the supply lines is possible, e.g. from lightning or the switching of large loads. The connector for the 24 VDC external power supply is not protected against strong electromagnetic pulses. Make sure that any cables liable to lightning strikes are fitted with suitable overvoltage protection.

Repairs

⚠ WARNING
Repairs only by authorized qualified personnel Repairs may only be carried out by authorized qualified personnel. Unauthorized opening of and improper repairs to the device may result in substantial damage to equipment or risk of personal injury to the user.

4.2 Connecting readers

Prior to installation and commissioning

NOTICE
Installation/removal with the power off Wire up the PC/PG and modules to be connected only when the power is off. Make sure that the power supply is turned off when installing/uninstalling the devices.

Interfaces

①	RP-TNC interfaces for connecting up to four or eight external antennas [1]	⑤	CLP interface (Configuration license plug) CLP
②	LED operating display	⑥	Ethernet interface X1 LAN (RJ45)
③	Power supply interface (NEC Class 2) X80 (terminal block)	⑦	Ground connection
④	DI/DQ interface X10 (terminal block)	⑧	Reset button RESET

Figure 4-1 Structure of the reader

[1] Depending on the reader used

Requirement

- The reader has been mounted.
- Only wire the reader when the supply voltage is switched off.

Procedure

Follow the steps below to connect a reader:

1. Connect the reader to a PC/PG or a switch using an Ethernet cable.
   Use an Ethernet connecting cable with two-sided RJ45 plug.
2. Connect one or more external antennas to the reader.
3. Connect the reader to the power supply using the connecting cable.

The reader is ready for operation when the "RUN" LED is lit/flashes green. If the "RUN" LED is flashing, the reader is waiting for a connection. If the "RUN" LED lights up statically, the connection to the user application (XML or OPC UA) was established.

You will find detailed information on installation as well as order data (reader, pre-assembled connecting cables, antennas, wide-range power supply, etc.) in the "SIMATIC RF600" system manual.

Addressing and configuring

5

Synchronize device time

Note that the time of the device clock corresponds to UTC time and cannot be adjusted to time zones. It is recommended to synchronize the time with an NTP server to obtain unique time information. If the device is disconnected from the power supply, the time continues to run for approx. 48 hours. Then the time is reset and must be synchronized.

5.1 Assigning the IP address / device name

To achieve optimum communication between the PC/PG and reader, you need to assign a unique IP address or device name to each individual reader.

The RF690R/RF695R readers ship with the factory-set IP address "192.168.0.254". You can change the IP addresses using SINEC PNI and the WBM. Changing the IP address using SINEC PNI is described below. Alternatively, you can change the IP address using a CLP.

Note

Support of option "12"

When the address is assigned via DHCP, the option "12" (hostname) is also supported. The hostname can be taken from the SNMP variable "sysName".

The variable can be written using SNMP tools or the WBM.

Requirement

SINEC PNI is installed and the reader is connected and running. You can find the SINEC PNI on the pages of the "Siemens Industry Online Support (https://support.industry.siemens.com/cs/ww/en/ps/26672/dl)".

Procedure

Follow the steps below to assign a new, unique IP address and a unique device name to the reader:

1. Start SINEC PNI.
2. In the "Settings" menu, select the "Network adapter" via which the reader is connected to the PC/PG.
3. Make sure that the "Discovery > PROFINET devices (DCP)" is selected.
   Note: Note that the "Read I&M data" function may take some time if there are numerous devices in the network.
4. Switch to the "Device list" menu.

5. Click the "Start network scan" button in the toolbar.
   Reaction: The network is scanned for connected devices and all recognized devices are displayed in the device list.
6. Select the desired reader in the device list.
7. Click the "Change device configuration" button in the toolbar.
   Reaction: The "Device configuration" window opens.
8. Enter a new, unique IP address for the reader in the "IP address" input box.
9. Enter the subnet mask of your network in the "Subnet mask" input box.
10. Switch to the "PROFINET" tab, if necessary.
11. Enter a device name in the "PROFINET device name" input box.
12. Click the "Load, only" button to transfer the settings to the reader.

Result: The reader is assigned the new IP address, subnet mask and a new device name.

Device flash test using SINEC PNI

Using the node flash test, you can identify the reader quickly and simply by having the LEDs of the device flash. This function is particularly helpful if several devices are connected to the network.

Follow the steps below to identify the relevant reader using the flash test:

1. Select the desired module from the device list in the "Device list" menu.
2. Select the menu command "Device management > Device availability > Flashing LED" in the toolbar.
   Reaction: The LEDs on the selected reader flash.
3. Click the "Stop" button to stop the flashing.

Configuring with the WBM

6

The readers are equipped with a Web server that provides Web Based Management (WBM) to the Web client for configuring the readers. The WBM can be called via the Web browser of a PC/PG.

The WBM server provides the Web client (PC/PG) with the parameter data of the reader and accepts parameter changes from the Web client. Note that changed parameter values are not automatically transferred to the reader. You must always manually transfer changes in the configuration to the reader.

Hereafter, the term "WBM" is used to represent the WBM interface displayed in the Web browser.

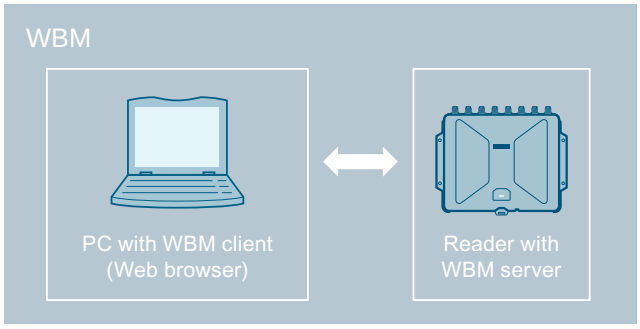


Figure 6-1 Design and function of the WBM

6.1 Starting WBM

Requirement

The reader is connected, turned on and ready for operation ("RUN" LED is lit or flashing green) and the relevant reader has been assigned an IP address.

You can call WBM using the versions of the following Web browsers current at the time of publication of this manual: Microsoft Edge, Mozilla Firefox and Google Chrome. The user interface of the WBM is designed for a screen resolution of at least 1366 x 786 pixels.

NOTICE
Aborting diagnostics due to open browser tabs Note that some web browsers stop updating web pages when other tabs are in the foreground. This behavior may cause ongoing diagnostics to be ended.

Procedure

Follow the steps below to start the WBM:

1. Start your Web browser.
2. Enter the IP address of the reader preceded by "https://" in the address field of your browser.
3. Confirm your entry by pressing the <Enter> key.

Note

Updating HTTPS certificates

On first login, the HTTPS certificate of the reader is classed as not secure. Note that the certificate stored in the reader is only intended to ensure that the initial connection establishment to the reader is encrypted. Confirm that the certificate is secure ("Advanced" button). Then transfer your own, secure certificate to the reader.

Result: The WBM of the reader opens.

Note

The connection to the reader cannot be established

If no connection can be established to the reader, check the following points:

- Make sure that all cables are correctly connected.
- Make sure that the Ethernet connection has been established ("LINK" LED lights up green).
- Make sure that the reader has started up ("RUN" LED lights up/flashes green).
- Check the IP addresses of the PC/PG and the reader as well as the subnet mask. Both IP addresses must be located in the same subnet.
- Make sure that the connection is not blocked by a firewall.
- Check the connection between the PC/PG and reader using a ping request.

Logging into the WBM for the first time

When logging into WBM for the first time, a popup window appears prompting you to log in with the default user "admin".

1. Select the desired user interface language from the drop-down list.
2. Enter the preset default user name "admin" in the "User" text box.
3. Enter the preset default password "admin" in the "Password" text box.

4. Click the "Log in" button.
   Reaction: The popup window is updated and you are prompted to change the default password for the "admin" user.
   Note: Alternatively, you can use the "Disable authentication" button to disable authentication.

NOTICE
Security recommendation: Authentication To ensure that no unauthorized persons can access the reader settings, we recommend that you enable the authentication and create new user profiles. You should also read the information under the "Passwords" heading of the "Security recommendations (Page 11)" section. The authentication can only be enabled/disabled by an administrator. For more information on logging into WBM and creating/deleting user profiles, refer to the section "The "User management" menu item (Page 83)".

5. If needed, select the primary interface connection over which you are operating the reader from the "Operational environment" drop-down list.
6. Enter your new password for the "admin" user in the "New password" text box.
7. In the "Confirm password" text box, enter the newly selected password again.
8. Click "Change".

Result: You are logged in to the WBM with the "admin" profile and can now set reader parameters.

Operational environment

Depending on the selected operational environment, the interfaces, communication channels, and associated services that match the mode are enabled beforehand. This restores the parameter values for the mode to the factory default values.

This function increases the security of your device because all other interfaces, communication channels and the associated services are disabled.

Regular login to the WBM

Depending on whether the authentication is enabled or disabled, you may have to log in with your user name and the corresponding password. The WBM homepage opens after logging in, even if authentication is disabled.

6.2 The WBM

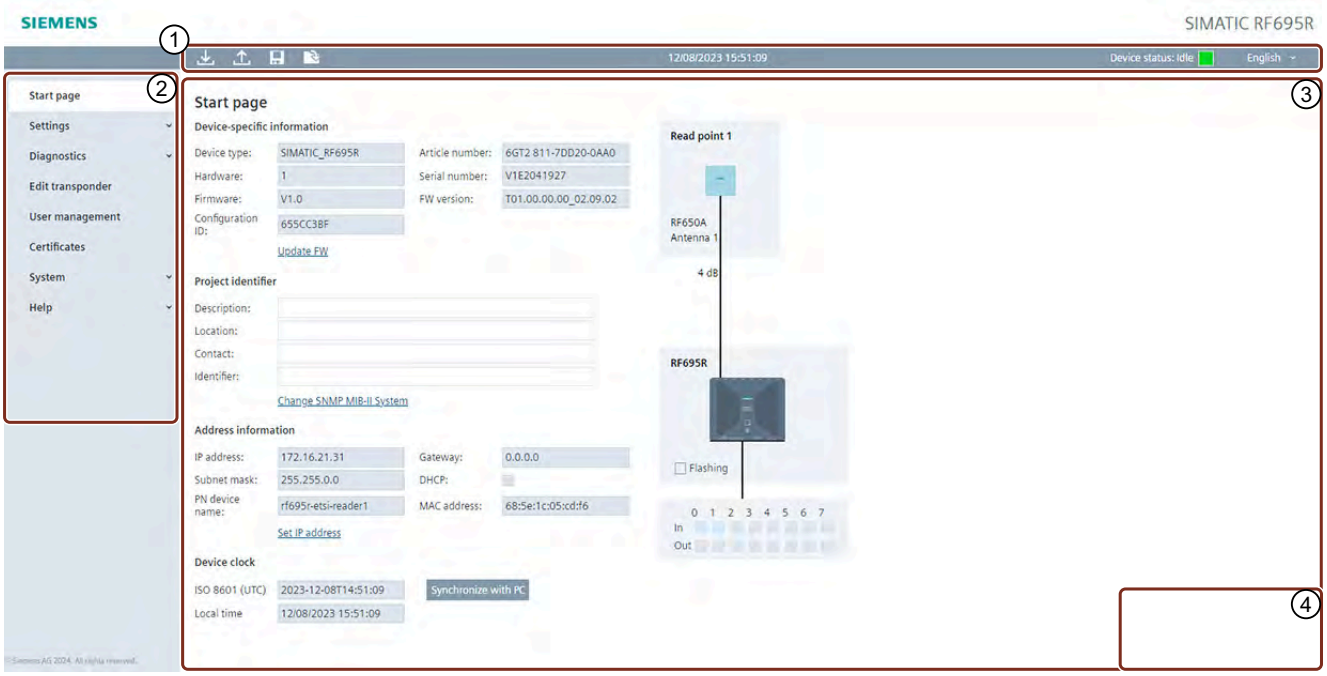
You can configure the readers using the WBM.

After you have created new user profiles, you need to log in with one of these user profiles when you restart the WBM.

NOTICE
Access to the reader Remember that simultaneous access to a reader using multiple WBM clients (web browsers) is possible but not recommended. If changes are made when two browsers are accessing a reader at the same time, this can lead to errors in the configuration or to an undesired result.

Layout of the WBM

After successful connection establishment to the reader and login (when authentication is enabled), the start window of the WBM appears:



The start window of the WBM is divided into 4 areas:

① Toolbar / status bar (including logout area)

② Menu tree

③ Main window

④ Message area

Figure 6-2 Start window of the WBM

Toolbar and status bar ①

Toolbar

On the left above the main window, there are four buttons for transferring/loading/storing the displayed configuration. You can also operate these buttons directly with key combinations.

Table 6-1 The toolbar of the WBM

Icon	Description
	Transfer configuration to reader With this button, you can transfer the configuration data set in the WBM to the reader. Key combination: Ctrl + L **Note** Please note that transferring a configuration can disrupt running user applications.
	Load configuration from reader Using this button, you can load the configuration data currently set on the reader into the WBM. Key combination: Ctrl + G **Note** Note that you cannot transfer any user profiles and passwords to other readers using the configuration file. After loading the configuration file into a new reader, you may need to enable authentication and create new user profiles and passwords.
	Save configuration as With this button, you can save the configuration data set in the WBM on the PC/PG. Key combination: Ctrl + S
	Load configuration from PC Using this button, you can load the configuration data stored on the PC/PG in the WBM. Remember that this data is only loaded in the WBM. To transfer the data to the reader, you also need to click the "Transfer configuration to reader" button. Key combination: Ctrl + O

Status bar

On the right above the main window there is the status bar with the following information:

- Date/time display of the reader
  Note that the time stamp is generated by the device clock (UTC time). This time is compared with the time format and time zone set on the PC/PG and displayed in the corresponding format.
- Display of the device status
  The following device states are possible:

	Idle The reader is ready for operation.
	Running The reader is running and has established a connection.
	Running The reader is running and has established multiple connections (e.g. via OPC UA and XML).
	Error (idle) The reader is ready for operation and there is an error.
	Error (running) The reader is running, has established a connection and there is an error.
	Error (running) The reader is running, has established multiple connections and there is an error.

- Drop-down list for selecting the user interface language
- Logged in user (when authentication is active)
- Display of the time until the automatic logout as well as a drop-down list for selecting the time interval
- Logout area (with active authentication)

Change notes in the user interface

Deviations between the settings in the user interface of the WBM and the configuration stored on the connected reader are shown with a symbol in the user interface. If a value is changed in the WBM interface, the relevant field is marked by a symbol. In addition, the tab, if applicable, as well as the menu command in which the changed value is located and the "Transfer configuration to reader" button are also marked by a symbol. A distinction is made between the following symbols:

- This symbol indicates that it is a simple change.
- This symbol indicates that, at the same time the reader is being accessed via the WBM, a change was made by another application. To make sure that this change is not lost, you should load the configuration from the reader.
- This symbol indicates that it is a change that leads to a restart of the reader when it is transferred.

Menu tree ②

The menu tree is located in the left margin of the WBM. The currently selected menu command is highlighted in color.

The following table provides an overview of the menu commands and the functions they provide.

Table 6-2 The menu structure of the WBM

Menu commands		Functions
Start page		• System overview • Viewing device-specific information • Entering customer-specific plant designation
Settings		
	General	• Selecting a country profile and channels • Enabling/disabling categories of logbook events • Advanced settings (e.g. modulation scheme)
	Communication	Making communication settings Activating/deactivating and configuring communication interfaces
	Read points	• Defining read points and assigning antennas • Specifying antenna parameters • Setting algorithms to improve reading quality • Assigning tag fields • Assigning filters • Setting triggers
	Tag fields	Creating and editing tag fields
	Filters	Creating and editing filters
	DI/DQ	Configure digital inputs/outputs
	Adjust antennas	Optimization of the antenna alignment
	Activation power	Detect activation power
Diagnostics		
	Tag monitor	• Display of the transponder detection quality • Overview of the identified transponders • Display of the digital inputs/outputs
	Logbook	Overview of the logbook entries
	Service logbook	Information for service cases
	Syslog logbook	Overview of Syslog messages
Edit transponder		• Changing the EPC ID/UID • Read/write transponder data and tag fields • Locking transponder access • 'Destroying' transponders

Menu commands		Functions
User management		• Enabling/disabling authentication • Creating and deleting user profiles • Changing passwords • Configuring security settings
Certificates		• Importing HTTPS certificates • Importing OPC UA certificates
System		
	Device settings	• Updating the firmware • Resetting the reader • Downloading device description files • Configuring the CLP
Help		Documentation relevant for the reader
	Service and Support	Additional information on the reader
	Manual	Manual of the reader

When you are logged in with the "Read only" role, you can only read out information from the WBM. You can find a list of the restrictions in the section "The "User management" menu item (Page 83)".

Main window ③

The main window shows the contents of the selected menu commands. Here, you can configure the various menu-dependent parameters.

Message area ④

The message area displays all WBM-related error messages and warnings (e.g. transmission errors).

Operating the WBM via the keyboard

In addition to operating the WBM with the mouse, you can also control the interface objects/text boxes using the keyboard:

- TAB
  Jump to next interface object / text box
- SHIFT + TAB
  Jump to the previous interface object / text box

Apart from manual entry of values, you can also change values in the text boxes with the following buttons:

- Arrow up / down
  The value is increased or decreased by one increment.
- PgUp / PgDn
  The value is increased or decreased by ten increments.
- Home / End
  The value is set to the minimum or maximum value.

6.3 The menu items of the WBM

6.3.1 The "Start page" menu item

The "Start page" menu command is divided into the following areas:

- Device-specific information
- Project ID
- Address information
- Device clock
- Configuration display

Device-specific information

The first area contains device-specific information. The "Device type", "MLFB", "Hardware" and "Serial number" boxes are set at the factory. The content of the "Firmware" and "Firmware version" boxes depends on the firmware stored on the reader. Using the "Firmware update" link, you jump to the "System" menu command in which you can update firmware. The "Configuration ID" box contains a unique identifier for the configuration that was last enabled on the reader or loaded on the reader.

Project ID

The second area contains input boxes with which you can store your own device-specific information on the reader. Among other things, this is intended to make it easier to identify the individual readers. Via the link "SNMP MIB-II system", you jump to the "Communication" menu command, where you can view and change the MIB variables.

Address information

The third area contains all the important address information with which the PC/PG can reach the reader. You can assign the IP address and PN device name to the reader using the "SINEC PNI" tools. Via the link "IP Address" you jump to the "System settings" menu command in which you can also reassign the IP address.

Device clock

The device time according to ISO 8601 (UTC), as well as the calculated local time, is displayed in this area. With the "Synchronize with PC" button, you can synchronize the displayed local time with the time in your operating system.

Note

The device time always corresponds to UTC time (ISO 8601)

Note that the device time always corresponds to UTC time (ISO 8601) and cannot be adapted to time zones. Clicking the button transfers the local time stored in your operating system to the WBM. Because the time synchronized with the PC/PG is lost when the power supply is terminated, we recommend synchronizing the time with an NTP server.

Configuration display

The current configuration is shown to the right of the areas. The schematic representation contains information on the connected reader type and antennas as well as the antenna cables being used including the cable loss and on the digital inputs/outputs.

With the "Flash" check box, you can have the LEDs of the reader flash. This allows fast and simple visual identification of the connected reader.

Table 6-3 States of the digital inputs/outputs

	The input/output is not configured.
	The input/output is configured but not connected
	The input/output is configured and connected.

6.3.2 The "Settings - General" menu item

The "Settings - General" menu command is divided into the following areas:

- Country profile
- Channels
- Logbook settings
- Service logbook settings
- Advanced settings

Country profile

From the "Country profile" drop-down list, you can select the radio profile the reader will use. The "Channels" area is adapted depending on the radio profile selected. The radio profiles depend on the country or region. To ensure that the reader keeps to the local radio regulations, select the country profile belonging to your country. You can find

information on which country profile applies to you in the "List of country profiles (https://support.industry.siemens.com/cs/ww/en/view/109773353)".

NOTICE
Country profile depending on country-specific approvals (wireless approvals) Note that the RF600 readers are only permitted to be operated in countries for which there is a corresponding country-specific approval (wireless approval). Only use country profiles that are permitted for your region. If you select a country profile that is not suitable for your region, you may violate the local wireless guidelines.

Channels

The "Channels" area displays the channels with the frequencies of the selected country profile. Disable the check boxes of the channels that the reader should not use. Note that with FCC country profiles, the check boxes cannot be disabled.

Logbook settings

In the "Logbook settings" area, you can use the check boxes to decide which events are entered in the log. The logbook is structured as a ring buffer. Bear in mind that with a high degree of detail of the data, the ring buffer fills up more quickly which can have a negative effect on the performance of the device.

Table 6-4 Description of the parameters of the logbook

Parameter	Description
Logbook types	
COMMON	Messages relating to general events: e.g. reader startup, login to the WBM, ...
ERRORS	Error and alarm messages of the reader
GPIO	Changes to the digital inputs / digital outputs
FILTER	Transponders that were filtered out.
COMMANDS	Commands of the user application
EVENTS	Recording of all tag events (e.g. glimpsed, ...)
COMMANDS: Detail information	
Return value	Return values for the commands of the user application and for the written or read transponder data.
Call parameters	Call parameters for the commands of the user application
Extended acquired values	Additional data obtained when the transponder was identified (antenna, polarization, channel, etc.).
EPC ID in ASCII format	In addition to hexadecimal output, the EPC ID is also output in ASCII notation.
Status frames	Recording of status frames Can be switched off if the status frames are used as line monitoring. In this way, the logbook is kept free for user data.

Service logbook settings

In the "Service logbook settings" area, you can use the check boxes to decide which events are entered in the Service logbook. The Service logbook is structured as a ring buffer. Bear in

mind that with a high degree of detail of the data, the ring buffer fills up more quickly which can have a negative effect on the performance of the device.

Table 6-5 Description of service logbook parameters

Parameter	Description
Communication channels	
CMD_XML	Frames on the XML interface
CMD_WEB	Internal frames to the Web server
UHF_LOGIC	Internal frames to the UHF part of the reader
Detail information	
Line monitoring	Recording of the line monitoring frames (CMD_XML) for service information. Can be switched off to keep the logbook free for user data.

Advanced settings

The "Advanced settings" area contains various, specific reader parameters. The "Modulation scheme" and "Max. RSSI events" parameters are relevant for use in the logistics sector. Some of these parameters are intended exclusively for trained users and have a default setting for a standard configuration. Only adapt parameters with which you are familiar.

Table 6-6 Description of the parameters

Parameter	Description
Expected number of transponders	The number of transponders expected to be read with the reader. With this input, the reader can optimize the transponder identification. Enter the value as specific as possible. If there is a large discrepancy between expected and actual number of transponders, this will have a negative effect on the reading speed. Value range: 1 ... 1000 Increment: 1
Modulation scheme	Specification of the data transfer rate, the country profile and the coding: • Tx: Data transfer rate from reader to transponder • Rx: Data transfer rate from transponder to reader • Miller/FM0: Coding of the transponder signal Miller is used with the "Dense Reader Mode". The "Dense Reader Mode" allows the operation of neighboring reader systems on an identical frequency channel. If an FM0 profile is selected, neighboring readers can have greater influence on each other. Note that higher data transfer rates can result in an increased susceptibility to errors. You can find additional information on the modulation schemes in the following paragraph.
Carrier off delay [s]	Off delay of the carrier frequency in seconds. This mechanism can reduce the time required to access transponders that are accessed more than once in succession. This time specifies how long the UHF carrier frequency remains active after transponder access. During the off delay time, the reader is quicker when re-accessing transponders that have already been accessed. Value range: 0 ... 25.5 s Increment: 0.1 s

Parameter	Description
Cyclic antenna test	Enabling/disabling the cyclic antenna test If the cyclic antenna test is active, the reader checks whether or not the antennas are plugged in and connected to the reader. To do this, the antennas are accessed with minimum power. To make sure that antennas radiate power only when this is specifically required, you can disable the antenna test. Without the cyclic antenna test, an interrupted connection can only be detected when the antenna is accessed. Note that the cyclic antenna test does not work if the cable attenuation is ≥ 4 dB.
Verified write access	When this check box is selected, the data written to the transponders can be checked or verified again after data transfer. If the function is disabled, the data is not verified. The recommendation is to disable this function if some bits in the data to be changed are managed by the transponder itself (e.g. the UMI bit).
Max. RSSI events	If the check box is selected, the reader operates in a mode that is designed to generate the maximum number of RSSI events. Note that this function can lead to reduced transponder detection with very large transponder populations, which may mean that not all transponders can be detected. Requirement: The RSSI events of the XML or OPC UA interface have been activated.

Modulation schemes

The readers have different modulation schemes with different strengths. The schemes allowed according to ETSI are described below.

- Modulation scheme 241: A profile that provides a good compromise between detection speed, sensitivity and interference immunity.
- Modulation schemes 222 and 223: Characterized by the fastest transponder detection. To achieve this, however, the transponders must be well positioned within the antenna field and sources of interference must be excluded.
  Classic area of application: Generate as many events as possible
- Modulation scheme 223: Characterized by good antenna field coverage.
  Classic area of application: Detection of dynamic transponder populations in gate applications
- Modulation scheme 244: Has the highest interference immunity.
  Classic area of application: Neighboring readers whose antenna fields overlap.
- Modulation scheme 285: Offers the highest sensitivity for the detection of transponders in demanding environments and therefore has the best transponder detection. However, this comes at the expense of speed.
  Classic area of application: Detection of transponder populations in difficult environments.

The following table provides an overview of the possible modulation schemes.

Table 6-7 Modulation schemes

Modulation scheme	Data transfer rate and code	Radio guideline
103	Tx: 160 kbps Rx: 640 kbps FM0	FCC
222	Tx: 50 kbps Rx: 160 kbps Miller2	ETSI, FCC, CMIIT
223	Tx: 67 kbps Rx: 160 kbps Miller2	ETSI, FCC
241	Tx: 50 kbps Rx: 80 kbps Miller4	ETSI, FCC, CMIIT
244	Tx: 50 kbps Rx: 62.5 kbps Miller4	ETSI, FCC, CMIIT
285	Tx: 50 kbps Rx: 20 kbps Miller8	ETSI, FCC, CMIIT
323	Tx: 133 kbps Rx: 320 kbps Miller2	FCC
344	Tx: 133 kbps Rx: 160 kbps Miller4	FCC

6.3.3 The "Settings - Communication" menu item

The "Settings - Communication" menu command is divided into the following tabs.

- Network interfaces
- XML
- OPC UA

In the "Network interfaces" tab, you can activate/deactivate SNMP and NTP, as well as Syslog messages. In the "XML" tab, you can specify which data is sent via the XML interface. In the "OPC UA" tab you can enable and edit the OPC UA server function of the readers.

Access using the communication protocols

The readers have various network interfaces. This enables you to access the readers using different communication protocols. All of the communication protocols are enabled in the delivery state.

To avoid simultaneous control of the readers, all other communication protocols are locked as soon as one communication protocol has established a connection to the device. For security reasons, it is recommended to lock all communication protocols that are not used.

Exceptions are the "XML" and "OPC UA" communication protocols, which can be enabled for a parallel diagnostics access. The diagnostics or parallel communication has only read-access to the readers, however.

The "Network interfaces" tab

The "Network interfaces" tab is divided into the following areas:

- Network settings
- SNMP
- NTP
- Syslog messages

Network settings

In the "Network settings" area, you can enable/disable the network ports of the reader, allow/prohibit access via DCP protocols, and specify the Web server protocols via which communication with the WBM is allowed.

Table 6-8 Description of the network basic parameters

Parameter	Description
IP address	In this area, you can change the IP address, subnet mask, gateway and DNS server of the reader. As an alternative, the configuration of the IP data can be obtained from a DHCP server. Using the "Apply" button, you can transfer the address data changed in the WBM to the reader. You can use the "Update" button to update the address data displayed in the WBM or reload it from the reader. Note: When the address is assigned via DHCP, the option "12" (hostname) is also supported. The hostname can be taken from the SNMP variable "sysName". The variable can be written using SNMP tools.
LLDP protocol	In this area, you can enable/disable the LLDP communication protocol. LLDP is a protocol for monitoring the neighborhood.
DCP protocol	You can enable/disable the communication via DCP protocols in this area. You can specify whether read-only or write access to the reader is permitted via the DCP protocols. Depending on the decision made here, the factory settings may or may not be restored to the reader via SINEC PNI, for example.
Web server protocols	In this area, you can specify which web server protocols can be used to access the WBM of the reader. For security reasons, it is recommended that you disable the "HTTP" function, since this protocol does not encrypt the data.

Disable the "Allow write access" function if you want to ensure that no access to the reader takes place via DCP. This setting prevents the reader from being reset to factory settings via SINEC PNI, for example.

Note

Requirement for port statistics

You can read out port statistics via SNMP. Note that this is a totalizer for the entire module and cannot be interface-specific.

SNMP

In the "SNMP" area, you can enable/disable the network protocol. "SNMP" is a protocol for monitoring network components.

SNMP is deactivated at the factory. SNMPv1 is classified as non-secure. If possible, use only SNMPv3. If you do not use the protocol, we recommend that you disable the setting for security reasons.

Table 6-9 Description of the SNMP parameters

Parameter	Description
SNMPv1 settings	
Read community string	Input box for specifying the user name for read access to SNMP variables. This property is typically assigned the "public" value.
Write community string	Input box for specifying the user name for write access to SNMP variables. This property is typically assigned the "private" value. In this box, changes can only be made if write access was permitted. Write access is only possible for the SNMP variables "sysName", "sysLocation" and "sysContact" of the "system" group of MIB-II.
Allow write access	Check box to enable/disable write protection for SNMP variables.
SNMPv3 settings	
Users	List of created SNMP users. You can use the "Add user" button to create new user profiles and add them. You can use the "Delete" button to delete selected user profiles.
User properties	Input box and drop-down list Enter the name of the newly created user profile in the input box and assign the user read or read/write rights.
Authentication	Drop-down list and input boxes Specify whether user authentication is required and, if so, which protocol is used. If applicable, enter the authentication password of the newly created user profile in the input boxes.
Encryption	Drop-down list and input boxes Specify whether encryption is required for the SNMP communication of the user and, if so, which protocol is used. If applicable, enter the encryption password of the newly created user profile in the input boxes.
Save	You can use the button to save changes made to existing user profiles.
SNMP MIB-II system	

Parameter	Description
sysName	Boxes for reading ("Update") and changing ("Apply") the MIB-II system information. These are usually assigned via network management systems (e.g. SINEC PNI). Changes made here are displayed accordingly in the network management system, and vice versa. This information is only required when using network management systems.
sysLocation	
sysContact	

NTP

In the "NTP" area, you can enable/disable the network protocol. "NTP" is a protocol for synchronizing the UTC time in network systems.

When supplied from the factory this setting is disabled and it needs to be enabled here before using NTP for the first time.

Table 6-10 Description of the NTP parameters

Parameter	Description
Address of the NTP server x	Input box for entering the address or name of the NTP master server from which the various connected readers synchronize their time. Up to four NTP servers can be specified to compensate possible server failures.
Update interval in seconds	Input box for specifying the intervals at which the readers automatically synchronize their time.
Accept time from unsynchronized NTP server	Check box to ensure that the readers also accept the time from unsynchronized NTP servers.

Syslog messages

In the "Syslog messages" area, you can enable/disable the Syslog messages. When the Syslog function is enabled, the module generates Syslog messages at the preset UDP port according to RFC 5426. Syslog messages log information during access to the module as well as configuration changes. By default, these are saved in a logbook file and output in the "The "Diagnostics - Syslog logbook" menu item (Page 79)" menu. The logbook file is set up as a ring buffer. If all the entries in the logbook file are occupied, the next new entry deletes the oldest entry.

When supplied from the factory this setting is disabled and it may have to be enabled here.

Table 6-11 Description of the Syslog parameters

Parameter	Description
Syslog Server Address	Text box for entering the address or the name of the Syslog master server to which the Syslog messages are transferred.
Default port	Text box for entering the default port of the Syslog master server via which the Syslog messages are transferred.

The "XML" tab

The "XML" tab is divided into the following areas:

- XML settings
- Diagnostic events of XML channel (1-4)
- Selection of transferred data

XML settings

You can enable/disable the XML communication via the XML interface of the reader in the "XML settings" area. In addition, you can define which events are sent (to the user application) via which XML channel.

Table 6-12 Description of the XML basic parameters

Parameter	Description
Parallel	When the check box is selected, parallel diagnostic access via the XML channel can take place in addition to an existing connection, e.g. to an OPC UA user application. In the case of parallel access, note that the XML application only has read access to the reader. If parallel access was already enabled for another communication protocol (e.g. OPC UA), this option cannot be selected. If the function is disabled, access to the reader can only take place over a communication channel. If a connection is already established over a communication channel, an additional connection attempt from another communication channel is rejected.
Transaction-secured transmission	If the check box is selected, each frame (XML report) received from the user application of the reader is confirmed with a response frame. If no response frame is received by the reader within 10 seconds, it sends the report to the application again. Reports that are not transmitted are buffered in the reader. With this function, you can make sure that no frames from the reader are lost even if the connection is unstable (e.g. WLAN connection aborts occasionally). This function also allows batch operation of the reader when there is only a connection to the user application at certain times. The reader collects the frames and these can, when necessary, be called up using a PC application.
Events	
Glimpsed events [1)]	If the check box is selected, a tag event frame (GLIMPSED) is generated with the tag event data selected in the "Selection of transferred data" area each time a transponder is detected for the first time.
Observed events [1)]	If the check box is selected, a tag event frame (OBSERVED) with the tag event data selected in the "Selection of transferred data" area is generated each time a transponder has been reliably detected for "x" read cycles. You can set the number "x" in the section "The "Settings - Read points" menu item (Page 52)" in the area "Algorithmen > Smoothing > Observed Count" (default value = 1).
Lost events [1)]	If the check box is selected, a tag event frame (LOST) with the tag event data selected in the "Selection of transferred data" area is generated each time a transponder has not been detected for at least "x" read cycles. You can set the number "x" in the section "The "Settings - Read points" menu item (Page 52)" in the area "Algorithmen> Smoothing > Lost Count" (default value = 5).
IO events	If the check box is selected, information about all events (edge transitions) of the digital inputs/outputs is reported.

Parameter	Description
XML basic configuration	
XML channel (1-4)	Specification of the XML channels to be used for communication. You can use the input boxes to specify the TCP/IP ports of the respective XML channels.

1) Requirement: At least one "Transmitting read point" has been enabled (see following paragraph).

Diagnostic events of XML channel (1-4)

In the "Diagnostic events of XML channel" area, you can define which events are transmitted via the selected XML channel.

Table 6-13 Description of the diagnostic event data for the XML channels

Data	Description
Presence events 1)	If the check box is selected, information about presence states or changes is sent.
Last access events 1)	If the check box is selected, all access operations to the transponders, as well as the data read/written in the process, is sent.
Logbook events	If the check box is selected, all logbook entries are sent.
RSSI events 1)	If the check box is selected, information about the signal strength of the transponder responses is sent. If RSSI events are not enabled, only tag events of the "Observed" type are sent.
Transmitting read point	
Read point (1-4)	Selection of whether and from which read point the selected entries are transmitted. Glimpsed, Observed, Lost, Presence, Last access and RSSI events are only sent from the selected read point. IO and logbook events are always sent, even if no read point has been selected. The displayed names depend on the read point names specified in the "Read points" menu.

1) Requirement: At least one "Transmitting read point" has been enabled.

Selection of transferred data

In the "Selection of transferred data" area, you specify which data is transmitted with the respective message frame (Tag event, Tag command or RSSI event).

Table 6-14 Description of the event data "Tag events / Tag commands / RSSI events"

Data	Description
Tag events / tag commands Tag events signal identified transponders or when they have left the antenna field. Information on all enabled tag events is reported by the reader to the XML API interface. Tag commands report when a transponder command has been executed. Information on all enabled tag commands is reported by the reader to the XML API interface. The data enabled in this area also affects the message content of the XML commands/reply frames (e.g. with "writeTagID"). For this information to be transmitted, the corresponding events must be enabled in the "Diagnostic events of the XML channel" area.	
event	Event type / status description of the transponder (Glimpsed, Observed, Lost)
utcTime	Time of the event

Data	Description
antennaName	Name of the antenna that scanned the transponder.
rSSI	Signal strength of the transponder
rSSIMin	Minimum signal strength of the transponder
rSSIMax	Maximum signal strength of the transponder
power	Radiated power of the antenna at the time of the scan
tagPC	Protocol Control Word You can find more information in the "EPCglobal Specifications".
tagExtPC	Extended Protocol Control Word You can find more information in the "EPCglobal Specifications".
channel	Active send channel at the time of the identification
airRetry	Number of command repetitions on the air interface
commandRetry	Number of command repetitions
filterDataAvailable	Information as to whether the data to be used for filtering could be read from the transponder.
inventoried	Number of identifications of a transponder
phase	Phase with which the transponder was detected.
polarization	Polarization of the antenna at the time of the scan
RSSI events RSSI events provide information about the signal strength of the transponder responses. The number of these events is significantly higher than that of the tag events and they are sent per identification (inventory) and in some cases even per antenna. This results in a precise series of events during the identification process, however it leads to a lot of data traffic particularly if large numbers of transponders are passing through the antenna field. For this reason, we recommend that you only enable RSSI events when diagnostics is necessary. The data enabled in this area also affects the message content of the XML commands/reply frames (e.g. with "writeTagID"). In order for this information to be transmitted, the "RSSI events" check box in the "Diagnostic events of the XML channel" area must be selected.	
utcTime	Time of the event
antennaName	Name of the antenna that detected the transponder.
rSSI	Signal strength of the transponder
power	Radiated power of the antenna at the time of the scan
tagPC	Protocol Control Word You can find more information in the "EPCglobal Specifications".
tagExtPC	Extended Protocol Control Word You can find more information in the "EPCglobal Specifications".
channel	Channel with which the transponder was read.
phase	Phase with which the transponder was detected.
polarization	Polarization of the antenna at the time of the scan

Not every XML command/event report provides information about all the data enabled here. You can find information about which XML commands/event reports provide which data (as well as about the IO events) in the "XML programming for SIMATIC Ident (https://support.industry.siemens.com/cs/ww/en/ps/15081/man)" manual.

The "OPC UA" tab

The "OPC UA" tab is divided into the following areas:

- OPC UA settings
- Diagnostic events and items
- Security settings

OPC UA settings

In the "OPC UA settings" area, you can enable/disable the communication via the OPC UA interface and make basic settings for the OPC UA interface.

Table 6-15 Description of the OPC UA basic parameters

Parameter	Description
Parallel	When the check box is selected, parallel diagnostic access via the OPC UA channel can take place in addition to an existing connection, e.g. to an XML user application. In the case of parallel access, please note that the OPC UA client only has read access to the reader. If parallel access was already enabled for another communication protocol (e.g. XML), this option cannot be selected. If the function is disabled, access to the reader can only take place over a communication channel. If a connection is already established over a communication channel, an additional connection attempt from another communication channel is rejected.
Observed events	If the check box is selected, a tag event frame (OBSERVED) is generated each time a transponder has been reliably detected for "x" read cycles. You can set the number "x" in the section "The "Settings - Read points" menu item (Page 52)" in the area "Algorithmen > Smoothing > Observed Count" (default value = 1).
Lost events	If the check box is selected, a tag event frame (LOST) is generated each time a transponder is no longer detected for at least "x" read cycles. You can set the number "x" in the section "The "Settings - Read points" menu item (Page 52)" in the area "Algorithmen> Smoothing > Lost Count" (default value = 5).
Application name	Name of the OPC UA application of the server. The application name is required to identify the OPC UA namespace of the reader and should be unique for each reader within the project. The application name is part of the URL of the OPC UA server of the reader.
Default port	Here you can change the port number of the application. As default, port number 4840 is used, the standard TCP port for the OPC UA binary protocol.
Minimum sampling interval	Minimum sampling interval at which the reader samples the process data. The sampling interval is limited to a minimum value of 10 milliseconds, to reserve adequate time for other processes. Value range: 10 .. 50 ms Default setting: 50 ms
Minimum supported publishing interval	Minimum publishing interval supported by the server application at which the process data is published for logged on OPC UA clients. Lower values set by an OPC UA client are not taken into account. Value range: 10 .. 65535 ms Default setting: 50 ms

Diagnostic events and items

You can define which events are transmitted in the "Diagnostic events and items" area.

Table 6-16 Description of the diagnostic events

Data	Description
Presence events	If the check box is selected, information about presence states or changes is sent.
Last access events	If the check box is selected, all access operations to the transponders, as well as the data read/written in the process, is sent. Activation of this event is the prerequisite for values being entered in the OPC UA variables of the "Diagnostics" group.
Logbook events	If the check box is selected, all logbook entries are sent.
RSSI events	If the check box is selected, information about the signal strength of the transponder responses is sent. If RSSI events are not enabled, only tag events of the "Observed" type are sent.
Transmitting read point (logbook events)	
Read point (1-4)	Definition of the read point via which the logbook entries are transmitted. The displayed names depend on the interface names specified in the "Read points" menu.

Security settings

In the "Security settings" area, you can make security settings for the OPC UA interface.

Table 6-17 Description of the security parameters

Parameter	Description
Security profiles	Specification of the security profile and the access options for the UA server of the communication module and the server certificate. • No security • Basic 256Sha256 - Sign • Basic 256Sha256 - Sign & Encrypt • Aes128Sha256RsaOaep - Sign • Aes128Sha256RsaOaep - Sign & Encrypt • Aes256Sha256RsaPss - Sign • Aes256Sha256RsaPss - Sign & Encrypt "No security" corresponds to the security profile "None". This profile does not offer any security mechanisms (encryption). If a "Sign" profile is selected, the communication module only allows communication with signed frames using the respective hash algorithm. If a "Sign & Encrypt" profile is selected, the communication module only allows communication with signed and encrypted frames using the respective hash algorithm. The profiles are arranged in order of increasing security levels. It is recommended to use the highest security level (Aes256Sha256RsaPss).
Certificate verification	
Validate certificates	If the check box is selected, the reader generally checks the certificate of the communications partner. If the partner certificate is invalid or not trustworthy, communication is aborted.

Parameter	Description
Accept expired certificates	The reader generally checks the period of validity of the communication partner certificate. When the check box is selected, certificates are accepted and communication is set up even if the current internal reader time is outside the period of validity of the partner certificate.
No strict validation	If the check box is selected, a CA certificate for which no revocation list is stored can still be imported. In addition, the reader allows communication if the application URI of the communication partner is not identical to the application URI in its certificate. Regardless of these exceptions, to establish a connection at least the following requirements must be met: • If the partner certificate is not trustworthy, the reader must at least have stored a self-signed and accepted certificate of the partner. • If the partner certificate was issued by several CAs (Certification Authorities), all CA root certificates must be stored in the certificate store of the reader.
Allow anonymous access	If the check box is selected, the reader allows anonymous users access to the data of its OPC UA server. Anonymous users do not need to specify a user name/password when establishing a connection. If anonymous access is not allowed, an OPC UA client or a user must provide a valid user name/password combination of a user with OPC UA rights. A user with OPC UA rights can be created via the WBM. The user profile preinstalled in the factory (user name: "admin", password "admin") also has OPC UA rights. It is recommended that you disable anonymous access.
Compatible certificate verification according to OpenSSL 1	When the check box is selected, certificates are validated according to the OpenSSL 1 specifications. This means that certificates with the following characteristics can also be used: • End entity certificates that have the value "keyCertSign" set in the "Key Usage" extension. • CA certificates that have nothing marked as "critical" in the "Basic Constraints" extension. • CA certificates that do not have the "Key Usage" extension. If the check box is not selected, the validations take place according to the stricter OpenSSL 3 specifications. Certificates with the previously specified properties cannot be used in this case.
Certificate management	Link to the "System - Certificates" menu command, where you can view existing certificates, import new certificates, as well as create certificate signing requests and transfer certificate files and certificate key files to the reader.
Generate OPC UA server certificate	Button for creating an OPC UA server certificate. Among other things the server certificate serves to identify the OPC UA server to the OPC UA client. The security level of the certificate depends on the selected security profile. The OPC UA server certificate contains the application name, the security profile and the IP address of the reader. If any part of this information is changed, the server certificate needs to be recreated. Note: Note that the procedure can take several minutes.
Outdated security profiles	With this switch, you can show and use security profiles that are outdated and classed as not secure. These profiles should only be used if no other security profile can be used due to compatibility reasons.

6.3.4 The "Settings - Read points" menu item

With the "Settings - Read points" menu command, up to four logical read points can be defined. A logical read point is, for example, an incoming goods gate in logistics or a machine infeed on a production line. A read point, on the other hand, can be assigned one or more antennas required to cover the identification area of the read point.

The settings are structured identically for each read point and divided into the following areas:

- Basic settings
- Assigned antennas
- Algorithms
- Tag fields
- Filters
- Trigger

Note

Parallel operation of read points

Note that simultaneous read/write/inventory access to multiple read points leads to delays. The length of the delay depends on the time required for the command and the number of commands.

The symbol in the Read point tab indicates that the relevant read point is active and at least one antenna has been assigned to it.

Basic settings

You can assign a name to the read point in the input boxes (e.g. Location: "Incoming goods gate 5") and enter information on the location of the read point. Note that the name is required for addressing the readers via XML or OPC UA, whereas the location information is output for OPC UA scan events.

Assigned antennas

In the "Assigned antennas" area, you can assign each read point 1 to 8 antennas depending on the connected reader type. To assign an antenna to a read point, select the button for the desired antenna in the list. If an antenna has already been assigned to a read point, this

antenna is marked with the symbol and the assigned read point is named. To specify antenna parameters for the individual antennas, select the required antenna in the list.

Table 6-18 Description of the antenna parameters

Parameter	Description
Description	Input box for storing device-specific information. For example, the location of the antenna
Radiated power	Required radiated power that the antenna should output. Note: The unit (ERP/EIRP) depends on the selected country profile. The two input boxes (dBm/mW) are linked with each other. If the value in one of the input boxes is changed, the value of the other input box will be adapted automatically. The actual radiated power emitted may be lower due to other components and/or parameters. Value range: 5 ... 36 dBm | 3 ... 4000 mW [1)] Increment: 0.25 dBm | -- [1)] The maximum value depends on the reader type being used.
Gain	The antenna gain affects the actual radiated power and depends on the antenna used. Here, you can either select an antenna based on its name or enter the value of the antenna gain of an antenna being used directly. Value range: -15 ... 15 dBi Increment: 0.25 dBi
Cable loss	The cable loss affects the actual radiated power. The cable loss depends on the cable being used. Here, you can either select a cable based on its name or enter the value of the cable loss of a cable being used directly. Value range: 0 ... 63.75 dB Increment: 0.25 dB
Apply parameters to all antennas	Button that transfers the entered values (radiated power, gain, cable loss) of this antenna to all other antennas.
Effective radiated power	The actual radiated power is made up of the transmit power emitted by the reader, the cable loss and the antenna gain. It is possible that the target value for the radiated power is never reached in reality due to the use of long cables and antennas with low gain. Note: The unit (ERP/EIRP) depends on the selected country profile.
RSSI threshold	The RSSI threshold specifies the signal strength above which a transponder is identified. Only the transponders that reach the RSSI threshold are entered in the list of identified transponders. In reflecting environments (metal reflects UHF waves), transponders could be detected that are not located directly in the antenna field and therefore should not actually be "read". The RSSI value of these transponders is usually noticeably lower than the RSSI value of transponders located directly in antenna field. Such transponders can be filtered out with a suitable RSSI threshold value. In the "Diagnostics - Tag monitor" menu command, all the detected transponders are displayed along with their RSSI values. From the RSSI values of the transponders that should be read and the RSSI values of the transponders that should not be read, it is possible to derive an RSSI threshold value. Range of values: -100 ... 100 Increment: 0.01

Parameter	Description
Input attenuation	The input attenuation weakens the strength of received transponder signals at the reader input. Increasing the attenuation means that received weak transponder signals are no longer identified by the reader. This attenuation applies both to transponder responses as well as to signals of neighboring readers. Adapting this parameter helps to reduce disruptions caused by neighboring readers and transponder populations. Value range: 0 ... 6 dB Increment: 0.25 dB
Sensitivity of the antenna test	When the antenna cable breaks, the transmit power fed in by the reader to the cable is reflected. This effect is taken advantage of to detect an interruption of the connection, e.g. in case of a cable break. Under unfavorable ambient conditions (e.g. due to metallic, reflecting surfaces), the transmit power is reflected by the environment and reflected back to the antenna. This may result in unwanted antenna errors. Using this parameter, you can vary the sensitivity of the antenna test. The lower the dBr value set here, the less sensitive the antenna reacts to a reflected transmit power. If the value is set too high, it may be possible that interruptions in the connection are no longer detected. As of a transmit power of 27 dB, the protection function of the antenna is automatically enabled. The value is then automatically set to 5 dBr and the input box is grayed out. Value range: 0 ... 6 dBr Increment: 0.25 dBr
Polarization	The polarization indicates the alignment of the waves of the antenna and depends on the antenna being used. Most antennas have an unchangeable polarization.

Algorithms

With the algorithms, you can optimize the read/write settings to ensure reliable communication between reader and transponder.

Compared with other frequency bands (LF, HF), UHF RFID has the following special properties:

- Large distances in the range of several meters,
- Reflection of the waves on metal surfaces,
- Region-dependent bandwidth restricted by regulations

In conjunction with the fact that electromagnetic waves in the UHF range are not visible, this often leads to unwanted or incomprehensible responses in UHF systems. Typical, simple examples include:

- Not everything is read or even nothing is read.
- Reading works but writing does not.
- Transponders are identified that should not be identified at all.

There are often simple explanations for these responses and therefore usually also solutions. Algorithms are additional functions that help you to achieve the required functionality even

in difficult radio conditions. The following environmental conditions are possible causes of this response:

- There are several readers/antennas in a restricted space, e.g. every 3 to 5 meters along a production line (high reader density).
- The objects to be identified or the transponders are close together (the distance from each other is less than the antenna field).
- Highly metallic environment (e.g. loading gates with metal ramps)

If none of these conditions applies, the use of algorithms is usually unnecessary.

Note

Algorithms for trained users

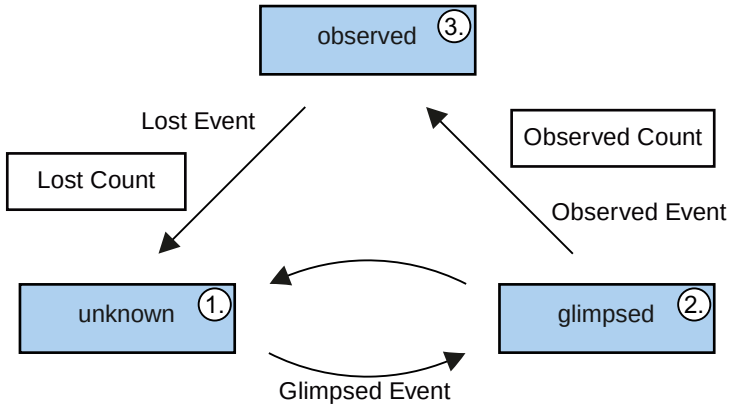
Note that the following algorithms were designed specifically for trained users. Settings in individual algorithms have effects on other algorithms. You should only work with the algorithms when you are aware of the interdependencies between the various algorithms and their purpose.

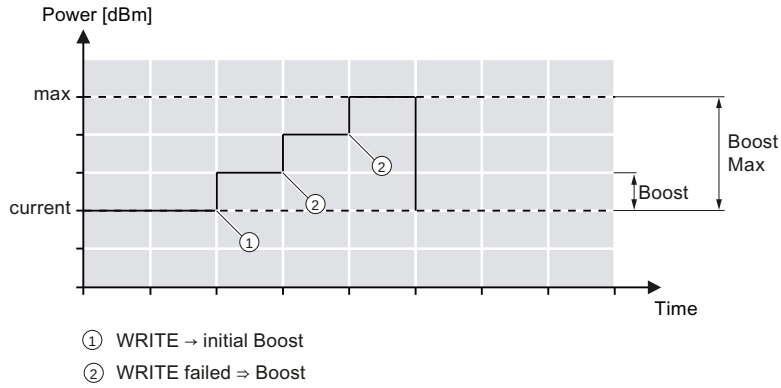
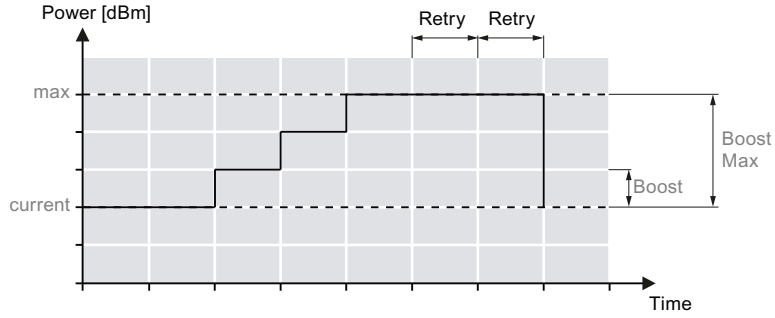
In the section "Application examples (Page 97)" you can find several examples of applications in which the use of the algorithms is described.

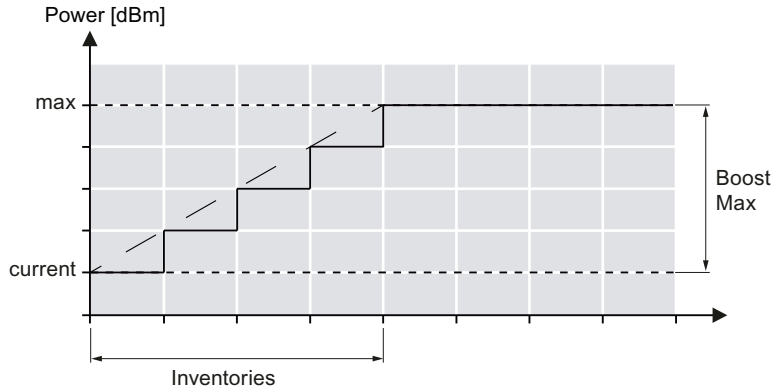
Click the "Adopt parameters from read point" button to transfer all parameters of the algorithms and the session from another read point to this read point.

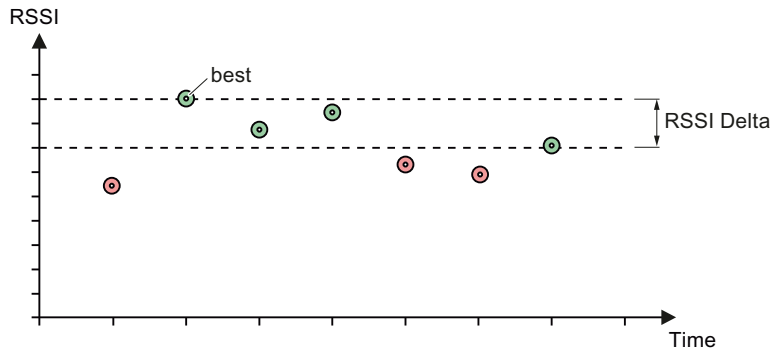
The icons of the algorithms indicate whether the algorithm is enabled ( ) or disabled ( ).

Table 6-19 Description of the algorithms

Algorithm	Description	
Smoothing	This algorithm ensures that only transponders that could be identified often enough will be reported as "reliably identified". Transponders that only appear briefly in the antenna field (e.g. due to overshoot) are filtered out.  	
	Observed Count	The value specifies the number of inventories in which a transponder needs to be identified before it is reported as "reliably identified" (observed). When entering the value "1", the transponder changes to the status "observed" during the first recognition. The "Glimpsed" event as well as the "Observed" event are generated in the process.
	Lost Count	The value specifies how often a transponder reported as "reliably identified" (observed) may no longer identified by cyclic inventories before it is reported as being "not identified" (lost). The value "0" indicates that the lost event is generated at the same time as the observed event. If you enter the maximum value "65535", transponders are never reported as "not identified" (lost).
	unknown	The transponder is unknown to the reader. The transponder has either never been identified or processing of the transponder data by the reader has been completed.
	glimpsed	Transponder was scanned the first time.
	observed	The transponder was identified reliably in at least "x" inventories. The number "x" is specified with "Observed Count".

Algorithm	Description	
Read/Write Power Ramp	The effect of this algorithm is to ensure that there is enough power available when executing a command (Read, Write, Lock, Kill). If a command fails to execute, it is repeated with an increased radiated power. The radiated power is increased step-by-step until the power is adequate to execute the command or until the specified maximum value is reached. Note that the read/write power ramp is based on the inventory power ramp and the power can only be increased up to the maximum possible radiated power. The value "Boost" corresponds to the initial power increase with write access. Read access is performed with the basic power without an initial boost. If a value is entered only for "Boost" and not for "Boost max", only write access with increased power are performed.  	
	Boost [dB]	The value specifies by how many dB the radiated power is increased in each step.
	Boost Max [dB]	The value specifies by how many dB the radiated power can be increased as maximum.
Command Retry	The effect of this algorithm is that commands are executed reliably. If the command fails to execute (Read, Write, Lock, Kill), it is repeated. The algorithm is linked to the "Read/Write Power Ramp" algorithm and is only started when no command could be executed despite the read/write power ramp. If no read/write power ramp is set, the commands are repeated with the current power.  	
	Retries	The value specifies how often the command is repeated with the specified maximum dB increase.

Algorithm	Description	
Inventory Power Ramp	This algorithm increases the radiated power automatically in steps if the specified number of expected transponders is not detected in each inventory. The power is increased until the required number of transponders is identified or until the specified maximum value is reached. This, for example, compensates for fluctuating radio conditions. This algorithm is only used when taking inventories. With read/write commands, the algorithm is not started.  The step size of the individual increases is calculated as follows: Boost max / Inventories Example: Boost max = 5 dB, Inventories = 10 ⇒ Step size = 0.5 dB	
	Expected Tags	The value specifies the minimum number of transponders that should be identified per read point in every inventory. If this value is not reached, the radiated power is increased.
	Boost Max [dB]	The value specifies by how many dB the radiated power can be increased as maximum.
	Inventories	The value specifies the number of inventories to be taken until the maximum radiated power is reached. If the specified number of transponders is identified before the maximum radiated power is reached, the radiated power is not increased to the maximum value.

Algorithm	Description	
RSSI Delta	The effect of this algorithm is that from a number "x" of identified transponders, only the "strongest" are reported. Transponders will only be reported as "reliably identified" if their RSSI value is at least as high as the RSSI value of the best identified transponder minus the RSSI Δ value. This algorithm is used only when taking inventories, for example with the XML commands "readTagIDs" and "readObservedTagIDs".  	
	RSSI Delta	The value specifies the maximum RSSI difference of the transponder compared with the highest RSSI value.
Black List	The effect of this algorithm is that transponders that have already been processed are hidden. This function is particularly useful at read points at which only one individual transponder or a few transponders should be identified but the antenna field is larger than the distance between the neighboring transponders. With suitable XML or control commands, these transponders can be included in the black list and therefore filtered out. For example because these transponders have already been identified or processed.	
	Size	The value specifies the maximum number of transponders (EPC IDs) that can be entered in the black list. The black list is a ring buffer with a configurable size. If all the entries in the black list are occupied, the next new entry deletes the oldest entry.

Sessions

The way in which sessions work is very complex and it is recommended that only trained users should use them. How they work is described in the "EPCglobal Specification (http://www.gs1.org)".

Sequence of the algorithms

The following table figure shows an overview of the algorithms over time.

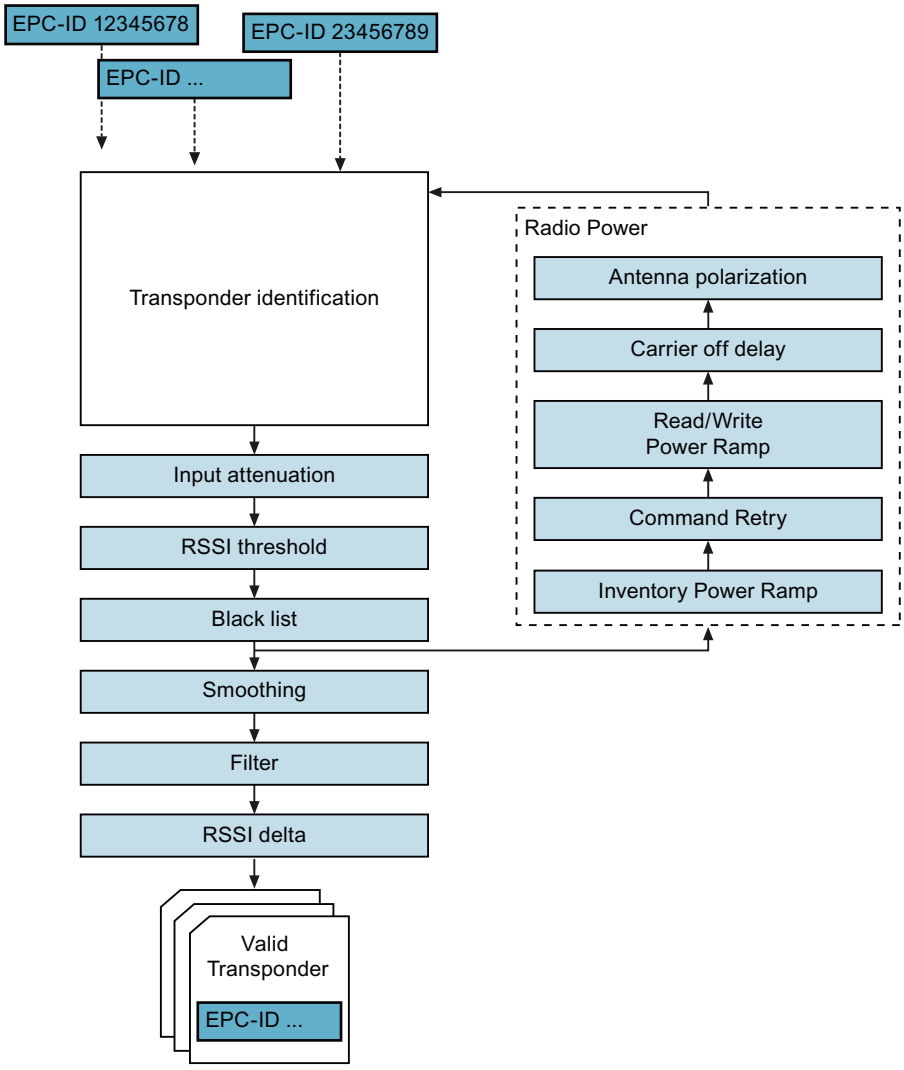


Figure 6-3 Sequence of the algorithms

Tag fields

In the "Tag fields" area, you can assign tag fields to a read point. Tag fields are logical memory areas of a transponder to which symbolic names are assigned. To read out the tag fields from every transponder automatically, these must be assigned to the read points. This additional data is sent along with the "TagEventReports" via the XML interface.

The content of the tag fields is output only via the tag events of the XML commands.

Table 6-20 Description of functions

Icon	Description
+	Assign tag field Click the button to assign existing tag fields to the read point. You can select the required tag field from the drop-down list.
	Create new tag field Click the button to create new tag fields. The button acts as a link to the "Settings - Tag fields" menu command.
×	Delete tag field Click the button to remove tag fields already assigned to the read point.
	Edit tag field Click the button to edit existing tag fields. The button acts as a link to the "Settings - Tag fields" menu command.

You can find more information on the tag fields in the section "The "Settings - Tag fields" menu item (Page 62)".

Filters

In the "Filters" area, you can assign filters to the read point. The data required for the comparison with the filter criteria is read out after reliable identification of a transponder. Depending on the filter evaluation (criterion applies / does not apply), identified transponders are either filtered out or not.

Table 6-21 Description of functions

Icon	Description
+	Assign filter Click the button to assign existing filters fields to the read point. You can select the required filter from the drop-down list.
	Create new filter Click the button to create new filters. The button acts as a link to the "Settings - Filters" menu command.
×	Remove filters Click the button to remove filters already assigned to the read point.
	Edit filters Click the button to edit existing filters. The button acts as a link to the "Settings - Filters" menu command.

You can find more information on filters in the section "The "Settings - Filters" menu item (Page 64)".

Trigger

In the "Trigger" area, you can specify the conditions that will trigger inventories. If a Inventory Power Ramp was configured, this is fired by triggers. If one of the assigned conditions

applies, inventories are taken. With "Trigger action" you can set whether the response to the trigger is that

- a specified/permanent number of inventories is taken or
- inventories are taken for a specified/permanent time [ms].

These conditions are also used with the XML command "triggerSource".

If you do not specify a trigger, you can trigger inventories using the corresponding XML commands ("triggerSource") or OPC UA commands ("Scan", "ScanStart").

Click the button + to specify up to two conditions. Click the button ✕ to remove already specified conditions.

Table 6-22 Description of the trigger conditions

Condition	Description
IO_LEVEL	With this condition, the reader continues to take inventories uninterrupted as long as the selected input/output is in the specified state "On" or "Off".
IO_EDGE	With this condition, the reader takes inventories once, as set under "Trigger action", as soon as the specified edge change occurs at the selected input.
CONTINUOUS	With this condition, the reader takes inventories continuously.
TIMER	With this condition, the reader takes inventories as set in "Trigger action". Following this, there is a pause with the value set in the "TIMER" box [ms].

6.3.5 The "Settings - Tag fields" menu item

In the "Settings - Tag fields" menu command, you can create and edit tag fields. Tag fields are logical memory areas of a transponder that have symbolic names. Memory areas are defined by a logical name, the memory bank, a memory address and the length. If a tag field is created and assigned to a read point, the data of every reliably identified transponder is read out automatically. This data is then reported to the user application with each "observed" "TagEventReport" via the XML interface or with the "LastScanData" and "LastAccessEvent" variables via the OPC UA interface, as long as the created tag fields could be read.

This page is divided into the following areas:

- Tag fields
- Tag field properties
- Transponder memory configuration

Tag fields

The "Tag fields" area contains a list of all tag fields that already exist. To edit a tag field, select the desired field name in the list. The selected field name is highlighted in color. Click the "Insert" button to create a new tag field. Click the "Delete" button to delete the selected tag field.

Tag field properties

In the "Tag field properties" area, you can adapt the parameters of the individual tag fields.

Table 6-23 Description of the parameters of the tag fields

Parameter	Description
Name	Input box for assigning a logical field name / descriptive title to the tag field.
Index	Input box for assigning a unique number to the tag field. Note: When older configurations that did not yet contain this parameter are imported, the index "-1" is automatically assigned to the respective tag fields.
Bank	Drop-down list for selecting the memory bank in which the memory area is located.
Start address	Input box for entering the start address within the selected memory area starting from which you want to read or change/write the data of the target transponder. You can use the button to change the input/output format (decimal or hexadecimal). Value range: 0 ... 65535 | 0 ... FFFF
Length	Input box for entering the data length within the selected memory area, starting from the start address, within which you want to read or change/write the data of the target transponder. You can use the button to change the input/output format (decimal or hexadecimal). Value range: 1 ... 1024 | 1 ... 400 bytes

Transponder memory configuration

The "Transponder memory configuration" area contains a graphic to illustrate the memory configuration and the memory areas of a UHF transponder.

Example

The production date of a product is stored on a transponder in the USER memory area (memory bank 3). The production date is located at address 10 and is 8 bytes long. The corresponding tag field is created and assigned to a read point. Following this, with each object/transponder identification (TagEvent "Observed"), the production date of the object is read out automatically and sent to the XML user application with the EPC ID of the transponder.

Explanation of the memory structure

The transponder memory is divided into four different memory banks.

Table 6-24 Description of the memory banks and areas

Memory type	Description
Memory banks	
USER (Bank 3)	Freely writable "USER" memory area Whether this memory area is available and, if so, how large it is, depends on the transponder type.
TID (Bank 2)	Specified by the manufacturer. The TID contains the class identifier and depending on the transponder type also the serial number of the transponder.

Memory type	Description
EPC (Bank 1)	Contains the EPC ID/UID, the protocol information (Protocol Control Word) and the CRC date of the transponder. You can write to the EPC memory area.
RESERVED (Bank 0)	Contains the access and kill password.
StoredPC	
EPC Length Field	Specifying the length of the EPC ID in words (16 bits)
UMI User Memory Indicator	Indication of whether the transponder contains a "USER" area. This bit is handled differently depending on the transponder type/manufacturer: • The reader can change the bit. • The transponder sets the bit if it has a "USER" area. • The transponder sets the bit if it has a "USER" area and the first words of it are not written with "0".
XPC_W1 Indicator	The transponder has an extended PC area.
NSIT Numbering System Identifier Toogle	Also referred to as a toggle bit. Determines the standard according to which the EPC ID is stored: • 0: EPCglobal • 1: ISO
RFU or AFI	Actual definition the numbering scheme: • RFU: according to EPCglobal • AFI: according to ISO

6.3.6 The "Settings - Filters" menu item

In the "Settings - Filters" menu command, you can create and edit filters. By using filters and depending on the filter criteria, you can separate out specific transponders that will not be processed any further.

This page is divided into the following areas:

- Filters
- Filter properties
- Filter example
- Transponder memory configuration

Filters

The "Filters" area contains a list of all created filters. To edit a filter profile, select the required filter in the list. The selected filter is highlighted in color. Click the "Insert" button to create a new filter. Click the "Delete" button to delete an existing selected filter.

Filter properties

In the "Filter properties" area, you can adapt the properties of the individual filters. To allow filter functions to take effect, the information defined in the filter must be read out from the transponders and compared with the filter criteria.

Table 6-25 Description of the filter properties

Property	Description
Name	Input box for assigning a logical filter name / descriptive title to the filter.
Bank	Drop-down list for selecting the memory bank in which the memory area is located.
Start address	Input box for entering the start address within the selected memory area starting from which the filter is checked. You can use the button to change the input/output format (decimal or hexadecimal). Value range: 0 ... 65535 | 0 ... FFFF
Length	Input box for entering the data length within the selected memory area, starting from the start address, within which the filter is checked. The value specified here affects the length of the input boxes "Mask" and "Filter criterion". You can use the button to change the input/output format (decimal or hexadecimal). Value range: 1 ... 1024 | 1 ... 400 bytes
Including filter	Check box for defining the condition that determines when the transponder is further processed. • Check mark set: The transponder data is transmitted if the filter criteria (filter criterion and mask) apply. • Check mark not set: The transponder data is transmitted if the filter criteria (filter criterion and mask) do not apply.
If filter criteria are not readable, discard transponder	Check box to specify what happens if the filter criteria (filter criterion and mask) could not be read, e.g. because the transponder was in the antenna field for too short a time. Depending on this setting, such transponders are ignored or processed further.
Filter criterion	Specifies which value the checked positions must contain (HEX representation). Possible characters: 0 ... 9, A ... F
Mask	Specifies which positions of the filter criterion should be checked. Example: 00FF → Bits 0 ... 8 of the criterion will be checked. Possible characters: 0 ... 9, A ... F
ASCII	The ASCII box shows the values of the filter criterion additionally in ASCII notation. You can edit the values both in HEX format and in ASCII format.

Filter example

In order to utilize the memory area of a transponder as efficiently as possible, Siemens RFID readers can perform the filtering bit-granularly. Which bits are included as criteria in the filtering and which are not, can be seen from the mask function. The filter example shows an example of how the mask function works.

Transponder memory configuration

The "Transponder memory configuration" area contains a graphic to illustrate the memory configuration and the memory areas. For a detailed description of the memory structure, refer to the section "The "Settings - Tag fields" menu item (Page 62)".

Explanation of how filters work

By using filters and depending on the filter criteria, you can separate out specific transponders that will not be processed any further. For example in environments with different product types, you can use filters to filter out the product types that are unimportant for the application or only identify the relevant product types. This is only possible if the information in the memory area of the transponder has been processed accordingly.

Example scenario

A forklift truck takes a pallet with goods from the conveyor belt to store it in the warehouse and in doing so drives through an RFID gate. Both the pallet and the products on the pallet are fitted with transponders. During the remainder of the process, only the information about the palette is necessary. Assign a filter-relevant ID to the transponder of the pallet. Write, for example, the value "3" to the 10th position (nibble) in the EPC ID of all pallet transponders. The EPC IDs of the goods transponders, on the other hand, must not have a "3" at the 10th position (nibble).

With a suitable filter, (value "3" at the 10th position of the EPC ID), you can now filter out all goods transponders and only identify and process the pallet transponders.

If a filter is active, the appropriate data is read out from each identified transponder and checked against the filter criteria. Depending on this check, transponders are discarded or processed further. A distinction can be made between inclusive and exclusive filtering.

Filter example "Pallet"

The following filter example shows how to create the suitable filter for the example scenario described.

1. Click the "Insert" button and select the newly created filter.
2. Name: Enter a name in the input box.
3. Bank: Select the memory area "1 - GEN2 - EPC" from the drop-down list.
4. Start address: Enter "8" in the input box as the start address.
   The EPC ID starts at the 4th byte of the "EPC" memory area. Each byte writes two positions of the EPC ID. To address the 10th digit (nibble) of the EPC ID, you must therefore select the 8th byte of the EPC memory area.
5. Length: Enter "1" in the input box as the number of bytes.
   Each byte contains two positions of the EPC ID. As only one digit (nibble) is relevant in this example, you must mask out the second digit (nibble) using the mask.
6. Filter criterion: Enter "03" in the input box as the criterion.
   As only the second digit (nibble) is relevant in this example (due to the mask), the first digit (nibble) of the criterion can also have any other value.

7. Mask: Enter the value "0F" in the input box.
   This specifies the positions in the EPC ID relevant for filtering out.
   Alternatively, you can also specify the value "03". This defines that the pallet transponder at the 10. The first digit (nibble) only needs to have the first two bits set. The remaining bits could then additionally be used for the differentiation of the goods transponders.
8. Inclusive/exclusive filters: Select the "Inclusive filter" filter type.

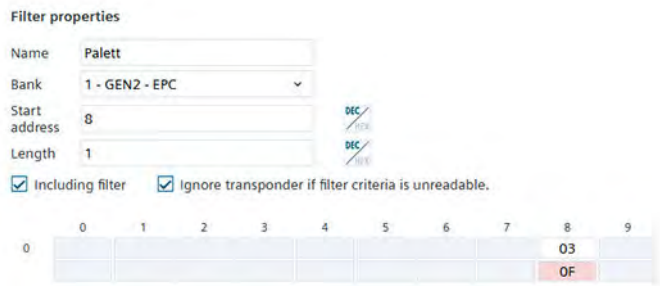


Figure 6-4 Example of a filter

9. Unreadable filter data: Select how the reader handles transponders whose filter data cannot be read out.
10. Enable the filter in the "Settings - Read points" menu command and transfer the configuration to the reader.

6.3.7 The menu command "Settings - DI/DO"

In the "Settings - DI/DQ" menu command, you can set the properties of the digital outputs and assign functions to the interface. The areas displayed depend on the basic settings.

Basic settings

In this area, you can select how the I/O interface is to be operated. Depending on your selection, the corresponding parameters are displayed in the following area.

The following options are available for the I/O interface:

- Input:
  An individual digital input (In 0) can be configured.
- Output:
  An individual digital output (Out 0) can be configured.
- IO-Link:
  Mode for connection of an IO-Link device with digital inputs/outputs. Up to eight digital inputs and outputs (In 0 ... 7 and Out 0 ... 7) can be configured.

NOTICE
Max. permissible current with operation via PoE
With operation via PoE, only digital inputs/outputs that require a maximum of 200 mA in total can be connected.

Depending on the selection made here, the "IO-Link process data assignment" areas and/or the outputs are displayed below.

IO-Link process data assignment

You can select the connected IO-Link device in the "IO-Link process data assignment" area via the "IO-Link device" drop-down list. If an IO-Link device from Siemens has been selected, the Mapping lines are automatically populated accordingly and the bit assignment is displayed. Otherwise, you need to perform the mapping manually. In the input/output mapping lines, you specify the 8 bits of the IO-Link device that should be accessed (by entering the number "1").

Digital inputs/outputs

You can define the following settings for each output in the basic settings:

Table 6-26 Description of the basic settings

Boxes	Description
Inactivity	State that the output adopts following device startup.
Reset time	Input box for entering the reset time after which the output automatically returns to the idle state. A value of 0 means that the state of the output is not influenced by this automatic function. Value range: 0 ... 65535 ms Increment: 1 ms

In the "Events" area, you can define events/conditions that cause a digital output to change to one of the following statuses:

Click the button + to add new events. Click the button ✕ to remove already specified events.

Table 6-27 Description of the events

Drop-down list	Description
1st to 3rd drop-down list	Selection of the event that causes the digital output in question to change its state: • Antenna error: If an antenna error occurs, the output is changed to the state specified here. Select the antenna and read point at which the event must occur. • Transponder detected: If a transponder is detected via the selected read point, the output is set to the state defined here. Select the read point at which the event must occur. • Input change: If the state at the selected digital input changes, the output is set to the state specified here. Select the input at which the change must occur as well as the type of state change. • Output change: If the state at the selected digital output changes, the output is set to the state specified here. Select the output at which the change must occur as well as the type of state change.
4th drop-down list	Selection of the state that the relevant digital output assumes: • On: The output is turned on. • Off: The output is turned off. • Inverted: The output changes its state starting from the state that is active at the moment the event occurs.

Note

Reaction time of the digital outputs

Note that the reaction time of the digital outputs (and inputs) depends on the reader load and is 50 - 100 ms on average.

Note the following properties of the digital outputs:

- The outputs are only changed once when the event occurs.
  The state of the individual outputs remains unchanged, even if the event is no longer or still present.
- If an output is changed when an antenna error is detected, this output remains unchanged even if the antenna error is rectified.

6.3.8 The "Settings - Adjust antenna" menu item

In the "Settings - Adjust antenna" menu command, you can optimize the antenna alignment.

This page is divided into the following areas:

- Basic settings
- Transponder list
- Alignment quality

NOTICE
Access to the reader Remember that simultaneous access to a reader using multiple WBM clients (web browsers) is possible but not recommended. If changes are made when two browsers are accessing a reader at the same time, this can lead to errors in the configuration or to an undesired result.

Procedure for optimizing the antenna alignment

Note

Do not optimize the antenna alignment during operation

We recommend that you do not optimize (▶) the antenna alignment during operation since this can disturb operation. Optimize the antenna alignment prior to putting the system into operation.

① Select the required antenna and start measuring the RSSI values.
② Select a transponder from the list.
③ Transfer the activation power to the "Radiated power" input box of the "Settings - Read points" menu command.
④ Adjust the antenna until you obtain the highest possible RSSI value.

Basic settings

In this area, you can select the antenna whose alignment needs to be optimized. With the "Start/Stop adjustment" buttons (▶/■), you can control the measurement of the RSSI values.

By clicking the ▶ button, a new measurement is started. Cyclic reads are performed on the selected antenna and the measured values obtained are displayed. Any existing measured values from a previous measurement are deleted. By clicking the ■ button, the measurement is stopped.

Transponder list

After you have started measuring the RSSI values, all the transponders identified by the reader are listed in the table. Select the "EPC ID in ASCII format" check box to display the EPC IDs of the transponders in ASCII representation.

Table 6-28 Displayed properties of the recognized transponders

Property	Description
EPC ID	ID of the identified transponder
RSSI	Last measured RSSI value of the transponder
RSSI min.	Minimum RSSI value of the transponder. Calculated over all successful inventories.
RSSI max.	Maximum RSSI value of the transponder. Calculated over all successful inventories.
Acquisition cycles	Number of inventory replies (scans) of this transponder.
Date/time	Time stamp when the transponder was observed the first time.

You can select the transponders individually in the table. The selected transponder is highlighted in color and its measured values shown in the "Alignment quality" area.

By clicking the "Apply" button, you transfer the power entered in the field to the "Radiated power" input box of the "Settings - Read points" menu command. This function is only available during the measurement and the change is transferred directly to the reader. To save this change fail-safe on the reader, you must stop the measurement and save it using the "Transfer configuration to reader" button.

Alignment quality

This area shows the measured RSSI values of the selected transponder. The bar chart shows the maximum measured and the current or last measured RSSI value of the transponder. The current RSSI value of the selected transponder is also displayed as a numeric value. The higher the value or the bar, the better the transponder was detected.

Alternatively, you can also display the number of transponders in the antenna field. You can use the arrow symbols to enlarge or reduce the window with the numerical RSSI value or the transponder count. This allows you to read the current RSSI value or the number of transponders in the antenna field even at long distance. By varying the antenna position, you can quickly and easily find the optimal alignment for the highest RSSI value and the optimal spatial coverage of the antenna field.

6.3.9 The "Settings - Activation power" menu item

With the "Settings - Activation power" menu command, you can detect the activation power of the various antennas. This function helps you to find the optimum radiated power with which the transponder will be reliably identified without generating overshoot.

This page is divided into the following areas:

- Basic settings
- Transponder list

- RSSI graph
- Measuring range settings

Procedure for determining the activation power

Note

Do not determine the activation power during operation

We recommend that you do not determine (▶) the activation power during operation since this can disturb operation.

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① Select the required read point and antenna, enter the required measuring range values and start the measurement.
② Select a transponder from the list.
③ Transfer the activation power to the "Radiated power" input box of the "Settings - Read points" menu command.

Basic settings

In this area, you can select the read point and antenna whose optimum activation power you want to measure.

With the "Start/Stop measurement" (▶/■) and "Delete display" (✕) buttons, you can control the power measurement.

By clicking the ▶ button, a new recording is started. Cyclic reads are performed on the selected antenna and the measured values obtained are displayed. Any existing measured values from a previous recording are deleted. By clicking the ■ button, the recording is stopped. Clicking the ✕ button deletes the currently displayed measured values.

Transponder list

After you have started the measurement, all the transponders identified by the reader are listed in the table (max. 500). Select the "EPC ID in ASCII format" check box to display the EPC IDs of the transponders in ASCII code.

Table 6-29 Displayed properties of the recognized transponders

Property	Description
☐ / ☑	Selection of the transponders to be displayed in the graph. Up to 10 transponders can be selected. The selected transponders are shown in the RSSI graph as continuous lines. Up to 10 further transponders from the list are displayed as dots.
EPC ID	ID of the identified transponder
Antenna	Antenna with which the transponder was detected.
Polarization	Polarization of the antenna
Power (min)	Minimum radiated power [dBm] of the antenna with which the transponder was identified.
Power	Radiated power [dBm] of the antenna with which the transponder was last identified.

Property	Description
RSSI	Last measured RSSI value of the transponder
RSSI min.	Minimum RSSI value of the transponder. Calculated over all successful inventories.
RSSI max.	Maximum RSSI value of the transponder. Calculated over all successful inventories.
Acquisition cycles	Number of inventory replies (scans) of this transponder.
Date/time	Time stamp when the transponder was observed the first time.

The value "Power (min)" of the transponder last selected in the transponder list is automatically transferred to the "Apply power" box with 2 dB added. By clicking the "Apply" button, you transfer the power entered in the field to the "Radiated power" input box of the "Settings - Read points" menu command. This function is only available during the measurement and the change is transferred directly to the reader. To save this change fail-safe on the reader, you must stop the measurement and save it using the "Transfer configuration to reader" button.

Note

Apply radiated power

The value entered automatically in the "Apply power" box corresponds to the minimum value with which the transponder was identified by the antenna (Power (min)) plus a power reserve of 2 dB. This value serves as a guideline and you can adapt it. To be sure that the antenna reliably detects the transponders regularly, we recommend that you accept the automatically adapted default value.

RSSI graph

The graph shows the course of the measured radiated power (dBm value) of the selected antenna over time (black line) and the RSSI values of all selected transponders (colored lines/dots). Using the icons, you can modify the display of the graph and adapt it to your needs.

Table 6-30 Buttons of the RSSI graph

Icon	Description
	Show/hide grid lines
	Change RSSI curve types Measured values are joined together either with a direct line or using horizontal and vertical lines.
	Change over the background color, between white and black
	Make RSSI graph moveable / fixed The RSSI graph is either fixed or the zero point of the graph can be moved holding down the left mouse button.
	Highlight RSSI measured values Display measured values with thin or thick points.

The current RSSI value of the last selected transponder or the number of valid transponders is also displayed as a numerical value. You can control the value that is displayed using the radio buttons. Using the arrow symbols, you can expand or reduce the window for the numeric RSSI value.

You can use the "Export" ( ) and "Import" ( ) buttons to export recorded data of the RSSI graph as a *.json file and re-import it a later time. This function enables recording of data for later analysis. During the import, make sure that the recorded data of the file corresponds to your configuration (reader type and connected antennas identical).

Measuring range settings

Using the input boxes in this area, you can influence the measurement. The parameters contained in this area are intended for trained users. For untrained users, we recommend using the default settings.

Table 6-31 Description of the measuring range parameters

Parameter	Description
Power from ... to ...	Specifies the range of values (dBm value) within which the measurement is made. The measurement starts at the "from" value and ends automatically as soon as the "to" value is reached.
Increment	Specifies the dB value by which the radiated power is increased step-by-step during the measurement.
Time interval	Specifies the time after which the radiated power is increased by the dB value increment during the measurement. Note that with a large number of transponders and inventory can take several seconds. Select a suitably large time interval to make sure that several inventories can be taken. If the time interval selected is too short this can mean that the power is increased already before the end of the inventories.

6.3.10 The "Diagnostics - Tag monitor" menu item

In the "Diagnostics - Tag monitor" menu command, you obtain an overview showing which transponder was identified and how well with which antenna.

This page is divided into the following areas:

- Basic settings
- Browser acquisition
- Transponder list
- RSSI graph
- Online status of digital inputs/outputs

NOTICE
Aborting diagnostics due to open browser tabs Note that some web browsers stop updating web pages when other tabs are in the foreground. This behavior may cause ongoing diagnostics to be ended.

Basic settings

In this area, you can select the read point and the antenna with the behavior you want to analyze. Depending on the reader being used or the connected antenna, you can also select the antenna polarization. With the "Start/Stop diagnostics" (▶/■), "Pause" (❚❚/❚▶) and "Delete display" (✕) buttons, you can control the diagnostics.

By clicking the ▶ button, a new diagnostics phase is started. Any existing measured values from previous diagnostics are deleted. If read procedures are triggered via the IT system or a digital input, the measured values are displayed in the table and in the RSSI graph. By clicking the ■ button, the diagnostics is stopped. Clicking the ✕ button deletes the currently displayed measured values. With the ❚❚ button, you can halt the display of the RSSI graphs while the diagnostics continues.

Browser acquisition

Note

Diagnostics in the commissioning phase / during normal operation

You can observe the behavior of the connected RFID system using the Tag monitor. Here a distinction is made between diagnostics during operation and diagnostics during the commissioning phase:

- Diagnostics during commissioning phase ("Continuous acquisition")
  To monitor the behavior of the RFID system during the commissioning phase, you can have the read access triggered continuously by the reader itself ("Continuous acquisition" enabled). Note that this type of diagnostics influences the acquisition triggers of any higher-level systems because continuous acquisition is given higher priority.
- Diagnostics during normal operation ("Single acquisition")
  To monitor the behavior of the RFID system during operation, you can trigger individual reads manually using the "Single acquisition" button. This requires "Continuous acquisition" to be disabled and read operations to be controlled during normal operation by a higher-level system (e.g. IT system) or by a digital input.

Note that diagnostics via the tag monitor influences current operation. In particular, continuous acquisition can lead to data being sent to the higher-level system, which can lead to unwanted behavior.

Follow the steps below to perform diagnostics during operation:

1. Disable the "Continuous acquisition" function if needed.
2. Start the diagnostics phase (▶).
3. Click the "Single acquisition" button.

Follow the steps below to perform diagnostics during the commissioning phase:

1. Enable the "Continuous acquisition" function.
2. Start the diagnostics phase (▶).

Transponder list

After you have started the diagnostics, all the transponders identified by the reader are listed in the table (max. 500).

Table 6-32 Output fields and displayed properties of the detected transponders

Output fields/ property	Description
Identified transponders	Number of transponders physically detected since the start of recording (max. 500).
Valid transponders	Number of identified transponders recognized by the reader as being valid since the start of the recording and that were possibly further processed.
Transponders in the antenna field	Number of transponders located in the antenna field of the reader or the antenna during the last inventory.
EPC ID in ASCII format	Check box to display the EPC IDs of the transponders in ASCII code.
☐ / ☑	Selection of the transponders to be displayed in the graph. Up to 10 transponders can be selected. The selected transponders are shown in the RSSI graph as continuous lines. Up to 10 further transponders from the list are displayed as dots.
✓ / ℹ / ☐	Filter status of the transponders • Transponder was identified and returned to the user application. • Transponder was identified but filtered out. Place the mouse over the symbol to find out which setting has filtered out the transponder. • Transponder was identified and not filtered out. However, the transponder has not yet been returned to the user application (e.g. due to the "Glimpsed" status).
EPC ID	ID of the identified transponder
Antenna	Antenna(s) with which the transponder was identified.
Power (min)	Minimum radiated power [dBm] of the antenna with which the transponder was identified.
Power	Radiated power [dBm] of the antenna with which the transponder was last identified.
RSSI	Last measured RSSI value of the transponder
RSSI min.	Minimum RSSI value of the transponder. Calculated over all successful inventories.
RSSI max.	Maximum RSSI value of the transponder. Calculated over all successful inventories.
Acquisition cycles	How often a transponder was detected by a reader or an antenna. An inventory means a scan on the air on one antenna and with one polarization. Not to be confused with number of inventories. With a single inventory, acquisition cycles for all antennas and with all polarizations are queried. If, for example, two antennas with three polarizations each are connected to a reader, an inventory contains six acquisition cycles.
Date/time	Time stamp when the transponder was observed the first time.

RSSI graph

The graph shows the course of the measured radiated power (dBm value) of the selected antenna over time (black line) and the RSSI values of all selected transponders (colored lines/ dots). Using the icons, you can modify the display of the graph and adapt it to your needs.

Table 6-33 Buttons of the RSSI graph

Icon	Description
	Show/hide grid lines
	Change RSSI curve types Measured values are joined together either with a direct line or using horizontal and vertical lines.
	Change over the background color, between white and black
	Make RSSI graph moveable / fixed The RSSI graph is either fixed or the zero point of the graph can be moved holding down the left mouse button.
	Highlight RSSI measured values Display measured values with thin or thick points.

In the "Recording time" input box, you can specify the duration of the recording shown in the graph. With the aid of the gray shift bar, you can move the recording time of the draft forwards or backwards.

The current RSSI value of the last selected transponder or the number of valid transponders is also displayed as a numerical value. You can control the value that is displayed using the radio buttons. Using the arrow symbols, you can expand or reduce the window for the numeric RSSI value.

You can use the "Export" ( ) and "Import" ( ) buttons to export recorded data of the RSSI graph as a *.json file and re-import it a later time. This function enables recording of data for later analysis. During the import, make sure that the recorded data of the file corresponds to your configuration (reader type and connected antennas identical).

Online status of digital inputs/outputs

In this area you can have these statuses of the digital inputs/outputs displayed. Inputs on a gray background indicate that the inputs are not connected. Outputs shown on a gray background indicate that the output is logically set to the value "0" or is turned off. If an input or an output is shown on a green background, this indicates logical status "1" or that the output is turned on or that a signal is applied to the input.

You can use this function to control the outputs. To make changes, you need to click the "Apply" button.

6.3.11 The "Diagnostics - Log" menu item

The log of the reader is displayed in the "Diagnostics - Logbook" menu command. All actions performed by the reader are documented in the logbook.

This page shows all message types that were selected in the menu command "Settings - General" in the "Logbook settings" area.

The entries contain the following properties:

Table 6-34 Displayed properties of the logbook messages

Property	Description
Date/time	Time stamp when the entry was made be the reader. Note that the time stamp is generated by the device clock (UTC time). This time is compared with the time format and time zone set on the PC/PG and displayed in the corresponding format.
Type	Type of message The message types that are signaled depends on the check boxes selected in the menu command "Settings - General" in the "Logbook settings" area.
Entry	Text of the message

You can use the buttons to control the entries:

- Update
  The logbook is read in again from the reader and the list updated. The logbook entries displayed include the most current data (200 KB).
- Total history
  The complete stored logbook of the reader is read in. The logbook entries displayed include all saved data (10 MB).
- Save
  The logbook read out by the reader is saved as a *.csv file on the PC/PG.
  Note that the time stamp is generated by the device clock according to ISO 8601 (UTC time).
- Reset
  All logbook entries are deleted in the reader.

With the "ISO 8601 (UTC)" check box, you can convert the data display in the "Date/Time" column to UTC time, identical to the output in the exported logbook.

With a large number of logbook entries in the history, it may take several minutes before these are displayed.

6.3.12 The "Diagnostics - Service Log" menu item

The service logbook of the reader is displayed in the "Diagnostics - Service Log" menu command. The logbook records internal processes of the reader and is required for service support by SIEMENS specialists. Only make settings on this page if you are instructed to do so by SIEMENS personnel. The logbook entries are also evaluated by SIEMENS personnel.

This page displays all the message types that have been defined in the "Settings - General > Service logbook settings" menu.

The entries contain the following properties:

Table 6-35 Displayed properties of the logbook messages

Property	Description
Date/time	Time stamp when the entry was made be the reader. Note that the time stamp is generated by the device clock (UTC time). This time is compared with the time format and time zone set on the PC/PG and displayed accordingly.
Type	Type of message The message types that are signaled depends on the check boxes selected in the menu command "Settings - General" in the "Logbook settings" area.
Entry	Text of the message

With the "Update", "Save as" and "Reset" buttons, you can control the entries:

- Update
  The logbook is read in again from the reader and the list updated. The logbook entries displayed include the most current data (200 KB).
- Total history
  The complete stored logbook of the reader is read in. The logbook entries displayed include all saved data (10 MB).
- Save
  The logbook read out by the reader is saved as a *.csv file on the PC/PG.
- Reset
  All logbook entries are deleted in the reader.
- Service file
  All diagnostics-related data from the reader is saved as an *.slf file on the PC/PG. The file only contains information relevant for Siemens service personnel and can only be evaluated by them.

With the "ISO 8601 (UTC)" check box, you can convert the data display in the "Date/Time" column to UTC time, identical to the output in the exported logbook.

With a large number of logbook entries in the history, it may take several minutes before these are displayed.

6.3.13 The "Diagnostics - Syslog logbook" menu item

The menu command "Diagnostics - Syslog logbook" displays the logbook of the Syslog messages when the Syslog function is enabled. This page can only be called by users with administrator rights.

This menu command documents all safety-related access operations to the reader and performed actions. You can find detailed information on the Syslog messages, their structure and contents in the section "Syslog messages".

With the "Update" button you can read in the entries from the reader again and update the list. The displayed log entries contain 128 KB of data.

6.3.14 The "Edit transponder" menu item

In the "Edit transponder" menu command, you can edit all the transponders identified from the read point that are currently located in the antenna field.

This page is divided into the following areas:

- Basic settings
- Transponder selection
- Write EPC ID
- Read/write
- VDA UM read
- Transponder memory configuration
- Lock
- Kill

Basic settings

In this area, you can select the read point with which transponders will be processed.

Select the "EPC ID in ASCII format" check box to display the EPC IDs of the transponders in ASCII code.

Transponder selection

The "EPC ID" list contains a list of the transponders recognized by the read point. To obtain or update list entries, click the "Scan" button. To edit a transponder, select the required EPC ID in the list. The selected EPC ID is highlighted in color. The selected EPC ID is also displayed in the "Selected EPC ID" box.

The EPC ID selected in the list is displayed in the "Selected EPC ID" box. In the "Password" input box, you can enter the access or kill password. You require these passwords to "Lock" or "Kill" the transponder. You specify the passwords in the section "The "Settings - Tag fields" menu item (Page 62)".

Write EPC ID

In the "New EPC ID" text box, you can enter the ID of the transponder. Click the "Copy selected EPC ID" button to transfer the EPC ID selected in the list to the text box. This allows you to change existing IDs both quickly and simply. Click the "Write" button to assign the new EPC ID to the transponder.

Read/write

In the "Read/write" area, you can read out and overwrite the memory areas. In so doing, you can access pre-defined addresses (tag fields) and existing memory areas. Using the parameters, you can adapt the memory area manually.

Table 6-36 Description of the parameters of the tag fields

Parameter	Description
Tag fields	List with pre-defined addresses.
Bank	Drop-down list for selecting the memory area to be read/written. The following properties relate to this setting.
Start address	Input box for entering the start address within the selected tag field starting from which you want to read or change/write the data of the target transponder. You can use the button to change the input/output format (decimal or hexadecimal). Value range: 0 ... 65535 | 0 ... FFFF
Length	Input box for entering the data length within the selected tag field, starting from the start address, within which you want to read or change/write the data of the target transponder. You can use the button to change the input/output format (decimal or hexadecimal). Value range: 1 ... 1024 | 1 ... 400 bytes
Data	Input/output boxes for the values (decimal or hexadecimal) Possible characters: 0 ... 9, A ... F
Initialize data	Button for initializing the data You can preset several data fields with a value using the input box and the initialization function.

Next to the list of tag fields, the data of the selected memory area is displayed (decimal or hexadecimal and in ASCII).

The "Read" button is used to read the data from the transponder. The data read from the transponder is highlighted in red to distinguish it from the data entered manually. If no values are displayed, this means that no values have yet been read from the transponder.

Click the "Write" button to transfer the changed data to the transponder.

Transponder memory configuration

The "Transponder memory configuration" area contains a graphic to illustrate the memory configuration and the memory areas of a UHF transponder.

Lock

In the "Lock" area, you can protect or unlock the memory areas (banks) as well as the access and kill password of the selected transponder. Select the "Permanent" check box to make the

setting irreversible. Click the "Apply" button to transfer the settings to the transponder. To enable/disable protection, you need to enter the access password in the "Password" input box.

NOTICE
Read/write protection of the passwords Note that passwords that are read/write protected can no longer be read out. We recommend that you note down the password.

Kill

In the "Kill" area, you can 'destroy' the entire memory of the transponder. Following a successful "kill", the transponder can no longer be identified by any RFID reader and is therefore no longer usable. To "destroy" the transponder, click the "Execute" button. To destroy the transponder, you need to enter the kill password in the "Password" input box. Note that with the default kill password, you cannot destroy a transponder. To destroy a transponder with the kill password, you first need to assign a kill password.

Explanation of the Lock command

With the factory setting, some UHF transponders are not protected by passwords in other words they are in the "open status". With these transponders, the memory banks 0 (kill / access password), 1 (EPC) and 3 (USER) can be modified with a write command. Memory bank 2 (TID) can normally not be changed since this is already locked by the manufacturer.

Some use cases, however require that writing is checked or completely prevented. The "Lock" command is used for this purpose. With this, you can lock individual or multiple memory areas. To be able to lock memory areas, you need to change the transponder to the "protected status" (access password ≠ 00000000). You create the access password using the predefined tag fields and can edit this using the "Read/write" area in this menu.

After you have changed the access password, you can still access the memory areas with the default password. To access the memory areas only with the access password, enable the read/write protection of the relevant memory area.

To lock the memory areas, two bits are used. These bits can also be combined with each other. Depending on the memory area this has the following different effects:

Table 6-37 Memory bank "1" (EPC) and memory bank "3" (USER)

Write protection	Permanent	Description
--	--	The associated memory area can be written in both the open status and in the protected status.
--	✓	The associated memory area can be written to permanently in both the open and in the protected status and must not be protected.
✓	--	The associated memory area can only be written to in the protected status.
✓	✓	The associated memory area cannot be written to in any of the statuses.

Table 6-38 Memory bank "0" (kill/access password)

Read/ write protection	Permanent	Description
--	--	The associated password memory area can be read and written to in both the open and in the protected status.
--	✓	The associated password memory area can be permanently read and written to in both the open and in the protected status and must not be protected.
✓	--	The associated password memory area can be read and written to only in the protected status.
✓	✓	The associated password memory area cannot be read or written to in any of the statuses.

Example

To ensure that the EPC ID cannot be modified by every user, you need to lock it. First assign an access password (≠ 00000000) and then lock memory bank 1 (EPC). You also need to lock the access password in memory bank 0 (access password) to make sure that no one can read out the access password and then change the EPC ID with it.

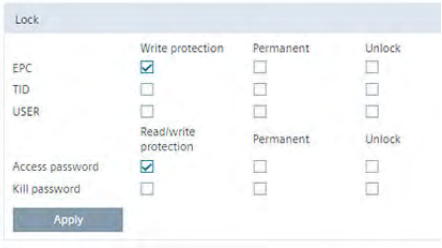


Figure 6-5 Example of locking the EPC ID

6.3.15 The "User management" menu item

In the "User management" menu command, you can enable/disable authentication, create, delete and edit user profiles and change passwords. Note that only users with the "Administrator" role can create new user profiles or edit other user profiles.

This page is divided into the following areas:

- User profiles
- User properties
- Password
- Roles
- Auto logout
- Authentication
- Security settings

User profiles

The "User profiles" area contains a list of all existing user profiles. Up to a maximum of 30 user profiles can be created. To edit a user profile, select the required user in the list. The selected user is highlighted in color.

Click the "Add user" button to create a new user. Click the "Delete" button to delete a selected user profile.

User properties

Enter the name of the newly created user profile in the "User" input field. You require the user and the password to log in to the WBM. This name cannot be edited subsequently.

In the "Name" input box, you can enter the name of the person or the name of the group that works with the user profile. In the "Description" input box, you can enter further information about the user profile.

Password

Enter the password of the user profile in the "Password" and "Repeat password" input boxes. You require the user name and the password to log in to the WBM. User passwords can be changed by the users themselves or an administrator.

NOTICE
Security recommendation: Passwords Refer to the notes in the "Authentication" section in section "Security recommendations (Page 11)".

If you lose your administrator password, you must reset the reader to the factory settings as described in the section "Restoring the factory settings for the hardware (Page 118)".

Roles

In the "Roles" area, you can assign roles to the user profile. Click the relevant check box to assign the required roles to the user profile. The "Administrator" role has all read/write rights.

- Administrator
  User profile with all read/write rights
- Read only
  Restricted user profile with read rights. With the "User" role, you cannot create new user profiles or edit other user profiles. Neither can you write to the reader.
- OPC UA
  Restricted user profile with OPC UA rights. With the "OPC UA" role, you can only establish the OPC UA connection. This role has no rights whatsoever in the WBM and it cannot be used to log into the WBM

Click the "Save" button to save the changes and to create the new user profile.

Auto logout

In this area, you can switch Auto logoff on/off and specify the time period after which you are automatically logged off of the WBM. This time period expires due to inactivity and is automatically reset to the value you have set. As soon as the configured time period has

expired, the connection to the reader is automatically disconnected. This ensures that the connection to the reader is not blocked by an inactive user and is shared with other users.

Authentication

You can activate/deactivate authentication in this area. Note that any user has all read/write privileges (administrator rights) when authentication is disabled. If authentication is switched on, all users must log on in the login area of the WBM to be able to make changes in the WBM.

NOTICE
Security recommendation: Authentication To ensure that no unauthorized persons can access the reader settings, we recommend that you enable the authentication and create new user profiles. Note that authentication can only be enabled/disabled by an administrator.

Security settings

In the "Security settings" area, you can define the conditions for the password policies, session timeout and Brute Force Prevention (BFP).

A brute force attack is an attack method in which password-protected access is decrypted through repeated and systematic input of user password variants and combinations using powerful computer systems and automated software. This attack method is able to process the maximum number of user password variants/combinations possible with high performance.

Brute Force Prevention (BFP) thwarts this attack method by limiting the number of invalid login attempts, significantly lowering the chance that brute force attacks will succeed. The Leaky Bucket algorithm is used for SIMATIC Ident devices. This means that the BFP level is incremented by the set value at each invalid login attempt. When the BFP level exceeds the defined BFP threshold, all further login attempts are ignored. At the same time, the BFP level is reduced by the set value every second.

Table 6-39 Description of the security settings

Parameter	Description
Password policy	
Minimum number of characters	In this input box, you can define the minimum number of characters required for passwords of the user profiles. Value range: 1 ... 32 Default setting: 8
Minimum number of character types	In this input box, you can define the minimum number of character types that must be used in passwords of the user profiles. Value range: 1 ... 4 Default setting: 3
Session timeout	

Parameter	Description
Duration	In this input box, you can define the duration after which the connection to the reader is disconnected automatically (e.g. due to a disconnected Ethernet connection). Value range: 30 ... 3600 s Default setting: 60 s
Brute Force Prevention (BFP)	
Increase of BFP level (per invalid login attempt)	In this input box, you can define the value by which the BFP level is increased at each incorrect password entry or each invalid login attempt. Value range: 1 ... 10000 Default setting: 200
BFP threshold	In this input box, you can define the BFP threshold. As soon as the BFP threshold is exceeded due to multiple incorrect password entries or invalid login attempts, all further login attempts are ignored. Only when the current BFP level drops below the BFP threshold can login attempts be made successfully again. Value range: 1 ... 10000 Default setting: 4000
Maximum value of the BFP level	In this input box, you can define the maximum value of the BFP level. The BFP level is increased maximally up to the value specified here. Value range: 1 ... 10000 Default setting: 10000
Reduction of BFP level (per second)	In this input box, you can define the value by which the current BFP level is reduced every second. The BFP level is reduced every second until the BFP level has dropped to "0". Value range: 1 ... 10000 s Default setting: 50 s
Number of invalid login attempts	Output box/display of the number of invalid login attempts that can be made within x seconds until the BFP threshold value is reached and, consequently, all further login attempts are ignored. The values in these boxes are calculated based on the input boxes "Increase of BFP level (per invalid login attempt)" and "BFP threshold" or "Increase of BFP level (per invalid login attempt)" and "Reduction of BFP level (per second)".
Maximum wait time	Output box/display of the maximum wait time that passes after the Maximum value of the BFP level was reached until login attempts can be made successfully again. Requires that no further invalid login attempts are made in the meantime. The value in this box is calculated based on the input boxes "Maximum value of the BFP level", "BFP threshold" and "Reduction of BFP level (per second)".

You can use the "Save" button to transfer the security settings changed in the WBM to the reader. You can use the "Update" button to update the security settings displayed in the WBM or reload it from the reader.

6.3.16 The "Certificates" menu item

In the "Certificates" menu item, you can view existing certificates, import new certificates, as well as create certificate signing requests and transfer certificate files and certificate key files to the reader.

Using certificates, you can integrate the reader in your specific security infrastructure. Certificates are used to check the identity of a person or a device, to authenticate a service or to encrypt files. You can create your own certificates or use official certificates created by a certification authority. You can import certificates that contain a certificate as well as a private key (PKCS#12). When you import certificates and the associated private keys in separate files, then both files must be coded either in "ASN.1" or "Base64".

Certificates always consist of a certificate file and a certificate key file that you must transfer to the reader. Remember that you first need to import the data into the reader before you can enable it.

Contact your administrative IT department for more information on the topic.

Parameter overview

Note that the parameters listed below depend on the selected certificate type and not all parameters are displayed for all certificate types.

Table 6-40 Description of the parameters

Parameter	Description
Certificate type	Selection of the certificate type Select the required certificate type from the drop-down list and click on the "Update" button to display the certificates matching the selected certificate type. • HTTPS certificate HTTPS certificate of the reader. • OPC UA server certificates OPC UA server certificate of the reader. • OPC UA client certificates OPC UA client certificates of the communication partner of the reader. • OPC UA CA certificates Root certificates from certification authorities. Certification authorities are organizations that issue signed certificates that were derived from their certificates for network participants. This means the CA certificates are root certificates for the client certificates. Client certificates for which a valid CA certificate exists are automatically accepted during connection setup. Note that these client certificates must contain the "authority key identifier extention" element of the CA certificate in the "subject key identifier" extension. • OPC UA issuer certificates Root certificates from certification authorities. Unlike CA certificates, client certificates derived from issuer certificates must still be accepted and permitted by an administrator with the "Accept" button. Note that the selection of the certificate type has an effect on the display of the subsequent parameters.
Update	Button for refreshing the certificates displayed in the list Updating loads and displays all certificates currently stored in the reader.

Parameter	Description
Certificates	List of all existing certificates The certificates included in this list with a black background are considered as trustworthy by the reader. To display details of a certificate, select the required certificate in the list. The selected certificate field is highlighted in color. Certificates displayed in red have not yet been classified as trustworthy. A client using such a certificate cannot yet establish a connection to the OPC UA partner. These certificates must still be accepted and permitted by an administrator with the "Accept" button. Certificates displayed in black have already been accepted and are classified as trustworthy. Depending on the selected certificate type, you can delete existing certificates. To do so, select the desired certificate in the list and click on the "Delete" button.
Certificate details	List with detail information on the selected certificate Detailed information about the certificate details is available in the X.509 specifications.
Blacklists	List of all blacklists This area is only displayed when the certificate types "CA certificates" or "Issuer certificates" were selected. A blacklist is issued by a certificate authority. In the case of strict validation, a revocation list must be stored for each CA certificate and issuer certificate. Blacklists give certification authorities the option to lock client certificates again that they have issued and signed. The certificates listed in a blacklist are locked for communication with the reader. To display the details of a blacklist, select the required blacklist in the list. The selected blacklist is highlighted in color. To delete blacklists from the list again, select the desired blacklist in the list and click on the "Delete" button.
Blacklist details	List with detail information on the selected blacklist Detailed information about the blacklist details is available in the X.509 specifications.

Parameter	Description
Importing a certificate	In this area, you can transfer the certificate files to the reader. Valid formats: • *.p12, *.pfx Binary file format, in which the certificate file and the private key are stored in a file. This file is usually protected by a password. Enter the password in the text box at the bottom. Note that this format can only be used for server certificates and can only contain one certificate. • *.cer, *.crt, *.der, *.pem Binary or text coded file format, in which the certificate file and the certificate key file are stored in separate files. Note that the server certificates necessarily require a separate certificate key file. For client certificates, CA certificates and issuer certificates, only the certificate file is specified. The certificate file and the certificate key file can be either binary or text coded. • *.crl Binary or text coded file format for blacklist files. In the case of strict validation, these revocation lists of CA certificates and issuer certificates are mandatory. In this case, first import the revocation list file and then the certificate file. If a matching blacklist has already been stored on the reader, a CA certificate or issuer certificate can be transferred by itself. Note that only files with the file extension *.crl can be used for blacklists. Once you have imported a server certificate, you still need to enable it.
Certificate signing request (CSR)	Button to create a certificate signing request. This area is only displayed when the certificate types "Server certificates" is selected. Click the "Create CSR" button to create a certificate signing request (CSR). The CSR file contains all relevant information of the installed server certificate. A CA (Certificate Authority) can create a signed, module-specific server certificate using this file that you can then import into this module. Make sure that the CommonName of the issuer is specified in the server certificate created.

Supported file formats

The following table provides an overview of the file formats that are supported by the various certificate types.

Table 6-41 Supported file formats of the various certificate types

Certificate types	Supported file formats
HTTPS OPC UA Server	*.p12 [1)] *.pfx [1)] *.pem *.cer *.crt *.der
OPC UA Client OPC UA CA OPC UA Issuer	*.pem *.cer *.crt *.der

[1)] Can only contain one certificate.

6.3.17 The "System - Device settings" menu item

In the "System - Device settings" menu command, you can perform firmware updates, restore the factory settings of the reader and transfer control files to the PC/PG. This page is divided into the following areas:

- Firmware update
- Reset
- Device description files
- CLP

Firmware update

- Update
  You can update the reader's firmware using the "Update" button. For a detailed description of the firmware update, refer to the section "Firmware update using the WBM (Page 115)".
- Check
  You can use the "Check" button to check the integrity of the firmware. You can use this function to check whether the firmware matches the firmware published by Siemens or whether it has possibly been maliciously modified.

Reset

In the "Reset" area, you can reset the reader to the factory settings or to the preset default values or restart it.

- Factory setting
  Using the "Factory settings" button, you can reset the reader to the factory configuration settings. When you restore the factory settings of the reader, all set configuration data, settings of the user management and address information are lost. After the reset, the reader is automatically restarted. Note that you then need to assign new IP addresses to the reader. Restoring the factory settings is only possible as an administrator. If you have lost your administrator password, you need to restore the factory settings of the reader as described in the "Restoring the factory settings for the hardware (Page 118)" section.
- Reboot
  You can use the "Reboot" button to restart the reader.
- Default values
  You can use the "Reboot" button to reset the parameter values of the reader to the factory configuration settings. When resetting to the default values, all set configuration data is lost, but user management settings and address information are retained.

Device description files

The OPC UA device description file current at the time of delivery is stored on the reader. Click the "Save to PC" button to transfer device description files to the connected PC/PG. You can use these files to integrate the readers into the OPC UA applications.

CLP

In this area, you can use the "Update" button to read out information on the data saved on the inserted CLP and configure the CLP.

The CLP is a storage medium that automatically synchronizes data. If an unwritten CLP (delivery state) is inserted in the reader, the reader automatically backs up the configuration data on the CLP during startup. In addition, changes to the configuration are automatically backed up directly on the CLP during operation. The firmware version of the reader can also be backed up. As soon as a firmware update is performed, the new firmware version is automatically transferred to the inserted CLP.

If a CLP is plugged in during operation, you must activate and deactivate the "Ignore CLP" function. If a written CLP is inserted in the reader, the reader automatically applies the configuration and the firmware of the inserted, written CLP during startup – even if the firmware on the CLP is older than the firmware stored on the reader. The prerequisite for this is that the configuration data was written by a compatible device type.

NOTICE
Prevention of outdated firmware being applied Note that a reader automatically applies the firmware of the inserted, written CLP during startup – even if the firmware on the CLP is older than the firmware stored on the reader. To prevent the reader firmware being overwritten by outdated firmware, check the firmware version stored on it before inserting a written CLP.

Table 6-42 Description of the parameters

Information/ parameter	Description
Status	Shows the current status of the CLP. • NOT PRESENT No CLP is inserted or no CLP was detected. • FACTORY The CLP has not yet been initialized or was reset to factory settings. • ACCEPTED The CLP contains data that is compatible with the reader because the device group and device type of the CLP correspond to the reader. • NOT ACCEPTED The CLP contains data that is incompatible with the reader because the device group and device type of the CLP do not correspond to the reader.
Device group	Shows the device group for which the CLP was initialized.
Device type	Shows the device type for which the CLP was initialized.
Device firmware version	Shows the firmware version of the device.
FW included	Shows whether a firmware update file (*.sfw) is stored on the CLP.
Info	Shows detailed information on the device with which the CLP was initialized.
Update	Button for updating the displayed information Updating reads out and displays the current information from the CLP.
Configure CLP	
Ignore CLP	When this function is enabled, the inserted CLP is ignored and all CLP functions are disabled again. One consequence of this is that the data is not automatically synchronized and no configuration data is transferred during startup.
Buttons	Buttons for running CLP functions • Factory settings The CLP is reset to factory settings. This deletes all data stored on the CLP. • Initialize The CLP is initialized. This transfers the data of the reader to the CLP. This has the result that the initialized CLP can only be used with devices of the same type (identical article number). • Secure unplug Perform this function before you remove/unplug the CLP.

6.3.18 The "Help" menu item

Service & Support

The "Help - Service & Support" menu command provides additional information on the RF690R/RF695R readers as well as links to relevant documents on the Siemens Industry Online Support pages. In addition, you can use a link to open the Readme OSS file with the copyright notices and license conditions for the open source software contained in this firmware.

Manual

You will find the configuration manual for the readers with the menu command "Help - Manual".

Programming

7

You can find information on programming the readers via XML, OPC UA, EtherNet/IP and Modbus/TCP in the respective manuals.

Link to the manuals (https://support.industry.siemens.com/cs/ww/en/ps/15081/man)

Application examples

8

This section describes some of the algorithms of the menu item "Settings - Read points" of the WBM based on application examples.

8.1 Minimizing radiated power

Description

In this example, the read points are mounted very close to each other. This makes it very likely that these read points will influence each other if the radiated power is too high. There is always approximately the same number of transponders (1-3 transponders) in the antenna fields of the read points.

The aim is to ensure that the transponders are reliably recognized without the read points interfering with each other and that write access can be made to them.

Algorithms used

- Inventory Power Ramp
- Read/Write Power Ramp

Requirement

The read points are mounted, equipped and aligned and during reading there are transponders in the antenna field.

Note that this example is only of practical use when the reader searches for transponders for a defined period. This is controlled by a light barrier which for example triggers a trigger "IO_LEVEL" or "IO_EDGE" or by parameter assignment of the period of an inventory command.

If there are no transponders in the antenna field during the reading process, the read point will always automatically use the maximum transmit power specified by the Power Ramp.

Parameter assignment

Follow the steps below to ensure that the transponders are reliably identified without the read points influencing each other.

1. With the menu command "Settings - Read points", in the area "Assigned antennas", specify the radiated power with which the antenna normally reads/writes.
   Using the "Settings - Activation power" menu, you can easily determine the appropriate radiated power at which the transponders are reliably detected.
2. In the area "Algorithms", enable the algorithm "Inventory Power Ramp" and specify the values as follows:
   - Expected Tags: Enter the minimum number of transponders that are always located in the antenna field (e.g. 2).
   - Boost max [dB]: Enter the value by which [dB] the radiated power may be increased as maximum (e.g. 4 dB).
   - Inventories: Enter the number of inventories to be taken until the maximum radiated power is reached (e.g. 4).
     Note: The higher the number of inventories the higher the likelihood that transponders will be identified with as low a radiated power as possible. However, the more inventories you specify, the longer it may take for the transponders to be detected in an unfavorable case.

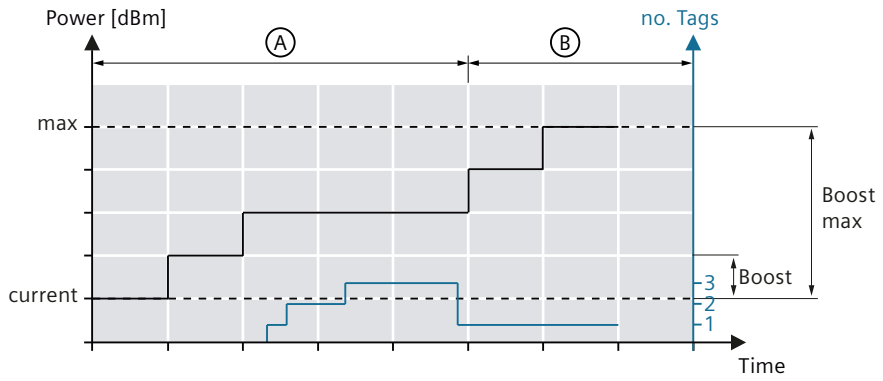


Figure 8-1 Access with minimum radiated power using the algorithms "Inventory Power Ramp" (A) and "Read/Write Power Ramp" (B)

3. In the area "Algorithms" activate the algorithm "Read/Write Power Ramp" and specify the values as follows:
   - Boost [dB]: Write access to a transponder normally requires more power than inventories. For this reason, you should enter a value here by how much [dB] the radiated power should be increased (in steps) if the execution of a command fails. With a "Write" command, the power is increased by this value directly the first time the command executes, with a "Read" command only when the command fails to execute.
   - Boost max [dB]: Enter by how much [dB] the radiated power may be increased as maximum (e.g. 6 dB). Here, you can also enter the value = 0, while you enter the value of 2 dB for "Boost [dB]". This brings about that only with a "Write" command, the power is increased once.

Note: The algorithm "Read/Write Power Ramp" is based on the algorithm "Inventory Power Ramp". This means that the dB values of the "Read/Write Power Ramp" are added to the existing values increased by the "Inventory Power Ramp".

8.2 Initiating trigger through light barrier

Description

In this example inventories will be taken as soon as an event occurs - e.g. when a forklift truck passes a light barrier (input 1). The aim is that all transponders on the palette of the forklift truck are reported as "reliably identified" if they are scanned often enough.

There are two different examples (A and B) in which a distinction is made as to when the inventories are ended again.

- Example A:
  The inventories are taken over a specified time of 10 seconds.
- Example B:
  The inventories are performed until the light barrier is no longer interrupted.

Algorithms used

- Smoothing

Requirement

The read points are mounted, equipped and aligned.

Parameter assignment - XML user

Follow the steps below to activate the "Smoothing" algorithm:

1. In the menu command "Settings - Read points" in the "Algorithms" area activate the "Smoothing" algorithm and specify the values as follows:
   - Observed Count: Here enter a value that guarantees reliable identification of the transponders and at the same time ensures that transponders identified by sporadic overshoots are sorted out. With the menu command "Diagnostics - Tag monitor" you can determine which value is ideal here for your application.
   - Lost Count: Enter the maximum value "65535" to ensure that all transponders are identified during the specified period and are also not reported more than once. This ensures that transponders located at the front of the palette do not leave the list of identified transponders while the forklift truck passes the read point and these transponders already leave the antenna field while inventories are still being taken.

Example A - further procedure

Follow the steps below to take inventories for 10 seconds if the light barrier is interrupted.

1. Create an If condition that then always invokes the command "getObservedTagIDs" when the light barrier (input 1) is interrupted.
2. Specify the command parameters as follows;
   - value_duration = 10000
   - value_unit = Time

Example B - further procedure

Follow the steps below to take inventories if the light barrier is interrupted until the light barrier is no longer interrupted:

1. Create an If condition that then always invokes the "triggerSource" command (value_triggerMode = Start) when the light barrier (input 1) is interrupted.
2. Program the condition so that as soon as the light barrier is no longer interrupted, the transponder data is fetched by the command "getObservedTagIDs".
   - value_duration = 0
   - value_unit = Time

   As an alternative the transponder data can also be fetched automatically using "Events".
3. End the If condition or the command with "value_triggerMode = Stop".

8.3 Filtering out transponders detected due to overshoot

Description

In this example there are production lines close together that are controlled by read points. This can lead to the read point of a production line identifying the transponders of the other production line due to overshoot.

The aim is that the read point involved recognizes these transponders and filters them out.

Antenna parameters and algorithms used

- RSSI threshold
- Smoothing
- RSSI delta

Requirement

The read points are mounted, equipped and aligned.

Note that this example is only of practical use when the reader searches for transponders for a defined period. This is controlled by a light barrier which for example triggers a trigger "IO_LEVEL" or "IO_EDGE" or by parameter assignment of the period of an inventory command.

Parameter assignment

Follow the steps below to ensure that the transponders identified due to overshoot are filtered out.

1. With the menu command "Diagnostics - Tag monitor" determine the RSSI values of the identified transponders of the read point involved.
   Based on the RSSI values you can determine when transponders are identified due to overshoot. Transponders identified due to overshoot normally have significantly poorer RSSI values than transponders located in the regular antenna field.
2. Use the "RSSI threshold" antenna parameter contained in the "Settings - Read point" menu command and the "Smoothing" algorithm to filter out those transponders that are detected due to overreach.
3. Check your settings using the menu command "Diagnostics - Tag monitor".

The "RSSI delta" algorithm

As an alternative to the antenna parameters named above, you can also use the algorithm "RSSI delta". When using this algorithm, you must, however, remember the following points:

- This algorithm is only useful for applications that use time-limited inventories.
- Transponders found to be valid once, must remain valid for the remaining time. To do this in "Smoothing" the value of "Lost Count" must be set to the maximum value "65535".
- The evaluation of which transponders are valid, may only take place at the end of the taking of inventories.

For this reason we recommend to limit overshoots first with the antenna parameters and algorithms listed above.

Error messages

9

You have the following options for error analysis of the modules:

- Via error messages of the WBM
- Via XML error messages
- Via OPC UA error messages

These alternative methods are described below.

9.1 Reading out error messages using the WBM

All the diagnostic and error messages of the reader are entered in the "Logbook" if a check mark was set for "ERRORS" in the WBM configuration in "Settings - General". You will find further information on the "Log" in the section "The "Diagnostics - Log" menu item (Page 77)".

9.2 XML/WBM error messages

Note that the error LED ("ERROR") of the reader always flashes in the event of error messages. You can read out the error via the XML error codes or via the WBM.

The XML/WBM error codes are listed in the following table. The WBM error codes only contain the errors relevant to the readers. You can find all other error codes in the Ident Profile manual.

Table 9-1 Error messages of the readers

"ERROR" LED	XML (hex)	WBM logbook (hex)	Error description
2 Hz	0x11	0xE1FE0100	Cannot write to the memory of the transponder. • Transponder memory is defective. • Transponder EEPROM was written too frequently and has reached the end of its service life.
2 Hz	0x12	0xE1FE0200	Presence error The transponder is no longer within the transmission window of the reader. The command was not or only partially executed. Read command: There is no valid data in "IDENT_DATA". Write command: The transponder that has just left the antenna field contains an incomplete data record. Possible causes: • Operating distance between reader and transponder is not being maintained. • Configuration error: The data record to be processed is too large (in dynamic mode).

"ERROR" LED	XML (hex)	WBM logbook (hex)	Error description
2 Hz	0x13	0xE1FE0300	Address error The address area of the transponder has been exceeded. Possible causes: • Start address of the command start has been incorrectly set. • Wrong transponder type • The area to be written to is write-protected.
2 Hz	0x1A	0xE1FE0A00	The transponder is read/write-protected.
2 Hz	0x91	0xE1FE8100	The transponder is not responding.
2 Hz	0x92	0xE1FE8200	The transponder password is incorrect. Access is denied.
2 Hz	0x93	0xE1FE8300	The verification of the written transponder data has failed.
2 Hz	0x94	0xE1FE8400	General transponder error
2 Hz	0x95	0xE1FE8500	The transponder has too little power to execute the command.
2 Hz	0x22	0xE2FE0200	More transponders are located in the transmission window than can be processed at the same time by the reader.
2 Hz	0xA1	0xE2FE8100	There is no transponder with the required EPC ID/UID in the transmission window or there is no transponder at all in the antenna field.
2 Hz	0xA2	0xE2FE8200	The requested data is not available.
2 Hz	0xA3	0xE2FE8300	CRC error in reader-transponder communication.
2 Hz	0xA4	0xE2FE8400	The selected antenna is not enabled.
2 Hz	0xA5	0xE2FE8500	The selected frequency is not enabled.
2 Hz	0xA6	0xE2FE8600	The carrier signal is not activated.
2 Hz	0xA7	0xE2FE8700	There is more than one transponder in the transmission window.
2 Hz	0xA8	0xE2FE8800	General radio protocol error
4 Hz	0x41	0xE4FE0100	Warning in the event of low power supply The power supply is very close to the low limit.
4 Hz	0x43	0xE4FE0300	Antenna error • The antenna or the antenna cable is defective. • Error in the connection to the reader; the reader is not answering (in PROFIBUS operation). – The cable between the communications module and reader is wired incorrectly or there is a cable break – The 24 V supply voltage is not connected or is turned off or has failed briefly – Automatic fuse on the communications module has blown – Hardware defective – Another reader is in the vicinity and is active – There is a reflecting metal surface in the vicinity that is disrupting the antenna field Possible corrective measures: - Reduce radiated power of antenna. - Change antenna alignment. Avoid parallel alignment of antenna/metal. - Use antenna cable with greater attenuation. - Install attenuator between antenna and reader. – Execute "init_run" after correcting the error

"ERROR" LED	XML (hex)	WBM logbook (hex)	Error description
2 Hz	0x44	0xE4FE0400	The buffer on the communications module or reader is not adequate to store the command temporarily.
2 Hz	0x45	0xE4FE0500	The buffer on the communications module or reader is not adequate to store the data temporarily.
2 Hz	0x46	0xE4FE0600	The command is not permitted in this status or is not supported. Possible cause: • "INIT" was chained. • Command repetition was started without "Presence mode".
2 Hz	0x47	0xE4FE0700	Startup message from reader/communications module The reader or communications module was off and has not yet received a "Reset_Reader" ("WRITE-CONFIG") command. • Execute "INIT" • The same physical address in the "IID_HW_CONNECT" parameter is being used more than once. Check your "IID_HW_CONNECT" parameter settings. • Check connection to the reader • The baud rate was switched over but power has not yet been cycled
2 Hz	0xC1	0xE4FE8100	The specified tag field of the transponder is unknown.
2 Hz	0xCA	0xE4FE8A00	General error
2 Hz	0xFA7	0xE4FE8A00	Unknown read point
2 Hz	0xCB	0xE4FE8B00	No or bad configuration data/parameters were transferred. Possible cause: • You are accessing a read point that is not configured.
--	0xCC	0xE4FE8C00	• Communication error between Ident profile and communications module. Handshake error. – UDT of this communications module is overwritten by other program sections – Check parameter settings of communications module in the UDT – Check the Ident profile command that caused this error – Start "INIT" after correcting the error • Backplane bus / PROFIBUS DP / PROFINET error occurred This error is only indicated when "Watchdog" has been enabled in the PROFIBUS configuration. – Backplane bus / PROFIBUS DP / PROFINET bus connection was interrupted (wire break on the bus; bus connector on the communications module was briefly unplugged) – Backplane bus / PROFIBUS DP / PROFINET master no longer addressing communications module – Execute "INIT" – The communications module has detected a frame interruption on the bus. The backplane bus, PROFIBUS or PROFINET may have been reconfigured (e.g. with HW Config or TIA Portal)

"ERROR" LED	XML (hex)	WBM logbook (hex)	Error description
2 Hz	0xCD	0xE4FE8D00	• Communication error of the communications module/reader – Connector contact problem on the communications module/reader – Hardware of the communications module/reader has a defect; → Send in communications module/reader for repair – Start "INIT" after correcting the error • Monitoring error of the communications module/reader – Program execution error on the communications module / reader – Turn the power supply of the communications module/reader off and on again – Start "INIT" after correcting the error • Firmware error Possible cause: The firmware update was not run completely.
2 Hz	0xCE	0xE4FE8E00	The current command was aborted by the "WRITE-CONFIG" ("INIT" or "RESET") command or the bus connector was pulled. Possible causes: • Communication with the transponder was aborted by "INIT". • This error can only be reported if there is an "INIT" or "RESET".
2 Hz	0x51	0xE5FE0100	Incorrect sequence number order (SN) on the reader/communications module.
--	0x52	0xE5FE0200	Incorrect sequence number order (SN) in the Ident profile
2 Hz	0x54	0xE5FE0400	Invalid data block number (DBN) on the reader/communications module
--	0x55	0xE5FE0500	Invalid data block number (DBN) in the Ident profile
2 Hz	0x56	0xE5FE0600	Invalid data block length (DBL) on the reader/communications module
--	0x57	0xE5FE0700	Invalid data block length (DBL) in the Ident profile
2 Hz	0x58	0xE5FE0800	The previous command is still active or the buffer is full. A new command was sent to the reader or communications module although the last command is still active. • The active command can only be aborted with "INIT". • Before a new command can be started, "DONE bit = 1" must be set (exception: "INIT"). • Two Ident profile calls had the same "HW_ID", "CM_CHANNEL" and "LADDR" parameter settings. • Two Ident profile calls are using the same pointer. • After eliminating the error, an "INIT" must be executed. • When working with command repetition (e.g., fixed code transponder), no data is being fetched from the transponder. The data buffer on the reader/communications module has overflowed. Transponder data has been lost.
--	0x59	0xE5FE0900	The reader/communications module runs a hardware reset ("INIT_ACTIVE" set to "1"). The Ident profile expects an "INIT" (bit 15 in the cyclic control word).
--	0x5A	0xE5FE0A00	The "CMD" command code and the relevant acknowledgement do not match. This can be a software error or synchronization error that cannot occur in normal operation.
--	0x5B	0xE5FE0B00	Incorrect sequence of acknowledgement frames (TDB / DBN)
--	0x5C	0xE5FE0C00	Synchronization error (incorrect increment of AC_H / AC_L and CC_H / CC_L in the cyclic control word). "INIT" had to be executed.

"ERROR" LED	XML (hex)	WBM logbook (hex)	Error description
--	--	0xE5FE8100	Communications error between reader and communications module Access denied
--	--	0xE5FE8200	Communications error between reader and communications module Resource is occupied
--	--	0xE5FE8300	Communications error between reader and communications module Functional error of the serial interface
--	--	0xE5FE8400	Communications error between reader and communications module Other faults/errors
2 Hz	0x61	0xE6FE0100	Unknown command An uninterpretable XML command was sent to the reader or the Ident profile sends an uninterpretable command to the reader. Possible causes: • The "AdvancedCmd" block was supplied with an incorrect "CMD". • The "CMD" input of the "AdvancedCmd" block was overwritten.
--	0x62	0xE6FE0200	Invalid command index (CI)
2 Hz	0x63	0xE6FE0300	• A parameter of an XML command has an invalid value or the parameter assignment of the communications module or the reader was incorrect. Possible causes / action to be taken: – Check the parameters in the Ident profile. – Check the relevant XML command. – Check the parameter assignment in HW Config / STEP 7 (TIA Portal). – The "WRITE-CONFIG" command has incorrect parameter settings. – After a startup, the reader or communications module has still not received an "INIT". • The parameter assignment of the reader or communications module on PROFIBUS/PROFINET was incorrect and the command cannot be executed. Possible causes / action to be taken: – Length of the input/output areas is too small for the cyclic I/O word. – Check whether you have used the correct GSD file. – User data length set with the command (e.g. "READ") is too high. • Error when processing the command. Possible causes / action to be taken: – The data in "AdvancedCmd" or "IID_CMD_STRUCT" is incorrect (e.g. "WRITE" command with length = 0). Check "AdvancedCmd" or "IID_CMD_STRUCT" and execute an "INIT". – The hardware of the reader/communications module is defective. The reader or communications module receives bad data with an "INIT". – The AB byte does not match the user data length. • The wrong reset block was selected. Possible causes / action to be taken: – Regardless of the selected reader system, use the "Reset_Reader" function block.

"ERROR" LED	XML (hex)	WBM logbook (hex)	Error description
--	0x64	0xE6FE0400	Presence error A transponder has passed through the transmission window of a reader without being processed. • This error message is not reported immediately. Instead, the reader or communications module waits for the next write / read command. This command is replied to immediately with this error and the write/read command is not executed. The next command is executed normally again by the reader/communications module. • You can reset this error status using an "INIT". • Bit 2 is set in the "OPT1" parameter and there is no transponder in the transmission window.
--	0x65	0xE6FE0500	An error has occurred that makes a Reset_Reader ("WRITE-CONFIG" with "Config = 3") necessary. Possible causes / action to be taken: • The "WRITE-CONFIG" command is incorrect. • After eliminating the error, execute an "INIT". • Check the "IID_HW_CONNECT" parameter.
--	0x66	0xE6FE0600	The reset timer has expired.
2 Hz	0xE1	0xE6FE8100	A parameter is missing.
2 Hz	0xE2	0xE6FE8200	The parameter has an invalid format.
2 Hz	0xE3	0xE6FE8300	The parameter type is invalid.
2 Hz	0xE4	0xE6FE8400	Unknown parameter.
2 Hz	0xE5	0xE6FE8500	The command or the frame has an invalid format.
2 Hz	0xE6	0xE6FE8600	The inventory command failed.
2 Hz	0xE7	0xE6FE8700	Read access to the transponder failed.
2 Hz	0xE8	0xE6FE8800	Write access to the transponder failed.
2 Hz	0xE9	0xE6FE8900	Writing the EPC ID/UID on the transponder has failed.
2 Hz	0xEA	0xE6FE8A00	Enabling write protection on the transponder failed.
2 Hz	0xEB	0xE6FE8B00	The "Kill" command failed.
2 Hz	0x71	0xE7FE0100	In this status, only the "Reset_Reader" command ("WRITE-CONFIG") is permitted.
--	0x72	0xE7FE0200	The "CMD" command code is not permitted.
--	0x73	0xE7FE0300	The "LEN_DATA" parameter of the command is too long and does not match the global data reserved within the send data buffer (TXBUF).
--	0x74	0xE7FE0400	The receive data buffer (RXBUF) or the send data buffer (TXBUF) is too small, the buffer created at TXBUF/RXBUF does not have the correct data types or the parameter "LEN_DATA" as a negative value. Possible cause / action to be taken: • Check whether the buffers TXBUF/RXBUF are at least as large as specified in LEN_DATA. • With S7-1200/1500: – In the Ident profile, only an "Array of Byte" may be created for TXBUF and RXBUF. – In the "Reader_Status" block, only an "Array of Byte" or the corresponding data types ("IID_TAG_STATUS_XX_XXX" or "IID_READER_STATUS_XX_XXX") may be created
--	0x75	0xE7FE0500	Error message that informs you that only an "INIT" command is permitted as the next command. All other commands are rejected.

"ERROR" LED	XML (hex)	WBM logbook (hex)	Error description
--	0x76	0xE7FE0600	Wrong index Permitted index is in the ranges "101 ... 108" and "-20401 ... -20418".
--	0x77	0xE7FE0700	The reader or communications module does not respond to "INIT" ("INIT_ACTIVE" is expected in the cyclic status message). The next steps: • Check the address parameter "LADDR".
--	0x78	0xE7FE0800	Timeout during "INIT" (60 seconds according to "TC3WG9")
--	0x97	0xE7FE0900	Command repetition is not supported.
--	0x7A	0xE7FE0A00	Error during the transfer of the PDU (Protocol Data Unit).

"--" means that the error is not displayed by the LEDs.

9.3 OPC UA error messages

The following table lists the OPC UA-specific error codes.

Table 9-2 OPC UA error messages of the readers

AutoId status	XML/ LED (hex)	OPC UA status	AutoId text	Error description
1	0xCA	good	MISC_ERROR_ TOTAL	General error
1	0xCD	good	MISC_ERROR_ TOTAL	• Firmware error Possible cause: The firmware update was not run completely. • Internal communications error of the communications module/reader – Connector contact problem on the communications module/reader – Hardware of the communications module/reader has a defect; → Send in communications module/ reader for repair – Start "INIT" after correcting the error • Internal monitoring error of the communications module/reader – Program execution error on the communications module / reader – Turn the power supply of the communications module/reader off and on again – Start "INIT" after correcting the error
1	0xE6	good	MISC_ERROR_ TOTAL	The inventory command failed.

AutoId status	XML/ LED (hex)	OPC UA status	AutoId text	Error description
1	0xEA	good	MISC_ERROR_ TOTAL	Enabling write protection on the transponder failed.
1	0xEB	good	MISC_ERROR_ TOTAL	The "Kill" command failed.
3	0x1A	good	PERMISSON_ ERROR	The transponder is read/write-protected.
4	0x92	good	PASSWORD_ ERROR	The transponder password is incorrect. Access is denied.
5	0x63	Bad Invalid Argument / good	REGION_NOT_ FOUND_ERROR	All commands: A parameter of an OPC UA command has an invalid value. "ReadTag"/"WriteTag" command: The addressed memory area is not available in the current transponder.
7	0x13	good	OUT_OF_RANGE_ ERROR	Address error The address area of the transponder has been exceeded. Possible causes: • Start address of the command start has been incorrectly set. • Wrong transponder type • The area to be written to is write-protected.
7	0xA2	good	OUT_OF_ RANGE_ERROR	The requested data is not available.
8	0xA1	good	NO_ IDENTIFIER	There is no transponder with the required EPC ID/UID in the transmission window or there is no transponder at all in the antenna field.
9	0x22	good	MULTIPLE_ IDENTIFIERS	More transponders are located in the transmission window than can be processed at the same time by the reader.
9	0xA7	good	MULTIPLE_ IDENTIFIERS	There is more than one transponder in the transmission window.
10	0xE7	good	READ_ ERROR	Read access to the transponder failed.
14	0xE8	good	WRITE_ ERROR	Write access to the transponder failed.
14	0xE9	good	WRITE_ ERROR	Writing the EPC ID/UID on the transponder has failed.
15	0x61	good	NOT_SUPPORTED_ BY_DEVICE	Unknown command An uninterpretable XML command was sent to the reader or the Ident profile sends an uninterpretable command to the reader. Possible causes: • The "AdvancedCmd" block was supplied with an incorrect "CMD". • The "CMD" input of the "AdvancedCmd" block was overwritten.
17	--	good	DEVICE_NOT_ READY	The specified read point is not active because no antenna was assigned to it.

AutoId status	XML/ LED (hex)	OPC UA status	AutoId text	Error description
18	0xC1	good	INVALID_ CONFIGURATION	The specified tag field of the transponder is unknown.
19	0x91	good	RF_COMMUNICATION_ ERROR	The transponder is not responding.
19	0x93	good	RF_COMMUNICATION_ ERROR	The verification of the written transponder data has failed.
19	0x94	good	RF_COMMUNICATION_ ERROR	General transponder error
19	0x95	good	RF_COMMUNICATION_ ERROR	The transponder has too little power to execute the command.
19	0xA3	good	RF_COMMUNICATION_ ERROR	The transponder signals a CRC error.
19	0xA5	good	RF_COMMUNICATION_ ERROR	The selected frequency is not enabled.
19	0xA6	good	RF_COMMUNICATION_ ERROR	The carrier signal is not activated.
19	0xA8	good	RF_COMMUNICATION_ ERROR	General radio protocol error
20	0x41	good	DEVICE_FAULT	Fault in power supply The power supply is very close to the low limit.
20	0x43	good	DEVICE_FAULT	Antenna error • The antenna or the antenna cable is defective. • Error in the connection to the reader; the reader is not answering (in PROFIBUS operation). – The cable between the communications module and reader is wired incorrectly or there is a cable break – The 24 V supply voltage is not connected or is turned off or has failed briefly – Automatic fuse on the communications module has blown – Hardware defective – Another reader is in the vicinity and is active – There is a reflecting metal surface in the vicinity that is disrupting the antenna field – Execute "init_run" after correcting the error
--	0xA4	--	--	The selected antenna is not enabled.
--	0x46	OpcUa_ BadInvalid-State	--	The command is not permitted in this status or is not supported. Possible cause: • "INIT" was chained. • Command repetition was started without "Presence mode".

AutoId status	XML/ LED (hex)	OPC UA status	AutoId text	Error description
--	0xCB	OpcUa_ BadOutOf Range / Op-cUa_ Bad Configura-tion Error	--	No or bad configuration data/parameters were transferred. Possible cause: • You are accessing a read point that is not configured.
--	0xE1	BadInvalid Argument	--	A parameter is missing.
--	0xE2	BadInvalid Argument	--	The parameter has an invalid format.
--	0xE3	BadInvalid Argument	--	The parameter type is invalid.
--	0xE4	BadInvalid Argument	--	Unknown parameter.
--	0xE5	Bad	--	The command or the frame has an invalid format.

Service and maintenance

10

10.1 Diagnostics

You have the following diagnostics options available for the reader:

- via the LED status displays of the reader
- Via SNMP
- Using the WBM
- Via XML
- Via OPC UA

These alternative methods are described below.

10.1.1 Diagnostics via the LED display

The LED operating display of the RF690R/RF695R readers is illustrated in detail in the following figure.

LED operating display

- RUN — Shows whether the reader is ready for operation.
- ERROR — Indicates whether an error has occurred.
- LINK — Indicates that there is a connection via the Ethernet interface.

Figure 10-1 LED display of the RF690R/RF695R readers

LED operating display

The operating status of the reader is indicated by the "RUN" and "ERROR" LEDs. The LEDs can adopt the colors green or red and the states off, on or flashing:

Table 10-1 Display of operating statuses

RUN	ERROR	LINK	Meaning
		--	The device is turned off.
			• The device is starting up. • Flash test for reader identification.
	--	--	The device is ready for operation. The connection to the application is not established.
	--	--	The device is working. The connection to the application is established. • XML application: The "hostGreetings" command was received. • OPC UA: Connection to the client is established.
--		--	There is an error. The LED flashes as long as the error/alarm is present, but at least for 10 seconds. You can find more information on error messages in the section "Error messages (Page 103)".

--: Not relevant

10.1.2 Diagnostics via SNMP

Using SNMP, you have extensive diagnostics options for the network functions of the reader. The following diagnostics options (MIBs) are supported by the readers:

- RFC 2863: IF-MIB
- RFC 3418: SNMPv2-MIB
- RFC 4022: TCP-MIB
- RFC 4113: UDP-MIB
- RFC 4292: IP-MIB
- SIEMENS:
  - AUTOMATION-SN-SYSTEM-MIB
  - AUTOMATION-SYSTEM-MIB
  - IEC-62439-2-MIB
  - IEEE 802.1AB 2005 LLDP-MIB
  - LLDP-EXT-DOT1-MIB
  - LLDP-EXT-DOT3-MIB
  - LLDP-EXT-PNO-MIB

You can find the MIB files corresponding to the readers on the pages of Siemens Industry Online Support, and you can find information on the MIB files under "Network management

diagnostics and configuration with SNMP (https://support.industry.siemens.com/cs/ww/en/view/103949062)".

The readers support the SNMPv3 and SNMPv1 protocols. SNMPv1 is deactivated ex works, classified as insecure and should not be used. Deactivate SNMP if it is not required or use SNMPv3 if possible. You can find information on SNMP in the section "MIB files (https://support.industry.siemens.com/cs/ww/en/view/67637278)".
You can find detailed information on using SNMP and, in particular, on the structure of the Automation MIB in the diagnostics manual "PROFINET user organization (https://www.profibus.com/download/profinet-specification/)".

10.1.3 Diagnostics using the WBM

A wide variety of diagnostics options are available via the WBM.

Using the "Tag monitor", you can read out how well which transponder was identified using which antenna. In the "Logbook" menu, you can find all the diagnostics messages of the communication module that have occurred. The "Service logbook" helps SIEMENS specialists to analyze errors. You can read out the Syslog messages of the connected readers via the "Syslog logbook".

You can find more information on the "Logbook" in the section "The "Diagnostics - Log" menu item (Page 77)".

10.1.4 Diagnostics via XML

Comprehensive diagnostics options are available to you via XML. You can find detailed information on the diagnostics options in the manual "XML programming for SIMATIC Ident (https://support.industry.siemens.com/cs/ww/en/ps/15081/man)".

10.1.5 Diagnostics over OPC UA

Comprehensive diagnostics options are available to you via OPC UA. The various diagnostics options are described below.

Using the OPC UA variables and events, you can read out diagnostic messages and query them in a targeted manner. You can find detailed information on the diagnostics options in the manual "OPC UA for SIMATIC Ident (https://support.industry.siemens.com/cs/ww/en/ps/15081/man)".

10.2 Firmware update using the WBM

The firmware of the reader is updated via the WBM.

Requirements

- The reader is connected to the PC/PG via Industrial Ethernet.
- The reader was disconnected from runtime operation.
- All user applications are closed.
- The required update file (*.sfw) is stored locally.

Procedure

Follow the steps below to run a firmware update using the WBM:

1. Start your Web browser.
2. Enter the IP address of the reader in the address field of your browser.
3. Log into the WBM if you have not already done so.
   Note that you as "User" can only run a firmware update if the reader is in the "Idle" status.
4. Click the "System - Device settings" menu command.
5. In the "Firmware update" area, click the "Select firmware file" icon .
6. Select the update file (*.sfw).
7. Click the "Open" button.
8. Click the "Update" button.

Result: The firmware is updated. The update process is indicated in the information bar.

After the update has completed, the reader is restarted. The reader is ready for operation when the "RUN" LED is lit green/flashing. Note that the startup process takes approx. 1 minute after a firmware update.

The updated firmware is active following the restart.

10.3 Factory settings

You can reset the configuration of the reader to the factory settings at any time. To reset to the factory settings, you have the following options available:

- Using the WBM
- Via SINEC PNI
- Via XML
- On the hardware side using the Reset button

These alternative methods are described below.

10.3.1 Restoring the factory settings via WBM

Requirement

The reader is connected to the PC/PG via Industrial Ethernet.

Procedure

Follow the steps below to reset all settings to the factory settings using the WBM:

1. Start your Web browser.
2. Enter the IP address of the reader in the address field of your browser.
3. Log into the WBM if you have not already done so.
   Note: If you have forgotten the administrator password, you need to reset the reader via the XML interface.
4. Click the "System" menu command.
5. In the "Reset" area, click the "Reset" button.

Result: The reader is reset to the original factory settings. The restore process is indicated in the information bar.

Note that if you reset to factory settings, the reader is assigned the factory default IP address "192.168.0.254". You can only recognize when the restore process is completed based on the "RUN" LED. After the reset, the reader is restarted. The reader is ready for operation when the "RUN" LED is lit green/flashing.

After restarting the reader, you may need to assign a new IP address or a new device name to the reader.

10.3.2 Reset the factory setting with SINEC PNI

Requirement

- The reader is connected to the PC/PG via Industrial Ethernet.
- The "Allow write access" check box must be set in the WBM ("Communication > Network interfaces" menu).

Procedure

Proceed as follows to reset all settings to the factory settings using SINEC PNI:

1. Start SINEC PNI.
2. Click the "Start network scan" button in the toolbar.
   Reaction: The network is scanned for connected devices and all recognized devices are displayed in the device list.

3. Select the desired reader in the device list.
4. Click the "Reset device" entry in the "Device management" drop-down list in the toolbar.

Result: The reader is reset to the original factory settings.

Note that if you reset to factory settings, the reader is assigned the factory default IP address "192.168.0.254". You can only recognize when the restore process is completed based on the "RUN" LED. After the reset, the reader is restarted. The reader is ready for operation when the "RUN" LED is lit green/flashing.

After restarting the reader, you may need to assign a new IP address or a new device name to the reader.

10.3.3 Restoring the factory settings via XML

Via the XML interface, you can use the command "resetDevice" to reset all settings to the factory settings. The prerequisite is that the "Parallel" check box was not set in the WBM ("Communication > XML" menu).

Note that if you reset to factory settings, the reader is assigned the factory default IP address "192.168.0.254". You can only recognize when the restore process is completed based on the "RUN" LED. After the reset, the reader is restarted. The reader is ready for operation when the "RUN" LED is lit green/flashing.

After restarting the reader, you may need to assign a new IP address or a new device name to the reader.

10.3.4 Restoring the factory settings for the hardware

Requirement

The reader has been disconnected from the power supply.

Procedure

Follow the steps below to reset all settings to the factory settings using the Reset button:

1. Press the reset button (RESET) and keep it pressed.

Figure 10-2 Reset button of the reader

2. With the Reset button pressed, connect the reader to the power supply (24 V DC or PoE).
3. Wait approx. 60 seconds.
4. Release the Reset button.

Result: The reader was reset to its original factory settings.

Note that if you reset to factory settings, the reader is assigned the factory default IP address "192.168.0.254".

After restarting the reader, you may need to assign a new IP address or a new device name to the reader.

10.4 Module replacement

NOTICE
Backing up the configuration Prior to replacing the module, make sure that you back up the configuration stored on the reader so that you can transfer this to the newly connected reader following module replacement.

NOTICE
Loading a configuration Note that you cannot transfer any user profiles and passwords to other readers using the configuration file. After loading the configuration file into a new reader, you may need to enable authentication and create new user profiles and passwords.

To back up the current configuration of the reader and to restore it on the newly connected reader following module replacement, you have the following options:

- Using the WBM or XML-API as an *.xml file on your PC/PG
- Using the WBM and the CLP

These alternative methods are described below.

10.4.1 Backup configuration data

Table 10-2 Properties and requirements for the backup options

Backup options	Properties
Backup via the WBM (as *.xml file)	• Configuration data is saved independently of the project ⇒ The download to the reader can be performed manually using the WBM or using the XML API by the user application. • Older configuration versions can be saved (versioning) ⇒ The updating and versioning of the configuration versions needs to be started and managed manually by you yourself. • Option of copying for other readers of the same type
Backup with the help of a CLP	• Backup of the current configuration including the IP address, the users created, the certificates and the firmware • Saving takes place automatically • Option of copying for other readers of the same type

Backup via the WBM (*.xml file)

On the upper toolbar of the WBM, there are two buttons for loading and saving configurations. Using these buttons, you can back up configurations, re-load them and transfer them to other readers.

Note

Loading a configuration

Note that you cannot transfer any user profiles and passwords to other readers using the configuration file. After loading the configuration file into a new reader, you may need to enable authentication and create new user profiles and passwords.

Backup with the help of a CLP

The CLP is a storage medium that automatically synchronizes data. If an unwritten CLP (delivery state) is inserted in the reader, the reader automatically backs up the configuration data on the CLP during startup. In addition, changes to the configuration are automatically backed up directly on the CLP during operation. The firmware version of the reader can also be backed up. As soon as a firmware update is performed, the new firmware version is automatically transferred to the inserted CLP.

If a CLP is plugged in during operation, you must activate and deactivate the "Ignore CLP" function. If a written CLP is inserted in the reader, the reader automatically applies the configuration and the firmware of the inserted, written CLP during startup – even if the firmware on the CLP is older than the firmware stored on the reader. The prerequisite for this is that the configuration data was written by a compatible device type.

NOTICE
Prevention of outdated firmware being applied Note that a reader automatically applies the firmware of the inserted, written CLP during startup – even if the firmware on the CLP is older than the firmware stored on the reader. To prevent the reader firmware being overwritten by outdated firmware, check the firmware version stored on it before inserting a written CLP.

You can find more information on saving and loading the configuration via a CLP in the section "The "System - Device settings" menu item (Page 90)".

10.4.2 Replacing a module

Before replacing a module

NOTICE
Installation/removal with the power off Wire up the modules to be connected only when the power is off. Make sure that the power supply is turned off when installing/uninstalling the devices.

Before replacing the module, back up the configuration of the reader so that you can transfer this to the new reader.

Procedure (*.xml file)

Follow the steps below to replace a reader and transfer an existing configuration via *.xml file:

1. Make sure that the reader is disconnected from the power supply.
2. Pull the cable from the reader.
3. Remove the reader from its mounting.
4. Mount the new reader.
5. Connect the reader to the PC/PG using the supplied Ethernet cable.
6. If necessary, connect the reader to one or more external antennas.
7. Connect the reader to the power supply using the connecting cable.
   Wait until the reader has started up and is ready for operation ("RUN" LED is lit/flashes green).

8. Assign the reader a unique IP address and a unique device name.
9. Load the configuration to the reader.

Procedure (CLP)

Proceed as follows to replace a reader and transfer an existing configuration via CLP:

1. Make sure that the reader is disconnected from the power supply.
2. Pull the cable from the reader.
3. Remove the reader from its mounting.
4. Mount the new reader.
5. Insert the CLP written with a compatible configuration into the CLP interface of the reader.
6. If necessary, connect the reader to one or more external antennas.
7. Connect the reader to the PC/PG using the existing Ethernet cable (PoE if necessary).
8. Connect the reader to the power supply using the connecting cable.

Wait until the reader has started up and is ready for operation ("RUN" LED is lit/flashes green). The reader automatically applies the configuration and the firmware of the inserted, written CLP during startup.

Appendix A

A.1 List of the country profiles (approvals)

You can find a list of the currently available country approvals of the RF600 readers, as well as the setting required in the WBM for the country profile with the following link: List of country profiles (https://support.industry.siemens.com/cs/ww/en/view/109773353)

A.2 Encryption methods (ciphers)

The following tables list the encryption methods (ciphers) that the device uses.

SSL

Table A-1 Supported encryption methods (cipher suites) for web servers (HTTPS; WBM)

Category	Method	Value (hex)	Enabled by default
Cipher suite	TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256	0xC02F	✓
Cipher suite	TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384	0xC030	✓
Cipher suite	TLS_ECDHE_ECDSA_WITH_AES256_GCM_SHA384	0xC02C	✓
Cipher suite	TLS_AES_256_GCM_SHA384	0x1302	✓
Cipher suite	TLS_CHACHA20_POLY1305_SHA256	0x1303	✓
Cipher suite	TLS_AES_128_GCM_SHA256	0x1301	✓
Cipher suite	TLS_AES_128_CCM_SHA256	0x1304	✓
Protocol version	TLSv1.2	--	✓
Protocol version	TLSv1.3	--	✓

SNMP

Table A-2 Supported encryption methods (cipher suites) for SNMP server

Category	Method	Value (hex)	Enabled by default
Authentication	HMAC-MD5-96	--	--
Authentication	HMAC-SHA-96	--	--
Encryption	aes128-cbc	--	--
Encryption	des-cbc	--	--

OPC UA

Table A-3 Supported encryption methods (cipher suites) for OPC UA server

Category	Method	Value (hex)	Enabled by de-fault
Security policy	None	--	--
Security policy	Basic128Rsa15	--	--
Security policy	Basic256	--	--
Security policy	Basic256Sha256	--	--
Security policy	Aes128-Sha256_RsaOaep	--	--
Security policy	Aes256Sha256RsaPss	--	✓
Transport profile	UA-TCP	--	✓

Planning and setting up UHF read points

B

Due to their comparatively large operating range, RFID UHF systems (frequency range 865 - 928 MHz) place higher demands on planning, commissioning and operation than HF systems (frequency range 13.56 MHz). This section describes important rules for the preparation and implementation of RFID UHF systems. Special features of the various reader systems are discussed.

B.1 Technical basics

With UHF technology, the radio waves are generated from an energy source consisting of oscillating magnetic and electric fields. The following graphic shows the structure of a UHF system. One characteristic is the design of the transponder that differs greatly from the structure used in HF systems, e.g. the use of a dipole or helix antenna.

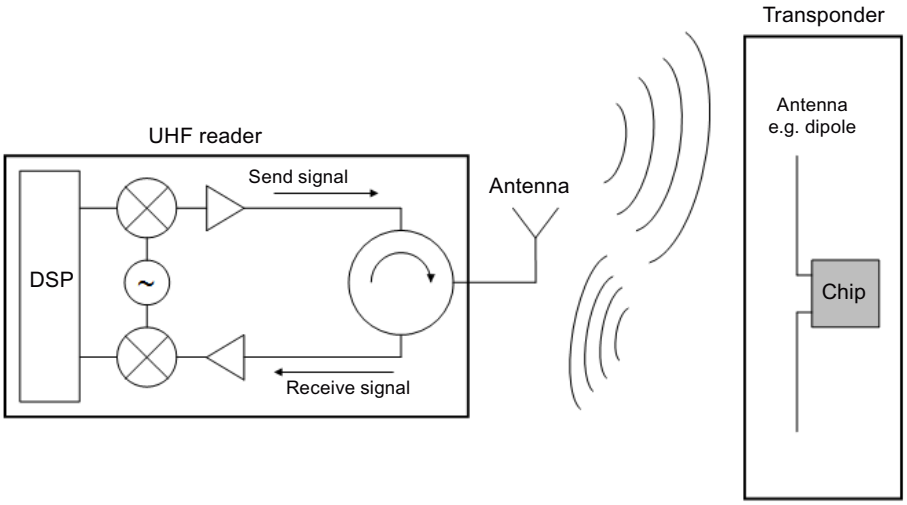


Figure B-1 Structure of a UHF RFID system

RSSI value

The RSSI value (Received Signal Strength Indicator) is the indicator that shows how well a system consisting of reader, antenna and transponder is set up and adapted. The RSSI value therefore shows the signal strength with which the transponder response was detected by the reader. The higher the value, the better the signal strength (according to the IEEE 802.11 standard).

With the RF61xR/RF650R/RF68xR readers, the RSSI value has no unit and is output as a one-byte value (0 to 255). With the RF69xR readers, on the other hand, the RSSI value reflects the dBm value and is output as a two-byte value (-10,000 to 0). In the WBM, the RSSI value is output divided by a factor of 100 (-100 ... 0). This also applies to the "RSSI threshold" and "RSSI delta" parameters.

For both reader groups, an RSSI value of 0 indicates that the transponder has not been detected.

The actual RSSI value depends on numerous parameters:

- Transponder type used
- Chip used in the transponder
- Antenna used
- Set transmit power
- Distance between antenna and transponder
- Polarization of the antenna and alignment of the transponder to the antenna
- Set country profile and selected channels
- reflections,
- noise level in the channel used and in neighboring channels

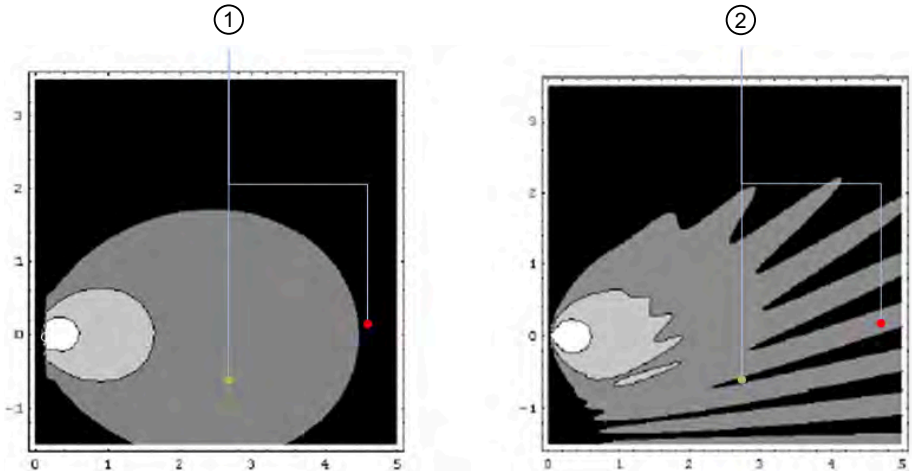
The RSSI value is important for the automatic evaluation of the read point and for filtering. A simple comparison of the RSSI values of two transponders is nevertheless not possible because the values are influenced by the transponder tolerances and the non-homogeneous antenna field (see paragraph below). This means that it is possible that a transponder positioned closer to the RFID antenna has a lower RSSI value than a transponder much further away.

Propagation of the antenna field

The waves do not propagate as a homogeneous field, there is superposition of the waves that can cause the following effects:

- Overshoots and field gaps due to obliteration of two waves
  These are caused by reflection and the resulting propagation on different paths (comparable with fading effects on the car radio, e.g. noise when the vehicle is standing)
- Generation of overshoots due to reflecting objects and surfaces

This can be illustrated by comparing it with a "hall of mirrors". The signal transmitted by the reader is reflected (several times) by metallic objects such as housings, steel supports or grilles and this can lead to unwanted effects and read errors. Is also possible that a transponder is not identified although it is located in the assumed direct identification range of the reader. It can also happen that a transponder moving outside the antenna field is read out due to overshoots.



① Identification situation with two transponders in an ideal radio/antenna field

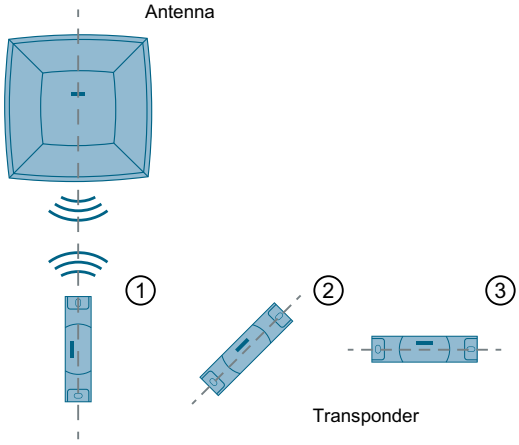
② Identification situation with two transponders in a real radio/antenna field with reflections that can lead to obliteration and overshoots

Figure B-2 Propagation of UHF RFID antenna fields

Properties of the transmitting antenna

Depending on their design, UHF RFID antennas provide different properties. They differ in the polarization and antenna gain.

The direction of the electrical field component of an electromagnetic wave and the alignment of the antenna decide the polarization of the radiation. A distinction is made between linear and circular polarization of an antenna. With linear polarization you achieve the maximum read/write distances when the polarization axes of the antenna and transponder are parallel to each other. As the deviation increases, the received power deteriorates.



① Polarization axes parallel: approx. 100 % range

② Polarization axis turned through 45°: approx. 50% range

③ Polarization axis turned through 90°: approx. 10% range

Figure B-3 Effect of the polarization axes on the read/write distance with linear antennas

Linear antennas can only be used if the alignment of the transponder is defined. On the other hand, one advantage of linear antennas is that they react less sensitively to reflections. This restriction does not apply with circular polarization. Circular antennas can also be used with differing alignments of the transponder and achieve constant results. It has been shown that with a defined transponder alignment, the linear antenna usually provides better results.

B.2 Implementation of UHF RFID installations

The use of UHF RFID systems requires careful planning and preparation to avoid problems during commissioning and operation. Therefore, make sure that the person commissioning the UHF RFID system is sufficiently trained.

Before you start with the installation and setup, you should determine the conditions that are required of your RFID system based on the operating environment. You should select the appropriate RFID components according to the requirements and think about the right configuration for your RFID system in advance.

B.2.1 Planning

B.2.1.1 Selecting the RFID components

When selecting the right RFID components, consider the following minimum criteria:

- Integration into the control or IT environment
- Requirements for degrees of protection
- Size of the identification range
- Type, number, position and alignment of the transponders in the antenna field
- Reflecting and absorbent materials in the vicinity of the reader antenna
- Distance between the antenna or the reader and the transponder (read/write distance)

The following application examples illustrate the requirements for specific use cases and provide suitable solutions:

- Read point in a conveyor system in confined installation conditions:
  A container should be transported in a conveyor system. Information on the next transport section is contained in a transponder which is attached to the side of the container.
  Possible configuration: RF610R or possibly RF615R with integrated internal antenna and a compact, external antenna (e.g. RF615A, RF622A)
- Four read points along the production line:
  A product needs to be processed by different machines along the production line. The information for this is contained on a transponder attached to the product that must be read out at each machine.
  Possible configuration: RF680R with four antennas (e.g. RF615A, RF622A, RF680A)

- Read point on a production line with a predominantly metallic environment:
  A product needs to be processed by different machines along the production line. The information for this is contained on a transponder attached to the product that must be read out at each machine.
  Possible configuration: RF685R with integrated adaptive antenna
- RFID gate at the incoming goods / outgoing goods department:
  Several transponders are located on different packaging of products on a pallet. These need to be identified when passing through the RFID gate.
  Possible configuration: RF690R with four circular antennas (e.g. RF650A, RF662A, depending on the required radiated power) or RF695R with eight antennas

B.2.1.2 Notes on setup and configuration

General notes on setup

- Use circularly polarized antennas for alternating and linearly polarized antennas for fixed transponder alignments.
- Only use as many antennas as are necessary to cover the target area. Avoid overlapping antenna fields.
- Select the correct cable lengths. Avoid unnecessarily long cables.

Notes on reader configuration for use in logistics

- Menu command "Settings - General"
  - Select the right modulation scheme for your application.
  - If required, activate the "Max. RSSI events" parameter (SIMATIC RF69xR) to generate the maximum number of RSSI events. Note that this function can lead to reduced transponder detection with very large transponder populations.
- Menu command "Settings - Read points"
  - If necessary, use parameters/algorithms such as "RSSI threshold", "Smoothing", "RSSI delta" and "Blacklist" if there are several transponders in the target area.
  - If possible, use triggers to perform targeted detection and to avoid mutual interference between reader antennas.

B.2.1.3 Static or dynamic detection

When detecting transponders, a distinction is made between static and dynamic detection. With static detection, the inventories are performed while the transponders are stationary. With dynamic detection, the inventories are performed while the transponders are moving through the antenna field (e.g. on a conveyor system or when passing through a gate). Select the right modulation scheme for your application. Particularly in dynamic operation, the correct selection of the modulation scheme is crucial for fast and complete transponder detection.

The maximum permissible movement speed of the transponders during dynamic detection depends on various components:

- Number of transponders to be detected
- Transponder types used
- Minimum distance between the objects
- Size, shape and nature of the transported objects

The basic rule is that the more transponders are to be detected during a passage, the slower they must cross the antenna field.

As dead spots cannot be ruled out, in a few applications it may make sense to use dynamic detection rather than static detection. If static identification is necessary, the antenna field can e virtually dynamized with the RF685R antenna or RF680A.

B.2.1.4 Read/Write operations (triggering)

To read or write transponder data, you can have readers perform permanent read/write processes or trigger read/write processes. The following reasons speak in favor of the targeted triggering of read/write processes:

- Due to the reduced number of read/write operations, the number of process errors is reduced, allowing them to be identified more quickly.
- Targeted read/write operations reduce the possibility of antenna fields interfering with each other. This increases process reliability in plants, particularly when there is a high reader density.

B.2.1.5 Decoupling third-party RFID systems

When using RFID systems from different manufacturers, ensure that both systems are not active at the same time or are operated separately from each other. Ideally, you should not use RFID systems from different manufacturers in the same area.

B.2.2 Setting up read points

B.2.2.1 Basic configuration

Once the readers and antennas have been installed, connected and basically aligned, you need to perform basic configuration of the readers. Ideally, you can perform this configuration with the help of the reader's WBM. Alternatively, you can also configure the readers via XML or OPC UA.

The basic procedure for setting up read points using the reader's WBM is described below.

1. Adapt the parameters to suit your operating environment.
   Relevant menus: "Settings > General" and "Settings > Read points"
2. Align the reader or antenna optimally to the transponders so that the best possible RSSI values are achieved.
   Relevant menu: "Settings > Adjust antenna"
3. Determine the optimum radiation line.
   Relevant menus: "Settings > Activation power" and "Settings > Read points"
4. Check how well specific transponders are detected with a given antenna.
   Relevant menu: "Diagnostics > Tag monitor"

Test your configuration in a test phase. If necessary, repeat the setup of the read point or change relevant parameters in order to achieve the best possible RSSI values.

B.2.2.2 Adjust antennas

Follow the steps below to optimize the antenna alignment:

1. Position the object to be identified with a transponder in the target area of the read point.
2. Align the reader or the antenna so that its front points in the direction of the object (transponder) to be identified.
   Keep to the minimum distances between antennas and transponders to avoid antenna errors.
   When using linear antennas, make sure the polarization direction is correct.
3. In the "Settings - Adjust antenna" menu command, select the connected antenna and click the "Start adjustment" button (▶).
4. In the "Alignment quality" area, you can see the current (light blue) and maximum achieved (dark blue) RSSI values.

   Note

   Transponder is not identified

   If no transponder is identified, first increase the radiated power as described in the following section. Then repeat the antenna adjustment.

   Also check the polarization of your antenna. If the transponder always has the same alignment, the antenna polarization should be adapted accordingly. If the transponder moves or the alignment of the transponder varies, it is advisable to combine several antenna polarization types or to select a circular polarization.

5. Optimize the antenna adjustment until the maximum possible RSSI value is reached.
6. Secure the antenna.

If you are using several antennas, repeat the process for the other antennas. Only use as many antennas as are necessary to cover the target area. Ensure comprehensive coverage of the target area without the antenna fields of the different antennas overlapping.

Note that the RSSI value depends on the following parameters (cf. section "Technical basics (Page 125)").

B.2.2.3 Determining the radiated/activation power

You can set the radiated power in the "Settings - Read points" menu command. Select the radiated power so that the required transponders can be identified reliably but without overreach. In this case, the following applies: "as much as necessary, as little as possible".

In the "Settings - Activation power" menu command, you can find the optimum radiated power for reliable transponder access.

Detect activation power

Follow the steps below to detect the activation power:

1. In the "Settings - Activation power" menu command, select the connected antenna and click the "Start measurement" button.
2. In the transponder list, "Power (min)" column, you can see the required activation power. The value "Power (min)" of the transponder last selected in the transponder list is automatically transferred to the "Apply power" box with 2 dB added.

   Note

   Detect radiated power

   The value entered automatically in the "Apply power" box corresponds to the minimum value with which the transponder was identified by the antenna ("Power (min)") plus a power reserve of 2 dB. This value serves as a guideline and you can adapt it. To be sure that the antenna reliably detects the transponders regularly, we recommend that you accept the automatically adapted default value.

3. Click the "Apply" button to transfer the value entered in the "Radiated power" input box of the "Settings - Read points" menu command.
4. Click the icon to transfer the configuration to the reader.

B.2.3 Test phase

Metals and absorbent materials have a major influence on the functioning of UHF RFID systems. Since every environment has different conditions, we recommend that you run a test with all the objects to be identified for each read point. Include neighboring readers in these tests as well as scenarios for overshoots. Run through the tests an adequate number of times to make sure that any sporadically occurring influences on the antenna fields are also tested.

The final position of the transponder should only be decided after an adequately intensive test phase so that suitable variations can be tried out if errors occur.

B.3 Dealing with field disturbances

B.3.1 Field interference and possible solutions

The superposition of radio waves and reflection by conductive materials (in particular metal) can lead to weakening or strengthening of the antenna field at certain points in space. These effects can lead to disruptions when identifying transponders that can be distinguished as follows:

Table B-1 Causes of field interference and possible solutions

Description of the problem	Solution approaches / measures for rectification
Overshoots due to increasing field strength Transponders are detected that are actually beyond the read distance.	• Reduction of the radiated power • Determining the input attenuation • Use of UHF algorithms • Changing the antenna position • Shielding measures • Varying the antenna polarization • Use antennas with a lower gain • Use antennas with adjustable polarization
Lack of separation of transponders Transponders positioned close together are detected together although the application logic requires individual detection (for example to determine the positioning order). All transponders are within the read distance.	• Reduction of the radiated power • Use of UHF algorithms • Changing the antenna position • Shielding measures • Use antennas with a lower gain
Field obliteration Due to the superposition of waves, obliteration effects occur within the read distance.	• Varying the antenna polarization • Using additional antennas • Use of UHF algorithms • Changing the antenna position • Shielding measures • Use antennas with a lower gain
Mutual interference between readers Several readers influence or disturb each other during transponder identification.	• "Interconnect" neighboring readers so that they do not send at the same time • Enable intermissions ("Settings - General" menu command) • Channel management

Description of the problem	Solution approaches / measures for rectification
Reader ↔ transponder influence: A reader communicates with a transponder that is also in the identification area of another reader.	"Interconnect" neighboring readers so that they do not send at the same time
Other sources of disturbances that can lead to restriction of transponder identification. Other sources of disturbances can occur if there are devices with similar frequency bands (for example 900 MHz) in the vicinity of the reader. The diagnostics corresponds to the influence of one reader on another. Mobile phones can also disturb identification. This is the case if a reader of the type FCC or CMIIT is operated in Europe.	The disturbances can be eliminated by temporarily turning off the suspected source of interference or its shielding. Interference can also occur with devices in other frequency bands if these are located in the immediate vicinity of the RFID antenna (e.g. DECT telephone directly in front of the RFID antenna). Common industrial interference mechanisms, such as the harmonics of frequency converters or static discharge (ESD) can also cause disturbances.

Note

Occurrence of field interference

Note that the field interferences mentioned can also occur sporadically or in combination.

B.3.2 Measures for eliminating field disturbances

Using shields

To avoid reflections, you can fit UHF absorbent material. To do this, the absorbent material is mounted at various suspected reflection points until the field disturbance no longer occurs. Where possible, avoid the use of metal structures (for example housings) and use plastic instead.

Even with reader-to-reader influence, you can use absorbent plates or shielding sheets.

Channel management

To operate the readers, depending on the country profile, you have between four and fifty send channels available. It is preferable to make the channel assignments manually. This allows you to reduce reader-to reader influence and if applicable field obliteration.

Table B-2 Example of a channel plan according to ETSI

Reader	Reader 1	Reader 2	Reader 3	Reader 4	Reader 5	...
Transmission channel	4	10	7	13	4	...
Frequency (MHz)	865.7	866.9	866.3	867.5	865.7	...

Use of multiple antennas

If you do not find the ideal antenna position to be able to identify the transponders in the various positions and alignments, you have the option of using more antennas. Multiple antennas mounted at different positions enlarge the identification range.

Enabling send pauses

If too many neighboring readers send at the same time, this causes overload of the radio channels. In this case, enable the "Intermissions" function in the "Settings - General" menu command to improve read reliability.

Varying the antenna polarization

By using linear or circular antennas, you can reduce field obliteration. This improves the reader reliability in difficult radio conditions.

The RF685R readers and the RF680A antenna also offer the option of operating the (internal) antenna linearly vertically, linearly horizontally or circularly. If more than one polarization is enabled, the polarization is changed automatically with each inventory. This increases the probability of identification in difficult radio conditions.

Changing the antenna position

In difficult radio conditions (e.g. where there is a lot of metal) it is possible that the communication between transponders and readers is impaired. You can counter this by changing the position of the antenna relative to the transponder. This also changes the multipath propagation of the radio waves and obliteration is reduced or shifted.

Use of UHF algorithms

In the "Settings - Read points" menu command of the WBM, you will find various "Tools" in the "Algorithms" area that you can use to improve the read/write reliability.

Syslog messages

C

C.1 Structure of the Syslog messages

The Syslog server collects all logbook information of the devices and informs you about specific events. The Syslog messages are received from the Syslog server over the configured UDP port (default: 514) and sent according to RFC 5424 or RFC 5426.

Syslog messages log information during access to the device. Information can be status information, such as the origin of the message or a time stamp. The Syslog protocol prescribes a specified order and structure of the possible parameters. Syslog messages are structured as follows in accordance with RFC 5424:

Table C-1 Structure of the Syslog messages

Parameter	Explanation
HEADER	
PRI	Within PRI, the priority of the Syslog message is coded into Severity (severity of the message) and Facility (origin of the message).
VERSION	Version number of the Syslog specification.
TIMESTAMP	The device sends the time stamp in the format "2010-01-01T02:03:15.0003+02:00" as local time including time zone and adjustment for daylight saving time/standard time, if necessary.
HOSTNAME	References the source device with its name or IP address. IPv4 address according to RFC1035: Bytes in decimal form: XXX.XXX.XXX.XXX If there is no information, "-" is output.
APP-NAME	Device or application from which the message originates. This parameter is not used by the device and "-" is always output.
PROCID	The process ID is used to clearly identify the individual processes, for example, during analysis and troubleshooting. This parameter is not used by the device and "-" is always output.
MSGID	ID for identification of the message. This parameter is not used by the device and "-" is always output.
STRUCTURED-DATA	
timeQuality	The structured data element "timeQuality" provides information on the system time. The "tzKnown" parameter specifies whether the sender knows its time zone (value "1" = known; value "0" = unknown). The "isSynced" parameter specifies whether the sender is synchronized with a reliable external time source, e.g. via NTP (value "1" = synchronized; value "0" = not synchronized).
sysUpTime	The "sysUpTime" parameter is metainformation about the message. It specifies the time (in hundredths of a second) since the last reinitialization of the network management part of the system.
MSG	
MESSAGE	Message as ASCII string (English)

Note

Additional information

You can find additional information on the structure of Syslog messages and the meaning of the parameters in the RFCs:

https://tools.ietf.org/html/rfc5424

https://tools.ietf.org/html/rfc5426

C.2 Variables in Syslog messages

The variables are displayed in the "Syslog messages" section in the "Message text" field with curly brackets {variable}.

The output messages can contain the following variables:

Table C-2 Possible variables in Syslog messages

Variable	Description	Format	Possible values or example
{Ip address}	IPv4 address to RFC1035	%d.%d.%d.%d XXX.XXX.XXX.XXX	192.168.1.105
{Protocol}	Protocol used or service that has generated the event.	%s	TCP | WBM | PNIO | PB | EIP | XML | OPC | DHCP
{User name}	Character string (without spaces) that identifies the authenticated user based on the name.	%s	<name>
{Action user name} or {Destination user name}	Identifies the user based on his/her name. This is not the authenticated user.	%s	<First name>.<Name>
{Role}	Symbolic name for the group role.	%s	Administrator | User | OPC UA
{Time second}	Number of seconds	%d	44
{Max sessions}	Maximum number of sessions	%d	10
{Url}	URL of the Web server that was accessed.	%s	/Engineering/Reset2Factory?r=0.6856445562508033
{Config detail}	Character string (with spaces) for the configuration.	%s	Power

C.3 List of Syslog messages

This section describes the Syslog messages. The structure of the messages is based on IEC 62443-3-3.

Human user identification and authentication

Message text	{protocol}: User {user name} logged in from {ip address}.
Example	WBM: User admin logged in from 192.168.0.1.
Explanation	Valid logon information that is provided during logon.
Severity	Info
Facility	local0
Standard	IEC 62443-3-3 Reference: SR 1.1

Message text	{protocol}: User {user name} failed to log in from {ip address}.
Example	WBM: User admin failed to log in from 192.168.0.1.
Explanation	Incorrect user name or incorrect password specified during logon.
Severity	Error
Facility	local0
Standard	IEC 62443-3-3 Reference: SR 1.1

Message text	{protocol}: User {user name} logged out from {ip address}.
Example	WBM: User admin logged out from 192.168.0.1.
Explanation	User session completed - logged out.
Severity	Info
Facility	local0
Standard	IEC 62443-3-3 Reference: SR 1.1

Message text	{protocol}: Default user {user name} logged in from {ip address}.
Example	PNIO: Default user admin logged in from 192.168.0.1.
Explanation	Default user is logged on via the IP address.
Severity	Info
Facility	local0
Standard	IEC 62443-3-3 Reference: n/a (NERC-CIP 007-R5)

Account management

Message text	Authentication was enabled.
Example	Authentication was enabled.
Explanation	Authentication was enabled.
Severity	Notice
Facility	local0
Standard	IEC 62443-3-3 Reference: SR 1.3

Message text	Authentication was disabled.
Example	Authentication was disabled.

Explanation	Authentication was disabled.
Severity	Notice
Facility	local0
Standard	IEC 62443-3-3 Reference: SR 1.3

Message text	{Protocol}: User {User name} has changed the password.
Example	WBM: User admin has changed the password.
Explanation	User has changed the password.
Severity	Notice
Facility	local0
Standard	IEC 62443-3-3 Reference: SR 1.3

Message text	{Protocol}: User {User name} has changed the password of user {Destination user name}.
Example	WBM: User admin has changed the password of user user1.
Explanation	User has changed the password of another user.
Severity	Notice
Facility	local0
Standard	IEC 62443-3-3 Reference: SR 1.3

Message text	{Protocol}: User {User name} created user-account {Destination user name} with role {Role}.
Example	WBM: User admin created user-account admin2 with role Administrator.
Explanation	The administrator has created an account.
Severity	Notice
Facility	local0
Standard	IEC 62443-3-3 Reference: SR 1.3

Message text	{Protocol}: User {User name} deleted user-account {Destination user name}.
Example	WBM: User admin deleted user-account admin2.
Explanation	The administrator has deleted an existing account.
Severity	Notice
Facility	local0
Standard	IEC 62443-3-3 Reference: SR 1.3

Authorization enforcement

Message text	{Protocol}: User {User}: Access to url {url} denied.
Example	WBM: User admin: Access to url /Engineering/Reset2Factory?r=0.6856445562508033 denied.
Explanation	Access to Web resource was denied.
Severity	Error
Facility	local0
Standard	IEC 62443-3-3 Reference: SR 2.1

Unsuccessful login attempts

Message text	• Brute force protection activated. • Brute force protection deactivated.
Example	Brute force protection activated.
Explanation	With too many failed logon attempts, the corresponding user account is locked for a specific time.
Severity	Warning
Facility	local0
Standard	IEC 62443-3-3 Reference: SR 1.11

Session lock

Message text	{Protocol}: The session of user {User name} was closed after {Time second} seconds of inactivity.
Example	WBM: The session of user admin was closed after 310 seconds of inactivity.
Explanation	The current session was locked due to inactivity.
Severity	Warning
Facility	local0
Standard	IEC 62443-3-3 Reference: SR 2.5

Limiting the number of simultaneous sessions

Message text	{Protocol}: The maximum number of {Max sessions} concurrent login session exceeded.
Example	WBM: The maximum number of 10 concurrent login sessions exceeded.
Explanation	The maximum number of simultaneous sessions has been exceeded.
Severity	Warning
Facility	local0
Standard	IEC 62443-3-3 Reference: SR 2.7

Nonrepudiation

Message text	{Protocol}: User {User name} has changed configuration.
Example	OPC: User unknown has changed configuration.
Explanation	User has changed the entire configuration. User could not be found. The "unknown" user is always output.
Severity	Info
Facility	local0
Standard	IEC 62443-3-3 Reference: SR 2.12

Message text	{Protocol}: User {User name} has changed {Config detail} configuration.
Example	OPC: User admin has changed Power configuration.
Explanation	User has changed specific configuration.
Severity	Info
Facility	local0
Standard	IEC 62443-3-3 Reference: SR 2.12

Message text	{Protocol}: User {User name} has initiated a reset to factory defaults.
Example	WBM: User admin has initiated a reset to factory defaults.

Explanation	User has initiated a reset to factory settings.
Severity	Info
Facility	local0
Standard	IEC 62443-3-3 Reference: SR 2.12

Message text	{Protocol}: {IP address} was assigned to the device {Dest mac}.
Example	DHCP: 192.168.0.1 was assigned to the device 11:11:11:11:11:11:11:11.
Explanation	The DHCP server assigned an IP address to the DHCP client.
Severity	Info
Facility	local0
Standard	IEC 62443-3-3 Reference: SR 2.12

Software and information integrity

Message text	Configuration integrity verification failed.
Example	Configuration integrity verification failed.
Explanation	Configuration integrity verification failed.
Severity	Error
Facility	local0
Standard	IEC 62443-3-3 Reference: SR 3.4

Session integrity

Message text	{Protocol}: Session ID verification failed.
Example	WBM: Session ID verification failed.
Explanation	Session ID is invalid.
Severity	Error
Facility	local0
Standard	IEC 62443-3-3 Reference: SR 3.8

Data backup in automation system

Message text	{Protocol}: User {User name} created backup file.
Example	WBM / UDP / TCP / Telnet / SSH / Console / PNIO / PB / OPC: User maier created backup file.
Explanation	User has created a backup file.
Severity	Notice
Facility	local0
Standard	IEC 62443-3-3 Reference: SR 7.3

Message text	{Protocol}: User {User name} failed to create backup file.
Example	WBM / UDP / TCP / Telnet / SSH / Console / PNIO / PB / OPC: User <user name> failed to create backup file.
Explanation	Creation of backup file by user failed.
Severity	Error
Facility	local0
Standard	IEC 62443-3-3 Reference: SR 7.3

Restoration of the automation system

Message text	{Protocol}: User {User name} failed to apply backup file.
Example	WBM / UDP / TCP / Telnet / SSH / Console / PNIO / PB / OPC: User <user name> failed to apply backup file.
Explanation	Use of backup file by user failed.
Severity	Error
Facility	local0
Standard	IEC 62443-3-3 Reference: SR 7.4

Message text	{Protocol}: User {User name} applied backup file.
Example	WBM / UDP / TCP / Telnet / SSH / Console / PNIO / PB / OPC: User <user name> applied backup file.
Explanation	Backup file successfully used by user.
Severity	Notice
Facility	local0
Standard	IEC 62443-3-3 Reference: SR 7.4

Message text	{Protocol}: Firmware {Version} was activated.
Example	WBM: Firmware V2 was activated.
Explanation	Firmware successfully activated.
Severity	Notice
Facility	local0
Standard	IEC 62443-3-3 Reference: SR 7.4

Message text	{Protocol}: Firmware activation failed.
Example	WBM: Firmware activation failed.
Explanation	Firmware activation failed.
Severity	Error
Facility	local0
Standard	IEC 62443-3-3 Reference: SR 7.4

Service & Support

D

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